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AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP RESEARCH REPORT 212

Airports and Unmanned Aircraft Systems

Volume 2: Incorporating UAS into Airport Infrastructure—Planning Guidebook

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2020

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AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

ACRP was authorized in December 2003 as part of the Vision 100—Century of Aviation Reauthorization Act. The primary participants in the ACRP are (1) an independent governing board, the ACRP Oversight Committee (AOC), appointed by the Secretary of the U.S. Department of Transportation with representation from airport operating agencies, other stakeholders, and relevant industry organizations such as the Airports Council International-North America (ACI-NA), the American Association of Airport Executives (AAAE), the National Association of State Aviation Officials (NASAO), Airlines for America (A4A), and the Airport Consultants Council (ACC) as vital links to the airport community; (2) TRB as program manager and secretariat for the governing board; and (3) the FAA as program sponsor. In October 2005, the FAA executed a contract with the National Academy of Sciences formally initiating the program.

ACRP benefits from the cooperation and participation of airport professionals, air carriers, shippers, state and local government officials, equipment and service suppliers, other airport users, and research organizations. Each of these participants has different interests and responsibilities, and each is an integral part of this cooperative research effort.

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Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

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FOREWORD

By Theresia H. Schatz
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ACRP Research Report 212 provides guidance for airports on Unmanned Aircraft Systems (UAS) in the following areas:

- **Topic A—Managing UAS Operations in the Vicinity of an Airport** educates airport operators in best practices for managing non-airport-sponsored UAS and small UAS (sUAS) activities applicable to airports of all types and categories.
- **Topic B—Engaging Stakeholders in UAS** assists airport operators to effectively engage stakeholders regarding UAS. The guidance helps airport operators identify potential stakeholders, assess the positive and negative impacts that UAS operations may have on them, and determine the best strategy to exchange this information.
- **Topic C—Incorporating UAS into Airport Infrastructure and Planning** provides guidance on the planning, development, and integration required to review and implement near-term, mid-term, and long-term facility-use improvements needed to support UAS at an airport.
- **Topic D—Potential Use of UAS by Airport Operators** explores the use of UAS to enhance the efficiency of airport operations with guidance materials to help identify, evaluate, and select UAS-related technologies, including (1) identification and evaluation of the different use cases and the types of enablers needed to support each use case such as cost-benefit analysis, training, and certification and (2) a framework for safety management system application and identification of potential risks associated with UAS.

ACRP Research Report 212 is published in 3 volumes. Topics A and B have been incorporated in *Volume 1: Managing and Engaging Stakeholders on UAS in the Vicinity of Airports*, Topic C has been included in *Volume 2: Incorporating UAS into Airport Infrastructure—Planning Guidebook*, and Topic D has been included in *Volume 3: Potential Use of UAS by Airport Operators*. This report is supplemented by *ACRP Web-Only Document 42: Toolkits and Resource Library for Airports and Unmanned Aircraft Systems*, which can be found on the TRB website by searching for "ACRP Research Report 212." The guidance provided in *ACRP Research Report 212* expands upon the guidance provided in *ACRP Report 144: Unmanned Aircraft Systems (UAS) at Airports: A Primer*.

UAS activity continues to grow for recreational and non-recreational uses. Recreational uses include applications in photography, racing, and sport. Non-recreational uses include law enforcement, emergency response, media coverage, delivery services, surveying, and utility inspection. Many airport operators see the potential benefits of using UAS for inspections, wildlife hazard management, security management, and emergency response to increase efficiency and reduce cost. The rapid increase in UAS activity—coupled with the

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diverse stakeholders employing the technology and the evolving regulatory landscape—has also resulted in airports facing new challenges as they strive to provide users, tenants, and customers with a safe, secure, and predictable operating environment. Airports need guidance, tools, and other resources to effectively address UAS issues and integrate UAS into their day-to-day operations and planning. There was a need to build on *ACRP Report 144: Unmanned Aircraft Systems (UAS) at Airports: A Primer* and *ACRP Legal Research Digest 32: Evolving Law on Airport Implications by Unmanned Aerial Systems* to provide additional guidance and information related to UAS at airports.

A thorough literature review, which included outreach with UAS technology manufacturers, former air traffic controllers, airline pilots, and several technical UAS subject matter experts, was conducted. Case studies were conducted to evaluate guidance methods targeting audiences in a variety of stakeholder groups.

Research under ACRP Project 03-42 was led by Booz Allen Hamilton in association with Embry-Riddle Aeronautical University; Hogan Lovells; Kimley-Horn and Associates; Novel Engineering; Toltz, King, Duvall, Anderson, and Associates; Vanasse Hangen Brustlin, Inc.; and Astrid Aviation and Aerospace. The research identifies airport-specific infrastructure and facilities needed to support UAS and describes field demonstrations to test various use cases for potential use of UAS by airport operators.

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CHAPTER 1

Introduction

This guidebook provides airport industry practitioners with guidance for incorporating unmanned aircraft systems (UAS) into airport infrastructure and planning. Chapter 1 includes background information on this rapidly changing industry and explains the guidebook organization.

1.1 Background

It is anticipated that UAS activity within the airport environment will expand due to market demand, operational requirements, and changes in UAS specifications (i.e., size, weight and payload). To date, most small UAS, less than 55 lbs, operate outside of the airport environment. However, some public airports are currently supporting limited UAS operations (e.g., testing, agriculture, survey, photography, and racing). As a result, this guidebook provides suggested planning, operational, and infrastructure guidance to safely integrate existing and anticipated UAS operations into an airport environment (See Figure 1).

Unmanned Aircraft Systems:

"An unmanned aircraft system is an unmanned aircraft and the equipment necessary for the safe and efficient operation of that aircraft. An unmanned aircraft is a component of a UAS. It is defined by statute as an aircraft that is operated without the possibility of direct human intervention from within or on the aircraft (Public Law 112-95, Section 331(8))."

Figure 1. FAA definition of UAS.

To develop this guidebook, the research team reviewed:

- current regulatory guidance both within the United States and abroad;
- existing and anticipated UAS activity; and
- existing UAS performance requirements and operational needs.

The research team also consulted with various airport and regulatory agency stakeholders including airport management and operating personnel, International Civil Aviation Organization (ICAO), FAA and U.S. DOT personnel, state DOT personnel, UAS commercial operators, U.S. Department of Defense (U.S. DOD) personnel, and UAS test site representatives. Through this consultation, the research team documented stakeholder concerns.

This guidebook presents processes and methods to incorporate UAS into airport infrastructure planning based on current and forecast conditions and operating needs. However, the UAS industry continues to rapidly grow and evolve while the integration of autonomy in both ground vehicles and aircraft expands. New developments occur rapidly and governing agencies are under pressure to keep pace with the changes. Regulatory, industry, and research efforts to date continue to focus on UAS integration into the national airspace system (NAS) including controlled airspace, sense and avoid, beyond visual line of sight, and radio spectrum demand rather than ground infrastructure needs. Therefore, while this guidebook strives to provide relevant guidance to support UAS airport integration, users should recognize that the evolving nature of the aviation/aerospace industry itself will impact airport infrastructure needs and funding priorities. Thus, users should continue to monitor UAS developments and regulations while working with airport stakeholders and federal and state regulators to address existing and future needs, and opportunities related to UAS development.

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This planning guidebook is particularly applicable to smaller airports (non-hub and general aviation) without capacity issues. The planning approach could help these airports prepare for and attract UAS operations for additional revenue in the near term. Larger airports (large, medium and small hubs) are likely less inclined to be interested in attracting UAS operations in the near term. They may recognize benefits of UAS for airport surveys, construction monitoring, and wildlife control but UAS used for these activities are relatively small and need very little, if any, infrastructure. That said, larger airports will have to accommodate UAS as they are integrated into the commercial cargo and passenger aircraft fleet in the future. These types of UAS are still under development. At this time, it is anticipated that commercial cargo and passenger UAS will use infrastructure similar to that used by manned aircraft. However, anticipated differences will likely include communication/sensing equipment, fueling infrastructure, and airspace control procedures. Additional research will be needed as commercial cargo and passenger UAS evolve and their characteristics and operational needs are better understood.

1.2 Guidebook Organization

This guidebook contains a comprehensive set of information for airport infrastructure planning as it relates to integrating UAS operations. It is not meant to be read cover-to-cover. Rather, readers should familiarize themselves with the chapter summaries (below) and jump around to the relevant topics.

1. **Chapter 2: UAS Terminology and Classifications.** This chapter provides basic information on UAS classifications and UAS terminology.
2. **Chapter 3: Current Conditions.** This chapter highlights current UAS regulations and guidance related to airport infrastructure and planning as well as UAS commercial, civil and military forecasts of demand (as of the writing of this report). Finally, examples of recent UAS infrastructure planning efforts are provided.
3. **Chapter 4: Airport Opportunities, Issues, and Challenges.** This chapter highlights opportunities, issues, and concerns regarding UAS operational impacts on future airport planning, environmental and infrastructure needs as described by the industry, airport management, and regulatory organizations [i.e., FAA/U.S. DOT, U.S. DOD, the European Union Aviation Safety Agency (EASA), and ICAO].
4. **Chapter 5: Airport Infrastructure Planning for UAS.** This chapter provides planning guidance based on the current conditions described in Chapter 3 and airport opportunities, issues, and challenges discussed in Chapter 4.
5. **Chapter 6: Anticipated Future Conditions.** This chapter offers insights on future UAS demand and operational needs based on research and expected resolution of issues identified in the previous chapter. This chapter also highlights suggested infrastructure and planning strategies to address likely future conditions.
6. **Appendices:** Appendices A through G provide further detail into various aspects of airport planning for UAS referred to in this document.

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CHAPTER 2

UAS Terminology and Classifications

This chapter provides fundamental UAS terminology to facilitate use of this guidebook. One important distinction is the difference between the terms "UAS" and "unmanned aircraft." The term UAS collectively refers to an unmanned aircraft, the associated control system, and the system for communicating between the two (NASEM, 2018). The unmanned aircraft, ". . . an aircraft that is operated without the possibility of direct human intervention from within or on the aircraft" (FAA, 2018a) is a component of a UAS. Thus, in this guidebook, unmanned aircraft is used when discussing the actual aircraft and UAS is used when referring to the entire system. Also, UAS, in this guidebook, refers to UAS operated by a remote pilot as opposed to a fully autonomous UAS.

Few definitive definitions of UAS types or classifications exist. Some are defined by law but even those definitions can change over time. Additionally, different countries may use different terminology.

The FAA Reauthorization Act of 2018 includes definitions of the following types of UAS:

- Small Unmanned Aircraft. "The term 'small unmanned aircraft' means an unmanned aircraft weighing less than 55 pounds, including the weight of anything attached to or carried by the aircraft."
- Actively Tethered Unmanned Aircraft System. "The term 'actively tethered unmanned aircraft system' means an unmanned aircraft system in which the unmanned aircraft component—
  (A) weighs 4.4 pounds or less, including payload but not including the tether;
  (B) is physically attached to a ground station with a taut, appropriately load-rated tether that provides continuous power to the unmanned aircraft and is unlikely to be separated from the unmanned aircraft; and
  (C) is controlled and retrieved by such ground station through physical manipulation of the tether."

The FAA Modernization and Reform Act of 2012 also defined a model aircraft as ". . . an unmanned aircraft that is

(1) capable of sustained flight in the atmosphere;
(2) flown within visual line of sight of the person operating the aircraft; and
(3) flown for hobby or recreational purposes." (FAA, 2012a)

However, the FAA Reauthorization Act of 2018 repealed this definition (FAA, 2018a). Regardless, the term model aircraft, also referred to as hobby or toy UAS, is still used in the industry. Also, in this guidebook, the term 'non-model UAS' refers to those unmanned aircraft that are not flown for hobby or recreational purposes.

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ICAO uses the term "remotely piloted aircraft (RPA)" which means "an aircraft where the flying pilot is not on board the aircraft" (ICAO, 2011). According to ICAO, RPA is a subcategory of UAS. The other subcategory is an autonomous aircraft; "an unmanned aircraft that does not allow pilot intervention in the management of the flight" (ICAO, 2011).

UAS are also classified by size, speed, or mission but these classifications are not uniform across the industry.

For industry forecasting purposes, UAS are classified by mission or use such as military, commercial, or civil. These classifications also vary within the industry. Some UAS forecasts consider commercial activity a subset of civil activity, while others classify civil as only government activity. In some cases, civil includes research, training, and educational UAS activity.

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CHAPTER 3

Current Conditions

This chapter focuses on UAS existing conditions which relate to UAS and airport infrastructure planning. Section 3.1 summarizes current regulations. Section 3.2 discusses UAS forecasts, and Section 3.3 reviews the status of UAS infrastructure design standards. Finally, Section 3.4 describes recent UAS infrastructure planning efforts.

3.1 Current Regulations

In June 2016, the FAA published its Small UAS Rule (14 Code of Federal Regulations (CFR) Part 107). Part 107 provides clarity and a streamlined operational pathway for operators seeking to use small UAS commercially (e.g., approved business use, such as survey, photography, and real estate). Part 107, however, does not currently apply to air carrier operations, public aircraft (i.e., an aircraft used by government (U.S. or foreign) for non-military use), and exempted aircraft under 49 United States Code (U.S.C.) Section 44807.

While Part 107 broadly authorizes low-risk commercial small UAS operations in the United States, the rule contains several key operating restrictions to maintain the safety of the NAS and ensure that small UAS do not pose a threat to national security. Key operational restrictions in Part 107 include the following:

- Unmanned aircraft must weigh less than 55 lbs (25 kg) including payload.
- Visual line of sight (VLOS) operations only.
- Daylight-only operations (official sunrise to official sunset, local time). Civil twilight operations (30 minutes before official sunrise to 30 minutes after official sunset, local time) are approved when the small UAS is equipped with lighted anti-collision lights.
- Must yield right-of-way to other aircraft, manned or unmanned, public, and military aircraft.
- UAS may not operate over any persons not directly involved in the operation.
- Maximum airspeed of 100 mph (87 knots).
- Maximum altitude of 400 ft above ground level unless flown within a 400-ft radius of a structure and no higher than 400 ft above the structure's immediate uppermost limit.
- Minimum weather visibility of 3 statute miles from control station.
- No operations are allowed in Class A (18,000 ft and above) airspace.
- Operations in Class B, C, D, and E airspaces are allowed with the required Air Traffic Control (ATC) permission. ATC permission comes in the form of an airspace authorization.
- Operations in Class G airspace are allowed without ATC permission.

UAS that weigh 55 lbs or more and do not hold an airworthiness certificate will require a Part 11 exemption from various sections of the CFR and an accompanying Certificate of Waiver or Authorization (COA) as well as a Special Authority for Certain Unmanned Systems exemption (49 U.S.C. §44807). These exemptions were previously obtained under

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"Section 333" exemptions; however, Section 347 of the FAA Reauthorization Act of 2018 repealed Section 333 of the FAA Modernization and Reform Act of 2012.

Additional regulations not specific to UAS operations but that nonetheless may be applicable to UAS operations include Part 137 and Part 135. Part 137 prescribes regulations for aerial spraying and agricultural applications while Part 135 includes requirements for commercial air carrier and operator certification. While Parts 135 and 137 are currently being used now to regulate UAS operations, a more tailored regulatory approach for UAS will be needed in the future (Thipphavong et al., 2018; Allred, Eash, Freeland, Martinez, and Wishart, 2018).

On October 5, 2018, the President signed the FAA Reauthorization Act of 2018. The FAA is currently evaluating the impacts of the act and assessing how implementation will proceed. The 2018 Act impacts the entire aviation system for the next 5 years through 2023. Regarding UAS, the 2018 Act supports the continued development of the unmanned traffic management (UTM) system and will help enable beyond visual line of sight (VLOS) and package delivery UAS operations (McMahon, 2018). The 2018 Act also supports current and future test sites to foster the development of sense and avoid technology, in addition to educational development and training programs. Regarding model aircraft, the act "establishes new conditions for recreational use of drones and immediately repeals the Special Rule for Model Aircraft [Section 336]" (FAA, 2018a). As such, the act clears the way for FAA to enact a rule that requires remote identification for all (or most) UAS.

In February 2019, the FAA published a notice of proposed rulemaking. The proposed rule would establish performance-based standards and means of compliance for allowing small UAS operations over people. The comment period for this notice of proposed rulemaking closed on April 15, 2019.

The FAA also published Advance Notice of Proposed Rulemaking (ANPRM) on February 13, 2019, after requesting public comments on UAS-security related issues. ANPRM highlights safety and security concerns provided by homeland security, federal law enforcement, and national defense communities related to UAS. Specifically, public comments will be solicited on several operational limitations, airspace restrictions, hardware requirements, and associated remote identification or tracking technologies for UAS. ANPRM followed the publication of *UAS Identification and Tracking (UAS ID) Aviation Rulemaking Committee (ARC)* in 2017, which included recommendations on issues related to identifying and tracking drones in flight.

Most recently, in February 2019, the FAA published 8900.504 Expanded Unmanned Aircraft Systems Oversight (FAA, 2019). The implications of this notice to Flight Standards District Offices are not yet entirely clear. However, the notice indicates the FAA's interest in expanding its efforts to assure that UAS are being adequately studied and regulated within the NAS.

Another important regulatory development is that the FAA Office of Airports is updating the definition of "aeronautical activity"—as it pertains to airport access—to include certain UAS. This new definition will be included in the next update to FAA Order 5190.6B, FAA Airport Compliance Manual. Note, however, defining UAS operations as an aeronautical activity does not itself confer federal funding eligibility under the Airport Improvement Program (AIP). Eligibility or justification for AIP-funded facilities continues to be subject to the requirements of FAA Order 5100.38D, Airport Improvement Program Handbook, and FAA Advisory Circular (AC) 150/5000-17, Critical Aircraft and Regular Use Determination.

For additional information on current regulatory efforts, refer to the FAA website (www.faa.gov/uas).

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3.2 Current UAS Forecasts of Industry Demand

This section summarizes available UAS forecasts: specifically commercial, civil (i.e., academic, governmental, or research), and military demand. The commercial and civil markets are still in their infancy, and, therefore, are vulnerable to changes in market conditions, cost, regulations, safety concerns, and other technological changes. While these forecasts apply to national and global UAS activity, the trends provide localities with a starting point to conduct further research on the specific risks or opportunities they may expect to face with UAS.

Teal Group, an independent aerospace and defense research and analysis company, produces an annual market profile and forecast for global UAS demand, including world UAS production by region. Although consumer systems (i.e., recreational UAS or "model" UAS) worldwide represent approximately 50 percent of UAS currently in service, the most dynamic growth is expected in the commercial (i.e., business) sector. This sector is expected to overtake consumer demand, according to Teal Group, "commercial systems (will) surpass consumer systems in value by 2024 . . ." (Finnegan, 2015). Teal Group also anticipates a 5 percent compound annual growth rate for military drone production as well as 15.5 percent compound annual growth for worldwide commercial UAS production from 2017 through 2026. The United States currently dominates the market with approximately 50 to 60 percent of all UAS production. However, as technology expands, the U.S. market share is anticipated to decrease as other countries and businesses enter the market (Finnegan, 2017).

Unmanned Aircraft Systems (UAS): Commercial Outlook for a New Industry (2015) was a report prepared for Congress by the Congressional Research Service to explain the opportunities and likely uses for UAS within the United States and worldwide. The Congressional Report identified various sectors for continued commercial development including agriculture, real estate, construction, utilities, film making, local law enforcement, and public safety. Product development is expected to accelerate as FAA issues guidance regarding commercial UAS operations including allowing nighttime flights, beyond user/visual line of sight, and the possibility of one operator or computer/artificial intelligence system operating more than one UAS at a time. The Congressional Report highlights forecasts by IBISWorld, Deloitte, Teal Group, and Business Insider. "IBISWorld estimates that of the $3.3 billion in revenue generated in the United States by all UAS sales in 2015 (military, civil, and commercial), the civil and commercial segments will account for 3.8 percent or about $125 million" (Canis, 2015). "Deloitte estimates that this year [2015] about 300,000 non-military [commercial and civil] UAS will be sold worldwide . . . with projected revenues of $200–$400 million" (Canis, 2015). "[Teal Group] forecasted that the United States will account for 64% of research and development spending and 38% of military procurement spending [over the next decade (2015–2025)]" (Canis, 2015). "A commercial drone report from Business Insider predicts that the commercial and civil UAS market will pick up slack from declining military spending on drones, growing at a compound annual growth rate of 19% over the next 5 years, compared to growth in the U.S. military's drone spending of about 5%" (Canis, 2015). Thus, based upon the market forecast data outlined in the Congressional Report, civil and military UAS aircraft are expected to increase at an annual rate of approximately 11 percent and 5 percent, respectively, over the next 10 to 15 years.

The FAA Aerospace Forecasts, FY 2018–2038, predict that the number of non-model UAS units operating in 2022 could range from approximately 452,000 (base forecast) to 718,000 (high forecast). The number of non-model UAS operating in the United States in 2017 is estimated at approximately 111,000. Thus, predicted base and high FAA forecasts of UAS

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units anticipated to be actively operating in the United States in 2022 are approximately 4.1 and 6.5 times higher than that in 2017.

The VOLPE Centers forecast of UAS for the period 2014–2035 (U.S. DOT, 2013) estimates that UAS are anticipated to represent approximately 70 percent of the U.S. military fleet by 2035 with the primary user being the U.S. Department of the Army (DOA). This data supports current trends that unmanned aircraft will ultimately resemble and operate like current manned (i.e., pilot on board and controlling) aircraft. U.S. DOD already uses UAS that in size, weight, and operating criteria resemble small (less than 12,500 pounds) and medium (greater than 12,500 pounds) sized manned aircraft.

Historically, U.S. DOD has led research, training, and certification efforts related to new technology including UAS. Thus, it is anticipated that U.S. DOD UAS research and integration efforts will continue to spearhead UAS growth and changes in the U.S. aviation/aerospace regulatory environment.

3.3 UAS Airport Design Guidance Summary

This section describes existing U.S. and international guidance regarding UAS infrastructure design at airports.

3.3.1 United States Airfield Design Standards and UAS

While the FAA and U.S. DOD publish standards for airfield and heliport design, only the DOA has standards that specifically address UAS.

The FAA publishes various standards and recommendations for the design and layout of airside features at civil airports via ACs, orders, planning guidance letters, and standard operating procedures. The primary sources for civil airport airfield design criteria are found in AC 150/5300-13A, Airport Design (Change 1), (2014a); AC 150/5390-2C Heliport Design; and AC 150/5000-17 Critical Aircraft and Regular Use Determination. However, none of these ACs specifically address UAS operating and infrastructure needs. Further, based upon discussions with FAA personnel, there are no plans at the time of this writing for specific UAS airport infrastructure guidance to be published.

Regardless, because UAS are by law designated as aircraft, several FAA sources, including personnel from UAS Integration Office, Airports, Aviation Safety and Air Traffic Offices, both headquarters and regional offices, FAA Technical Center personnel, as well as Policy, International Affairs, and Environmental personnel, indicated that AC 150/5300-13A guidance may be relevant for determining infrastructure needs to support UAS operations. Similarly, since a substantial portion of UAS currently in operation and in design allow for vertical takeoff and landing (VTOL), AC 150/5390-2C guidance may be relevant for defining infrastructure needs to support VTOL UAS rotorcraft operations.

U.S. DOD publishes specific airfield design and operational standards outlined in *United Facilities Criteria (UFC) 3-260-01 Airfield and Heliport Planning and Design*. In addition to planning and design criteria, this document also contains guidance for construction, sustainment, restoration, and modernization of runways, taxiways, aprons, and other facilities on airport property. Standards and recommendations are provided for Navy, Air Force, Army, and Marine Corps facilities. UFC 3-260-01 does not specifically address UAS infrastructure design.

The DOA, however, provides UAS infrastructure design guidance in its *Engineering Technical Letter (ETL) No. 1110-3-510,* published in May 2013 and expired in May 2018. To date, no information is available regarding an update to this document. This document provides

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specific aviation complex planning and design criteria for UAS including runway and movement area design criteria such as airfield facility dimensions, lateral safety clearances, and imaginary surfaces for seven unmanned aircraft (RQ-4A/B Global Hawk, MQ-9A Reaper, MQ-1B Predator, MQ-1C Gray Eagle, MQ-5B Hunter, RQ-7A/B Shadow 200, and the MQ-8 Fire Scout).

The DOA ETL specifies the design requirements for facilities used by both manned and unmanned aircraft (dual use facilities) based on the critical aircraft(s) operating criteria (i.e., approach speed, wingspan, tail height, and landing gear) and a combination of dimensional standards outlined in UFC 3-260-01 and in ETL 1110-3-510, whichever is the most stringent. The ETL also provides required separation distances between runway centerline and UAS support equipment if the airfield does not conform with clearances required for UAS operations, specifically the Tactical Automated Landing System (TALS) and Tactical Automated Landing System—Tracking System (TALS-TS).

For UAS-only facilities, the DOA ETL supplements the airfield design criteria provided in UFC 3-260-01. Unlike manned/unmanned runways, lighting is not required for UAS-only runways. Also, taxiways are required for only some of the unmanned aircraft. The DOA ETL requires that UAS-only runways be marked with a "UAS" on each end of the runway and notes that these runways are not designed to support standard manned fixed wing or rotorcraft takeoff, landing, or taxiing operations.

The DOA ETL also defers to UFC 3-260-01 regarding helipad design criteria, specifically the limited-use helipad (50 ft by 50 ft). Four new layout diagrams were added with dimensions recommendations for specific helicopters (UH-60 and CH-47/CH-53) and those sharing similar characteristics. The DOA ETL states that the performance and clearance requirements are currently being developed for helicopter UAS. The DOA also published an update to UFC 3-260-01 *Airfield and Heliport Planning and Design* in May 2014 (U.S. Army, 2014). The main addition to UFC 3-260-01 is a new paragraph concerning elevated helipads. It defines what elevated helipads are, their location relative to the level of the roof, and recommended structure and design, including the recommendation for separate access points. Elevated helipads are relevant to UAS infrastructure planning because there are industry proposals to create UAS takeoff/landing sites on top of buildings, including parking structures.

However, AC 150/5390-2C also includes guidance on heliport site selection, and differentiates among various purposes (i.e., general aviation heliport, transport heliport, hospital heliports, helicopter facilities on airports) which, along with the detailed guidance provided in the military standards, forms a basis to determine how to modify the airside layout of the airport, if necessary, to accommodate manned and unmanned operations.

3.3.2 International UAS Infrastructure Standards

Based on discussions with international aviation professionals on November 1, 2018, and review of ICAO and EASA documentation, there are no specific international standards for UAS airport infrastructure. ICAO has not published airfield infrastructure standards specifically for UAS nor has Annex 14, Aerodromes, Volume I, Aerodrome Design and Operations, been updated to specifically address UAS. However, ICAO has published two documents that address UAS integration challenges including issues at airports: ICAO Circular 328, Unmanned Aircraft Systems (UAS), and Document 10019, Manual on Remotely Piloted Aircraft Systems (RPAS).

In 2007, during the second informal ICAO meeting on UAS, the need for a strategic guidance document to harmonize integration of UAS was recognized. ICAO Circular 328 was published in 2011. The purpose of ICAO Circular 328 was threefold: (1) present the ICAO perspective

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on integrating UAS into non-segregated airspace and at airports, (2) identify and take into consideration the differences between the operation of manned aircraft and unmanned aircraft, and (3) encourage collaboration and sharing of information between member states (ICAO, 2011).

ICAO Circular 328 noted that ICAO's existing Standards and Recommended Practices (SARPs) generally apply to UAS, since unmanned aircraft is considered an aircraft. That said, ICAO recognized that UAS-specific SARPs may be required to supplement the existing SARPs.

However, ICAO Circular 328 did not address commercial passenger UAS nor autonomous UAS. According to ICAO, UAS carrying passengers "will not, for the foreseeable future, have passengers on board for remuneration" (ICAO, 2011). Likewise, ICAO assumed that only remotely piloted aircraft could be integrated into civil aviation in the foreseeable future.

In terms of aerodromes, ICAO Circular 328 noted that integration of UAS at airports will be a challenge. Regardless, ICAO Circular 328 went on to state ". . . aerodrome standards should not be significantly changed, and the equipment developed for RPA [Remotely Piloted Aircraft] must be able to comply with existing provisions to the greatest extent practicable" (ICAO, 2011). Also, in terms of airports for unmanned aircraft operations only, ICAO Circular 328 stated that current design standards would apply, along with special alterations to accommodate UAS-specific issues. Circular 328 also noted areas of concern for aerodrome operations given the unique characteristics of RPAs:

a) Applicability of aerodrome signs and markings for RPA;
b) Integration of RPA with manned aircraft operations on the maneuvering area of an aerodrome;
c) Issues surrounding the ability of RPA to avoid collisions while maneuvering;
d) Issues surrounding the ability of RPA to follow ATC instructions in the air or on the maneuvering area (e.g., "follow green Cessna 172" or "cross behind Air France A320");
e) Applicability of instrument approach minima to RPA operations;
f) Necessity of RPA observers at aerodromes to assist the remote pilot with collision avoidance requirements;
g) Implications for aerodrome licensing requirements of RPA infrastructure, such as approach aids, ground handling vehicles, landing aids, and launch/recovery aids;
h) Rescue and firefighting requirements for RPA, if applicable;
i) RPA launch/recovery at sites other than aerodromes;
j) Integration of RPA with manned aircraft in the vicinity of an aerodrome; and
k) Aerodrome implications for RPA-specific equipment (e.g., remote pilot stations) (ICAO, 2011)

ICAO Circular 328 also discussed unique safety and security considerations relevant to aerodrome design. One such consideration was access to a remote pilot station. A remote pilot station is defined as "the component of the remote pilot aircraft system containing the equipment used to pilot the remotely piloted aircraft" (ICAO, 2011). Since the remote pilot station is separate from the aircraft itself, the station itself and installed equipment must be designed to withstand operational, environmental, and security conditions, including unlawful communications interference, as designated in the Remote Piloted Aircraft System (RPAS) certified flight manual. Both ICAO and EASA provide specific design, operating and equipment guidance related to remote pilot station standards (European Union Aviation Safety Agency, 2016). Further, both recommend that access control to the remote pilot station be to the same or greater level of security as currently applied to commercial aircraft.

ICAO also developed the *Manual on Remotely Piloted Aircraft Systems (RPAS)* to ". . . provide guidance on technical and operational issues applicable to the integration of RPA

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in non-segregated airspace and at aerodromes" (ICAO, n.d.). Chapter 15 of the manual identifies UAS integration issues with airports that should be considered by UAS stakeholders including airport operators. Integration issues are similar to those identified in ICAO Circular 328. The manual also notes that aerodrome emergency response plans should include guidance related to RPAS and operator safety management plans.

3.4 Recent Examples of UAS Infrastructure Planning

This section describes recent UAS infrastructure planning at airports and at droneports (i.e., airfield facilities supporting UAS-only operations).

3.4.1 UAS Infrastructure Planning at Airports

To date, UAS infrastructure planning and integration at public airports has been limited. The Eastern Oregon Regional Airport (PDT) Master Plan (October 2018) is the most robust example of recent UAS infrastructure planning. "The evaluation of UAS facility needs and operational issues as an element of the Eastern Oregon Regional Airport Master Plan represents the first known FAA-funded airport master plan in Oregon or the Northwest region to integrate UAS into conventional airport planning" (Century West Engineering, 2018).

Other planning efforts to incorporate UAS infrastructure at existing airports have been minimal likely due to lack of funding, interest, or local regulatory agency(ies) support. For example, the Airport Master Plan for the Silver Springs Airport (SPZ) in Nevada included limited reference to UAS. Since Nevada was selected as a designated UAS testing site, the SPZ Master Plan recommendations included constructing a designated area for a UAS apron, Aerial Operations Control Center and vehicle parking. The SPZ Master Plan also states that the ". . . UAS testing and training area should be segregated from manned aircraft movement areas . . ." (Armstrong Consultants, 2017). The 2017 Mankato Regional Airport (MKT) in Minnesota included limited UAS forecasts, airspace, and facility needs related to UAS flight training and agricultural use as well as recommended dual use (i.e., manned and unmanned) infrastructure improvements which were highlighted in the report and airport layout plan (ALP).

The City of Pendleton, Oregon, began the process to update the PDT Master Plan in 2015. A final draft of the PDT Master Plan was published in December 2016 and the final report was published in October 2018. The final report can be found on the City of Pendleton's website at https://pendleton.or.us/article/airport-master-plan. A summary of this plan can be found in Appendix A.

3.4.2 Droneport Planning

Droneport infrastructure varies widely based on the intended services. A droneport, which may vary in size and scope, is a facility designed specifically to support UAS operations, and in some cases UAS operations only, rather than manned aircraft (Daniels, 2018). Droneport planning ranges from a 'do more with less' design for delivery of materials in Africa to futuristic concepts for anticipated urban air mobility. Examples of droneport concepts and plans are highlighted in the following paragraphs.

The Norman Foster Foundation, a non-profit institution, is advancing the droneport project based on a design by British architect Foster + Partners. The droneport project will facilitate delivery of medical supplies and necessities to communities in Africa that are difficult

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to reach. Droneport infrastructure consists of a vaulted brick structure that the communities can build themselves. The design is flexible allowing multiple vaults to be linked together to meet community needs and evolving drone technology. A prototype was unveiled in 2016, and a pilot droneport is to be developed in Rwanda (Norman Foster Foundation, 2016).

The Eldorado Droneport and the USA Drone Port are designed to facilitate UAS testing, research, and training. The master plan for the Eldorado Droneport located near Boulder City, Nevada, intends to develop the following infrastructure:

- "50 acres zoned Light Manufacturing, accommodating R & D, testing and evaluation, assembly and manufacturing,
- Dronecube (60' x 60' x 30')—Configurable with geotextile surface landing pad,
- A/C temp office space,
- A/C pilots lounge,
- 500-foot runway, and
- Charging stations" (The Aerodrome, LLC, 2019).

The Naval Base Ventura County Point Mugu houses Navy unmanned patrol aircraft. The recently refurbished hangars provide maintenance and support for Navy UAS at a fully operational military airfield. They have the following infrastructure:

- Hangar for four MQ-4C Triton aircraft,
- Runway 3-21 (11,100 x 200 ft) and Runway 9-27 (5,500 x 200 ft), and
- Designated UAV Zones for proper airspace separation.

The USA Drone Port or National Unmanned Robotic Research and Development Center is located near Hazard, Kentucky. The USA Drone Port currently includes multiple flight areas and support facilities from asphalt runways to indoor flight testing and robotic manufacturing and printing. Currently, there are plans to expand beyond the three existing runways to provide:

- A secure on-site building that houses advanced manufacturing equipment;
- An indoor unmanned flight-testing facility that provides for all-weather testing 24 hours-a-day;
- Office space;
- Hangars;
- Computer labs;
- High-speed Internet access; and
- A pilot safety shelter (USA Drone Port, n.d.).

USA Drone Port is an example of the possibility of integrating UAS and manned aircraft into one airport. It includes a simplified layout with hangars and facilities for UAS:

- A 3,500-ft runway and circular landing pads;
- Solution shops as well as hangars; and
- Gigabit enabled office space, meeting areas, common areas and computer labs (USA Drone Port, n.d.).

Examples of the most futuristic UAS infrastructure include proposals for Skyports presented at the 2018 Uber Elevate Summit. Skyport concepts were developed by Uber partner architects to accommodate more than 4,000 electric vertical takeoff and landing (eVTOL) passengers per hour within a three-acre area (Dickey, 2018). The Mega Skyport by Corgan consists of modular components that could be built in open spaces or on top of structures such as parking garages or skyscrapers (Dickey, 2018). Another concept developed by Pickard Chilton and Arup is called the Sky Tower. Individual modules of the Sky Tower are adaptable in that they could be constructed vertically or horizontally (Pickard Chilton, n.d.).

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As demand and technology continue to evolve, future airports will likely expand beyond their traditional role as a transportation center to become an attraction in their own right. Future airports will fill this new role, as shown by international trends, by providing community areas, living spaces, attractions, and other amenities to support user changing needs and wants. This will also likely require existing infrastructure, such as ground level parking and garages, to be retrofitted to support new demand. Thus, airport planning in the future must consider the impacts that UAS, urban air mobility, automation, artificial intelligence, and sustainability will have on airport infrastructure, funding, and financial viability.

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CHAPTER 4

Airport Opportunities, Issues, and Challenges

Although a large segment of UAS activity will operate outside the existing airport environment, a portion of the UAS commercial, civil, and military market is expected to regularly use airport facilities and infrastructure. Currently there is no source of state or federal funding for these projects, but regulatory and capacity limitations at large commercial airports may drive UAS activity and encourage investment in nearby regional and general aviation reliever airports found just outside high population markets. Driving UAS demand to smaller airports with excess capacity and infrastructure allows for more efficient use of airspace and infrastructure while stimulating regional economic growth. Both small and large UAS can provide airports opportunities for revenue enhancement and growth. However, to effectively incorporate UAS into the airport environment, several issues and challenges must be addressed.

4.1 Opportunities

At least four direct airport benefits can be attributed to the development and growth of UAS demand. These airport opportunities are highlighted in the following sections and airports are encouraged to plan for these potential uses.

4.1.1 Benefits to Airport Owners and Operators

Use of UAS by the airport sponsor and their service providers for on-airport survey, construction, security and law enforcement, and wildlife management has and continues to be allowed by the FAA through FAR Part 107 and FAA operating waivers. With FAA approval, a UAS survey of the parking garage areas at Hartsfield-Jackson Atlanta International Airport was performed as part of design and construction. Also, following several hurricanes, staff at St. Pete-Clearwater International Airport used UAS to survey the airport property and infrastructure for damage and debris.

These examples illustrate that airports can effectively use UAS to improve the quality, efficiency, and economy of their operations. Other potential uses of UAS by airport sponsors could also include supporting "ground vehicle traffic management, collision avoidance, wildlife management, environmental monitoring and even small package delivery within the airport perimeter" (Matthews, Frisbie, and Cistone, 2017).

4.1.2 Benefits to Airport Customers and Tenants

Another opportunity exists with UAS operators becoming regular users and tenants of airports. Public use airports supporting UAS activity currently include: Southern California

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Logistics Airport, Killeen-Fort Hood Regional Airport, Golden Triangle Regional Airport, Syracuse Hancock International Airport (Neubauer et al., 2015), as well as Sebring Regional and Cape May County Airports. Although a substantial percentage of current UAS operations are associated with DOD operations, commercial and civil use of UAS within the airport environment is on the rise.

As technology improves, various forecasts suggest that existing airport tenants may expand and transition part of their existing aircraft fleet to UAS. For example:

- FedEx is testing UAS for large cargo transport as well as small UAS and ground automated vehicles for customer distribution directly from the airport rather than shipping to a local distribution facility.
- Boeing and Airbus are also evaluating UAS development to support short and long-haul commercial passenger demand.
- Uber Air is looking to apply UAS for air taxi and corporate passenger operations.
- The Orlando-Sanford International Airport was contacted about a UAS air taxi service operator who was interested in ferrying passengers between the airport and Walt Disney World Resorts.

Although UAS may replace some forms of existing technology and transportation, it creates a variety of new opportunities in transportation, automation, and robotics. Thus, given planned industry growth and facility needs, airports are well-suited to support long-term demand.

4.1.3 Benefits of Fixed and Multi-Rotor UAS

The majority of small model and non-model civilian UAS utilize single or multi-copter UAS models. However, fixed wing UAS offer several benefits compared to quadcopter UAS. In addition, facilities needed to support long-range fixed wing UAS already exist at airports throughout the world, including runways, taxiways, and aprons. Further, several existing fixed wing UAS also include tilt-rotor engines which allow the aircraft the flexibility to takeoff and land vertically or horizontally.

When evaluating the UAS mission, cost, payload, and time in the air, non-model/recreational users should consider the benefits of using fixed wing UAS over multi-rotor UAS.

- Multi-rotor UAS are slower than fixed wing UAS, and, therefore, will take longer to implement avoidance procedures with other aircraft or infrastructure.
- A fixed wing UAS can cover a lot more area in less time and with less power than their multi-rotor counterparts. Fixed wing small UAS have greater range and endurance than small multi-rotor UAS and are designed for being operated beyond the visual line of sight of the pilot, like inspecting pipeline and rails and monitoring crops. Beyond VLOS refers to operations that take the UAS farther than the pilot's or any other participating visual observer's direct vision of the UAS, with vision unaided by any device other than corrective lenses (Krause, 2018).
- A fixed wing UAS can operate on an existing paved or turf-type runway for takeoff and landing, so additional special or UAS-only infrastructure may be unnecessary. For example, a catapult type launch system, which may be used for multi-rotor or fixed wing UAS, is more expensive and time consuming to use. Further, the launch and recovery system must be set up in locations that are free from obstructions and away from structures. Existing airport airfield facilities are already free from potential obstructions (natural or manmade) to air navigation.
- Long endurance UAS (all fixed wing) can fly at higher altitudes and are quieter than the rotor copter UAS. Most research and high altitude (above 18,000 ft MSL) military UAS are fixed wing.

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- Fixed wing UAS have greater payload capacity to size compared with multi-rotor UAS counterparts. Although a fixed wing UAS may not be useful for door to door cargo distribution, it would be useful for longer range medical or other high value transports. In addition, a fixed wing design would support larger payload activities, such as passenger transport and large cargo transport.
- Fixed-wing UAS can operate in higher wind conditions compared to multi-rotor UAS. Higher wind conditions drain the batteries more quickly on a multi-rotor UAS thus requiring extra power to operate.
- Repair and maintenance costs associated with fixed wing UAS models are lower than those of multi-rotor UAS models (ATL, 2018).

Multi-rotor UAS will continue to be used in support of a variety of business and public use missions; however, as shown, fixed wing UAS provide several advantages over their multi-rotor counterparts. Since airports are already equipped with much of the infrastructure needed to support fixed wing UAS operations, needed infrastructure would likely be related to communications and navigational equipment. Increased use of fixed wing UAS will provide additional revenue enhancing opportunities to the airport and proportionally a better return on investment.

4.1.4 Potential Opportunities to Enhance Airport Revenue

Airport revenues as defined in FAA Order 5190.6B relate to any paid or due revenues to the airport sponsor for the use of airport property by aeronautical and nonaeronautical users, sale of airport property and resources, and airport tax revenues. Anticipated UAS related revenue enhancement opportunities may include construction and development of both aeronautical and nonaeronautical facilities to serve UAS tenants and customer demand. Suggested opportunities include construction of a distribution center within industrial park locations on or adjacent to airport property. Further, expanded development and use of UAS for distribution of various goods and services is anticipated to trigger development of various aeronautical and nonaeronautical support facilities (e.g., UAS parts, maintenance, and avionics), which may also be located on airport property.

4.2 Issues and Challenges

Although many analysts and entrepreneurs envision strong growth in UAS, several regulatory hurdles and issues must be addressed to allow safe integration of UAS into the NAS and airport environment. According to the U.S. Government Accountability Office (GAO), the top UAS issues and challenges that must be addressed are safety and privacy (Longley, 2018). Two major safety concerns include UAS versus manned aircraft and UAS versus humans. To address the various safety, security and privacy concerns, several governmental entities along with academia and industry are working together to develop new and updated regulations for UAS integration.

As part of the FAA Reauthorization Act of 2018, several changes were made including aeronautical testing for recreational UAS pilots, unmanned aircraft tracking and registration, privacy reporting and enforcement, and UAS threat management. Additional concerns highlighted by industry, airport operators and sponsors, FAA and other stakeholders include UAS and manned aircraft communication, airport charting, airport security and risk management, funding and compliance concerns, training needs, and tenant privacy concerns, which are discussed in greater detail in Appendix B as well as in Volume 1 of this report, *Managing and Engaging Stakeholders on UAS in the Vicinity of Airports*.

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CHAPTER 5

Airport Infrastructure Planning for UAS

Suggested guidance on how to evaluate UAS demand and integrate UAS into airport infrastructure planning is provided in this chapter. Each airport has its own set of unique advantages, needs, opportunities, and issues that should be considered when evaluating the viability of UAS activity at the airport. Therefore, although the following chapters provide methodologies for evaluating UAS demand, infrastructure, and operational requirements based upon the team's experience and expertise, without the existence of regulatory guidelines and standard practices, proposed UAS and airport guidance provided herein are educated strategies. Airport sponsors should work with their federal and state aviation program managers, airport stakeholders, consultants and UAS operators to design a plan that fits their specific needs and stay abreast of the evolving UAS industry, technology, guidance, and regulations.

Proposed guidance highlighted in this chapter is based on current conditions described in Chapter 3. Case studies were conducted to test proposed planning concepts in a "real world" environment. Input from airport management, stakeholders, FAA, state DOT, UAS operators and manufacturers, and academia were used to modify and/or create planning methodologies to address existing and potential UAS airport integration. Findings from Topic A—*Managing Unmanned Aircraft Systems in the Vicinity of Airports*, Topic B—*Engaging Stakeholders in UAS*, and Topic D—*Potential Use of UAS by Airport Operators*, were also considered.

The following sections provide suggested guidance for planning UAS integration into an airport environment. Accompanying "How To" tools consisting of quick reference flow charts highlighting key components of the proposed planning process are provided in Appendix E.

5.1 UAS and Airport Planning Documentation

According to FAA, "Airport planning is a systematic process used to establish guidelines for the efficient development of airports that is consistent with local, state and national goals. . . . (and) assure the effective use of airport resources . . . to satisfy aviation demand in a financially feasible manner" (FAA, 2018b). Airport planning includes national and state system plans, airport master plans for a specific airport, and even planning studies related to a specific airport need and/or demand. Since UAS represents only a portion of an airport's total demand and infrastructure needs, two types of planning documents could be developed to evaluate UAS integration: incorporation into an ongoing airport master plan or development of a stand-alone UAS integration plan document. Both options achieve the same goal—providing a plan to effectively and efficiently use the airport's physical and financial resources to support UAS. In developing the suggested planning processes,

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existing FAA and state DOT planning, environmental, design, and funding guidance was used where applicable.

5.1.1 UAS Integrated Airport Master Plan

An airport master plan provides the airport sponsor short- (5 years), mid- (10 years), and long- (20+ years) range planning infrastructure and operational recommendations based upon existing and forecast demand, regulatory requirements, and likely changes in the aviation/aerospace industry. Integrating UAS as part of an Airport Master Plan and Layout Plan allows UAS demand, infrastructure, noise levels, and operating criteria to be evaluated as part of the overall long-range planning process.

UAS integration into master plan efforts allows efficient and holistic airport development. Therefore, airports should consider planning for UAS even though current federal and state funding may not specifically identify UAS development as part of a traditional master plan. If an airport chooses to include UAS in its Airport Master Plan, UAS analysis should be provided in a separate chapter or appendix of the Airport Master Plan narrative report to allow for easy updates as the industry evolves. There are numerous overlaps between UAS and a traditional master plan that could be addressed in such a section. For example, infrastructure recommendations should also be included in the airport's long-range development program, including financial feasibility analysis, capital improvement plan, and ALP set, and could include plans for UAS in the same estimation process.

5.1.2 UAS Planning Study

A UAS planning study is a stand-alone study that focuses on identifying airport infrastructure, operational, and financial needs necessary to support UAS activity at the airport. While the UAS planning study, like the integrated master plan, also considers an airport's long-term infrastructure, planning and financial needs, and grant assurances, it usually does not include proposed airport development beyond that needed to support UAS demand unless identified for potential dual aviation use. The size and detail of a UAS planning study depends on the goals and financial resources of the sponsor since this study is not currently eligible for federal or state funding. A UAS planning study may or may not include updates to the ALP set and/or airport capital improvement program requests provided to the state. However, like the integrated master plan study, a financial feasibility analysis including a draft 20-year capital improvement program can be useful.

Typically, this type of study is initiated when there is an opportunity to support UAS activity at an airport, and, due to timing issues, funding for an airport master plan and ALP update is not available. A UAS planning study can follow the same steps, apart from the ALP, as outlined in FAA AC 150/5070-6B, Airport Master Plan, but tailored to the specific UAS needs of the airport.

5.2 Initial Needs Assessment and Pre-Planning

The goal of any airport planning effort is to provide the sponsor and community a framework for airport development that satisfies demand while efficiently addressing resource needs. Thus, the initial sponsor needs determination and pre-planning efforts are critical. As part of the needs determination and pre-planning process, the type of planning study and the level of detail needed will be identified. The level of detail may vary, but the following activities should be considered in the pre-planning process.

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- Consider the following questions:
  - What are the airport sponsor(s) and stakeholder's vision and objectives?
  - What is the current and anticipated UAS demand, including commercial, military, and civil?
  - What is the current and anticipated non-UAS activity?
  - What capital improvement funding is available to support UAS development?
  - What is the likely return on investment associated with UAS infrastructure development?
  - Does the community, local, and state government and airport stakeholder support UAS activity and development at the airport?
  - Which stakeholders need to participate in this process (e.g., FAA ADO personnel, state DOT aviation personnel, local legislative personnel, and airport tenants)?
  - What could be the airport's liability and insurance requirements related to UAS?
  - Has the airport applied or received a COA to support UAS activity?
- Analyze and evaluate current UAS industry opportunities and challenges, such as those highlighted in Chapter 4 of this guidebook, and
- Create an airport strengths, weaknesses, opportunities, and threats (SWOT) analysis as described in the following section.

SWOT Analysis. Development of a SWOT analysis is especially helpful in determining anticipated UAS demand and the viability of an airport to successfully support potential demand. A SWOT analysis is a strategic planning tool that can be used to address issues and opportunities that may be impacting demand. *ACRP Report 28: Marketing Guidebook for Small Airports*, *ACRP Report 18: Passenger Air Service Development Techniques*, and *ACRP Report 98: Understanding Airline and Passenger Choice in Multi-Airport Regions*, all provide resources for SWOT analysis.

The following factors may be included in an airport SWOT analysis as recommended by *ACRP WebResource 1* (Ward et al., 2017):

- Size of catchment area,
- Levels of demand,
- Major airport users including industrial park tenants likely to use or support UAS,
- Major employers and universities in the vicinity involved with UAS,
- Airport facilities to support UAS,
- Community support for UAS,
- Existing marketing efforts, and
- Potential capital improvement funding.

Typically, strengths and weaknesses are internal to the airport whereas opportunities and threats usually relate to external factors. Through a SWOT analysis, an airport can identify key strengths, best opportunities, needs to rectify weaknesses, and threats to monitor and address. Data related to the SWOT examination may be obtained through coordination with airport management; economic development and county representatives; and potential commercial, civil, and military UAS users. Table 1 highlights some of the potential strengths, weaknesses, opportunities, and threats that an airport may face related to UAS development.

Although there is a great deal of uncertainty related to integrating UAS into a public airport environment, substantial opportunities abound for UAS research and development, maintenance and repair, flight and ATC training, manufacturing, and air cargo. As the UAS industry grows, other opportunities will continue to arise. Sponsors can perform a business case analysis for each opportunity (e.g., air cargo, air taxi service, and on-site manufacturing) to determine the viability of proposed development which includes potential revenues and

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Table 1. SWOT analysis results for sample UAS integration factors.

Strengths	Weaknesses
Uncongested airspace	Limited airport authority or board support
Airport already equipped with COA	Legal and insurance liability issues
Designated aviation test site (UAS or non-UAS)	Non-commercial landing fees
Excess airfield capacity	Limited developable land due to environmental constraints and long-term land leases
Available hangars, apron, and/or administrative facilities	Class B or C controlled airspace (however, FAA has been providing waivers on a case-by-case basis)
On-airport developable land	Substantial manned military and/or commercial operations
On-site aircraft rescue and fire fighting (ARFF) and other emergency facilities	Limited airport operating staff
Military facilities at the airport or near the airport	Airport is not equipped with COA to support UAS activity
On-site maintenance, repair and overhaul facilities	Limited airfield/airport capacity
On-site ATC tower	Airspace constraints such as restricted airspace, military operating areas (MOAs), and congestion
On-site industrial park	Visibility of airport to potential UAS manufacturers and users
Compatible adjacent land use (e.g., agricultural, industrial, and commercial)	Existing tenant and stakeholder concerns
	Proximity to other airports
	Existing communication infrastructure
	Insufficient cash flow to support sale of bonds for infrastructure improvements

Opportunities	Threats
State and local economic goals that support UAS development	Lack of federal, state, and local funding for infrastructure improvements
Airport use of UAS to support operations, maintenance, and management	Large manned aircraft operations—ATC and aviation tenant coordination concerns
Skilled UAS trained workforce	Federal, state, and local laws, regulations, and ordinances—limited guidance
Academic organizations providing training/education related to UAS	Competition from nearby commercial and general aviation airports
Support DOD UAS needs and demands	Environmental and compatible land use concerns
Support UAS flight training and maintenance education in conjunction with regional universities and trade schools	Community privacy, security, and safety concerns
Potential UAS commercial and air taxi opportunities (urban air mobility)	Lack of skilled labor needed to support UAS and area has a hard time attracting and keeping skilled professionals
Potential opportunity to establish airport authority which may improve cash flow and funding for infrastructure improvements	Airport supports multijurisdictions with differing goals and objectives
Market connectivity	Lack of resources to market airport both in the United States and worldwide
Low business costs (non-union work force; low cost of living, low cost of land, and low rental rates; low operating costs)	

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costs, other benefits, potential regulatory barriers, stakeholder and community acceptance as well as potential airport liability.

UAS integrated master plan, UAS planning study, or other similar documentation will assist management in addressing these issues. If the analysis demonstrates demand and local support for UAS airport growth, it can be used to justify the airport's obtaining a COA as well as related planning, environmental, and infrastructure projects.

5.3 Suggested Planning Documents Scope

The scope of an integrated airport master plan or UAS planning study may include the following components based upon client need and budget. Additional detail is provided in the remaining sections of this chapter.

- **Airport Strategic Vision and Objectives**. This section highlights the airport sponsor and community's vision for the airport and UAS during the planning period.
- **Existing Airport Conditions.** Airport master plans and UAS planning studies should identify the existing conditions of the airport, airspace, land use, environmental constraints, and aviation activity including UAS.
- **UAS Forecasts**. Several UAS forecasts should be developed as part of any UAS analysis, including likely based unmanned aircrafts; UAS operational forecasts by type of operation (i.e., commercial or general aviation/civil, military, itinerant, or local aviation); as well as the anticipated unmanned aircraft fleet mix (e.g., DOD Groups 1–5) to identify possible activity during a specified planning period.
- **Critical Aircraft Analysis**. The critical aircraft analysis identifies the "most demanding aircraft type, or grouping of aircraft with similar characteristics, that make regular use (500 annual operations) of the airport" (FAA, 2017). An airport's critical aircraft may consist of a combination of UAS and traditional manned aircraft design specifications.
- **Airport Capacity Evaluation**. Although unmanned aircraft are not currently included in FAA airport capacity models, developing an airport capacity evaluation for both manned and unmanned aircraft can identify if the current airfield capacity is adequate to accommodate future operational demand. In other words, does the airport currently have the capacity to support the addition of UAS-related operations.
- **UAS Facility and Infrastructure Needs**. Existing airport facilities and infrastructure (i.e., airfield, landside, and support facilities) should be evaluated based upon existing and future aircraft, including UAS, demand needs.
- **UAS and Airport Operational Guidance**. Existing unmanned, manned, and ATC procedures should be considered and types of operational guidance to support the introduction of UAS into airport environment evaluated.
- **Alternatives Development**. Alternatives to address UAS facility and infrastructure needs and operational requirements are assessed.
- **Airport Compliance and Environmental and Land Use Analyses**. Overall impacts of UAS integration should be evaluated by considering on and off airport land use and zoning, airport compliance with grant assurances, and existing and potential environmental impacts.
- **Implementation Plan and Financial Feasibility Analysis**. Current airport planning guidance requires both a facilities implementation plan and financial feasibility analysis.
- **Airport Layout Plan Set**. A key product of an airport master plan is the APL set, which is a graphical representation of long-term airport development including proposed UAS development.

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- **Next Steps**. Following the initial planning efforts (integrated airport master plan or UAS planning study), other efforts may be needed to support UAS development including but not limited to National Environmental Policy Act (NEPA) documentation (e.g., environmental assessments), request for COA, coordination with local legislature, changes to local ordinances, research of potential alternative funding sources, and marketing initiatives.

5.4 Airport Strategic Vision and Objectives

An airport's strategic vision, objectives, and goals identify how the airport sponsor plans to attract and maintain various aviation/aerospace activities, including UAS, along with the role of the airport both within the community and within the state and national air transportation system. The SWOT analysis conducted as part of the pre-planning process should be refined and included in this section of the airport master plan or UAS planning study. The SWOT analysis includes an overview of current and anticipated regulatory and industry changes associated with UAS that may impact airport development.

5.5 Existing Conditions

The existing conditions section of a planning study typically includes various broad categories of historical and current information needed to support the planning analysis: a brief history of the airport; a summary of current physical facilities at the airport; the current airspace environment and operational procedures, including identifying potential obstructions to air navigation; environmental and land use and zoning data; as well as historical activity. The level and detail required will depend upon the goals of the airport sponsor.

Some existing conditions are particularly relevant to UAS. These include:

- If the airport already has a COA in support of UAS operations;
- Existing airport and airfield infrastructure that may support UAS—ATC tower, runways, taxiways, helipads, aprons, hangars, fuel facilities (e.g., electric charging stations), and utilities;
- Existing airspace classifications (Classes B, C, D, or E), nearby UAS designated operating areas, military operating areas, and ATC procedures;
- Aircraft approach and departure, ground movement, and communication procedures;
- Airport zoning, land use, and obstruction identification;
- Existing environmental conditions;
- Historical manned and unmanned activity (e.g., based aircraft, operations, and fleet mix);
- Historical financial and grant assurance data;
- Existing airfield and communications infrastructure; and
- Existing safety and security protocols for manned and unmanned aircraft operations.

With the integrated UAS master plan, most of this data will be acquired as part of the traditional master planning process. Whereas, the level of existing conditions assessment needed as part of the UAS planning study effort will depend specifically on the purpose and goal of the analysis.

5.6 Forecast Process

The purpose of developing aviation activity forecasts is to provide a realistic picture of potential activity and demand as well as anticipated infrastructure and operational needs over an established planning period, typically 20 years. To provide realistic forecasts, planners use historical data, existing operations and activity, current and forecast industry trends, changes

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in technology, socioeconomic trends, and other pertinent data to evaluate and predict likely aviation demand.

The FAA Aerospace Forecast, 2018–2038, defines UAS as either model or non-model aircraft or hobby and non-hobby, respectively. Non-model UAS activity is also defined as commercial operations, which includes all business, academic, and governmental operations except DOD activities. Military UAS activity is defined by Congress, U.S. DOD, U.S. DOT/FAA and the UAS industry as unmanned aircraft used for military/DOD activity and research. Civil UAS operations are defined as public aircraft operations. Thus, public/civil UAS operations, according to U.S. DOT and Congress, are those activities performed by or associated with governmental aircraft and/or activities (e.g., mosquito spraying, U.S. Forestry Service).

The FAA's Office of Aviation Policy and Plans Statistics and Forecast Branch (APO-110) indicated that FAA's aviation and airport activity forecasts [e.g., terminal area forecasts (TAF)] do consider UAS activity. Therefore, based upon these conversations, airport forecasts established as part of the integrated or stand-alone planning studies can be compared to the most recent airport TAF. In order to compare UAS and manned operations it is critical to identify common terminology. Thus, in this forecast and fleet mix analyses, definitions are based on those outlined for aircraft in the FAA Glossary for 2016 TAF (FAA, 2016) and applied as follows to UAS operations and activity:

- **Itinerant UAS Operations** are defined as operations performed by unmanned aircraft that land at an airport arriving from outside the airport area or depart from an airport and leave the airport area. These UAS operate beyond the 20-mile radius of the airport. These UAS operations include both visual and instrument UAS operations.
- **Local UAS Operations** are airport operations (visual and instrument) performed by an unmanned aircraft that remains in the local traffic pattern, executes simulated approaches or low passes at the airport, and performs operations to or from the same airport within a designated practice area (20-mile radius of the airport).
- **Commercial UAS Operations** are defined as unmanned aircraft carrying passengers or cargo for hire or compensation.
- **General Aviation and Civil UAS Operations** are defined as all private and commercial UAS used for hire, compensation, or governmental use. In other words, all UAS operations not designated as military, "model/recreational," or for the carriage of passengers and large cargo for compensation.
- **Military UAS Operations** are operations performed by military aircraft under the U.S. DOD and Defense Advanced Research Projects Agency.
- **Recreational, Model, Consumer, or Hobby UAS Operations** are defined as operations specifically used by hobbyists or consumers for recreational purposes only. These activities are discouraged from occurring within the public airport environment and, therefore, are not included as part of the UAS activity forecasts.

Although consumer/hobby use of UAS continues to be a large segment of the market, this type of UAS activity is unlikely to regularly occur within a traditional airport environment. Instead, military, commercial UAS, and general aviation/civil UAS activity are anticipated to drive UAS operations within a public airport environment. However, most UAS are currently limited to less than 55 lbs. Thus, commercial UAS passenger and cargo activity is unlikely to occur on a regular basis within the next decade unless substantive regulatory and technology changes occur. For this reason, it is expected that only a portion of industry and governmental UAS active units and operational forecasts will likely occur at an airport. However, as the size of the unmanned aerial vehicle increases to a size resembling current manned aircraft, increased use of existing airport infrastructure (i.e., runways, taxiways, aprons, and hangars) is likely.

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There are a variety of approved methods to forecast future demand: regression and trend analysis, share analysis, exponential smoothing, survey techniques, comparison to other airports, and range projections. However, all forecast methodologies require planners to collect and evaluate historical data, existing forecast data (governmental and industry forecasts), current demand and socioeconomic data, if relevant, and any other relevant data needed to establish realistic forecasts of demand.

Forecasts, depending upon the type of existing manned and unmanned aircraft and likely UAS operations, may include:

- Based aircraft forecasts;
- Commercial manned and UAS demand;
- Military manned and UAS demand;
- General aviation manned and UAS demand;
- Aircraft, manned and unmanned, fleet mix projections; and
- Local and itinerant operations.

Using historical and forecast data specific to the airport as well as UAS national and international forecasts of demand, planners can create two (e.g., base and high) or three (e.g., low, mid and high) probable forecasts using approved statistical methodologies to provide the sponsor a range of likely activity given the various unknown variables. This type of methodology was applied in the FAA Aerospace Forecast, 2018–2038, and was used in the airport case studies conducted for this report. While this type of "range" forecast is not the only methodology that could be used, it provides realistic estimates of demand based on current conditions and UAS industry forecasts as well as provides the sponsor flexibility to address demand. Appendix C suggests steps for the creation of UAS range forecasts.

5.7 Determining Critical Aircraft

Although UAS are aircraft under the law and fit the definition of "aeronautical use" under 49 U.S.C. §40102(a)(6), 14 CFR §1.1, UAS may not currently be designated as a critical aircraft or be part of the composite critical aircraft design requirements. However, if following the FAA's recommendation of 'right-sizing' airport infrastructure to support the most demanding aircraft using the facilities, technically UAS should be included. Congress has signaled its intent to treat UAS as aircraft through the FAA Modernization and Reform Act of 2012 and the FAA Reauthorization Act of 2018. Therefore, FAA's current stance may change in the future making it important for airports to consider how UAS might impact critical aircraft.

The critical aircraft may consist of a single aircraft or a composite of the most demanding characteristics of several aircraft (e.g., approach speed, wingspan, tail height, landing gear). It also represents the aircraft or group of aircraft that regularly uses the airport (approximately 500 operations annually) excluding touch-and-go operations. An operation is either a takeoff or landing (FAA, 2017).

The critical aircraft drives airport infrastructure design requirements such as runway pavement width, strength, and separation standards. FAA also advocates for "right-sizing" airport infrastructure and facilities to support the most critical aircraft, which may not be the airport critical aircraft, that regularly uses these assets.

Depending upon UAS demand and operational criteria, the future critical aircraft at an airport could consist of a combination of manned and unmanned as well as commercial and military aircraft. Review of military, civil, and commercial existing and prototype unmanned aircraft revealed that several fixed wing unmanned aircraft have longer wingspans than traditional

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airframes and require greater runway length for takeoff acceleration. A comparison of design dimensions between manned aircraft and unmanned aircraft which require airport infrastructure (e.g., runways, taxiways, and apron) is shown in Table 2.

Based upon FAA's own guidance, UAS that regularly uses airport facilities should be included as part of the critical aircraft analyses whether for the whole airport and/or for specific airfield facilities.

Table 2. Comparison of selected manned aircraft and UAS characteristics.

Aircraft Type	AAC	ADG	# of Engines	Approach Speed (knots)	Wingspan (feet)	Length (feet)	Tail Height (feet)	MTOW (lbs)
MH-65D Dolphin Helicopter	NA	NA	1	Max speed 119	Main Rotor: 39.1	44.29	12.8	20,900
F-16 Fighter Jet	D	II	1	141	32.8	49.3	16.7	37,500
Cessna Citation Jet (III, VI)	C	II	2	126	53.50	55.5	17.25	22,000
Ultralight UAS Airframe*	A, B or C	I	1	65.17 < 130.34	9 > 40	< 30	TBD	55 to 255
Light Sport UAS Airframe*	A, B or C	I or II	1	65.17 < 130.34	15 > 45	< 45	TBD	255–1,320
Small UAS Airframe/Rotorcraft*	A-E	II	1 to 2	86.90 < 173.8	< 79	< 60	TBD	1,320–12,500
Medium UAS Airframe*	A-E	III or IV	2	86.90 < 173.8	79 > 139	60 > 95	TBD	12,500–41,000
Large UAS Airframe*	C-E	IV, V or VI	2+	86.90 < 173.8	> 139	> 95	TBD	> 41,000
MQ-1 Predator	DOD Group 4		1	~69.7 Max speed 117	48.7	27	NA	2,282
Global Hawk RQ-4	DOD Group 5		1	~126.6 Max Speed 211	131.2	48	15.4	14,991
Reaper MQ-9	DOD Group 5		1	~97.2 Max Speed 162	65.6	36	NA	10,000
Luna*	DOD Group 3		1	38	13.8	7.5	NA	88.2
Fulmar*	DOD Group 2		1	~48.6 Max speed 81	9.8	4	NA	44.1

Notes: *Catapult takeoff and net recovery.

Drone approach speed estimated at 60% of max speed.

Sources: FAA Aircraft Characteristics Database, Updated 1/6/2018, U.S. DOD Report, Drone Report 2018, Military Drones Specifications, and Commercial Industry UAS Criteria, 2018

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5.8 Airport Capacity Evaluation

Airport capacity is evaluated to determine the potential impacts of forecast demand on existing facilities as well as support for various airfield improvements. FAA provides guidance and approved programs for determining airport capacity. However, neither addresses UAS. Further, based on discussions with FAA personnel, UAS should not be included as part of the capacity evaluation at this time because most UAS fall into aircraft capacity categories A and B (less than 12,500 lbs) and thus minimally impact airport capacity. Thus, the appropriateness of conducting a capacity assessment in the early stages of UAS planning efforts is questionable unless capacity issues associated with manned aircraft have already been identified. Then, the addition of UAS operations to an already potentially constrained environment should be considered.

5.9 UAS Facility and Infrastructure Needs

As part of any airport planning process, existing airport facilities and infrastructure (i.e., airfield, landside, and support facilities) are evaluated considering existing and future demand needs. The facilities analysis evaluates aviation issues and alternatives to address or mitigate those issues. Traditionally, the facilities analysis highlights infrastructure needs to support forecast demand. This data is then used to develop airport alternatives and recommended development during the established planning period.

As previously mentioned, there are no specific airport design criteria for UAS other than that provided by the DOA. However, per discussions with FAA and state aviation personnel, existing infrastructure and design guidance may be applied because UAS ultimately will likely use the same facilities as manned aircraft. Therefore, FAA design guidance, FAA AC 150/5300-13A; AC 150/5390-2C, Heliport Design; AC 150/5000-17 Critical Aircraft and Regular Use Determination; along with supplemental data obtained from the U.S. DOD can be used to determine airport UAS facility requirements.

Ultralight, Light Sport, and Small Aircraft UAS airframes, which range in weight from 55 to 12,500 lbs are predominantly used by U.S. military, governmental agencies, and research organizations and are not yet authorized for business-related use. These UAS may require runways, helipads or some other launch, recovery and control mechanisms to safely operate. As previously noted, existing UAS wingspan and rotorcraft blade lengths are often longer than comparable manned aircraft. Therefore, using design criteria outlined in FAA AC 150/5300-13A as well as DOD recommendations, UAS wingspan will be a critical factor for future airfield design.

Medium to large unmanned aircraft are currently being tested. Again, these unmanned aircraft resemble traditional aircraft and are being used for beyond VLOS testing for commercial air cargo, firefighting, medivac, and commercial passenger activity. Existing test models require traditional airport infrastructure to operate (including runways, taxiways, parking apron, helicopter landing pads, as well as expanded navigational and communication equipment).

Thus, where warranted, it is suggested that existing FAA and DOD design standards based upon comparably-sized conventional aircraft be used to establish suggested design criteria for airfield, aircraft storage, and other support facilities. Also, given that technology, industry, and consumer demand will continue to drive UAS development, demand, safety, cost, and operational efficiency, the suggested methodology includes incorporating flexibility, reviewing compliance requirements, and considering operational recommendations. The suggested

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Table 3. Aircraft approach category descriptions.

AAC	Approach Speed
A	Approach speed less than 91 knots
B	Approach speed 91 knots or more but less than 121 knots
C	Approach speed 121 knots or more but less than 141 knots
D	Approach speed 141 knots or more but less than 166 knots
E	Approach speed 166 knots or more

methodology to identify UAS infrastructure needs and operational improvements and evaluate compliance are outlined in the following subsections.

5.9.1 Airfield

Airfield facilities include runways, taxiways, aprons, safety areas, and other infrastructure. These facilities are designed to comply with requirements for specific reference codes (Airport Reference Code or ARC). ARC is based upon the most demanding (i.e., critical) aircraft design criteria (e.g., approach speed, wingspan, tail height, maximum takeoff weight, and landing gear configuration) and approach visibility minima.

Specific guidance used to establish airfield design needs for UAS was based upon FAA AC 150/5300-13A runway and taxiway design standards. These standards cover various elements of airport infrastructure and their functions. A summary of airfield standards is provided in the following paragraphs.

Runway design requirements [Runway Design Code (RDC)] are based upon the critical aircraft approach code (AAC), Airplane Design Group (ADG) and approach visibility minima. RDC is then used to identify runway width, centerline separation to taxiways, taxilanes and aircraft parking, shoulder width requirements, and runway safety area requirements. RDC factors are provided in Table 3 to Table 5 and may be used to identify runway dimensional needs to support UAS.

UAS fixed wing regularly utilize open fields or grass strips to support operations. If due to capacity constraints or need to segregate UAS and manned aircraft activities due to safety concerns, installation of a turf or grass strip on the airport is a cost-effective option. It is

Table 4. Airplane design group descriptions.

Group #	Tail Height (ft [m])	Wingspan (ft [m])
I	< 20′ (< 6 m)	< 49′ (< 15 m)
II	20′ - < 30′ (6 m - < 9 m)	49′ - < 79′ (15 m - < 24 m)
III	30′ - < 45′ (9 m - < 13.5 m)	79′ - < 118′ (24 m - < 36 m)
IV	45′ - < 60′ (13.5 m - < 18.5 m)	118′ - < 171′ (36 m - < 52 m)
V	60′ - < 66′ (18.5 m - < 20 m)	171′ - < 214′ (52 m - < 65 m)
VI	66′ - < 80′ (20 m - < 24.5 m)	214′ - < 262′ (65 m - < 80 m)

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Table 5. Visibility minimum descriptions.

RVR (ft) *	Instrument Flight Visibility Category (statute mile)
5000	Not lower than 1 mile
4000	Lower than 1 mile but not lower than 3/4 mile
2400	Lower than 3/4 mile but not lower than 1/2 mile
1600	Lower than 1/2 mile but not lower than 1/4 mile
1200	Lower than 1/4 mile
*Runway Visual Range (RVR) values are not exact equivalents.	

important to note that instrument procedures on turf or grass strips are not allowed unless with FAA Flight Standards approval. Turf runway criteria, as provided in FAA AC 150/5300-13A, are summarized below:

- **Recommended runway length**: Landing, takeoff, and accelerated stop distance requirements due to limited friction and terrain are longer than paved runways, therefore a factor of 1.2 should be added to all landing, takeoff, and accelerate stop distance associated with the most critical aircraft (manned or unmanned) using the runway.
- **Recommended runway width and safety areas**: The runway width is based upon the same classifications as a paved runway which are based upon dimensional criteria (i.e., approach speed, tail height, and wingspan) of the most demanding aircraft regularly using the runway. Turf runway safety areas also use the same dimensional criteria as those applied to paved runways.
- **Runway grading and compaction:** Turf runways must be kept well drained to support aircraft under all conditions. Therefore, it is required that turf runways be graded to provide at least a 2.0 percent slope away from the runway centerline for a minimum distance of 40 ft to both sides of the landing strip. AC 150/5300-13A also recommends a 5.0 percent slope from that point to the edge of the RSA to provide rapid drainage. The turf runway as well as supporting safety areas, taxiways, and aprons should be compacted to allow for the safe movement of airfield maintenance and emergency equipment as well as equipment specific to UAS operations, such as launch and recovery equipment as well as UAS personnel. AC 150/5300-13A recommends applying the same strength and compaction standards used for a paved runway's safety areas. Construction and compaction criteria are further outlined in FAA AC 150/5370-10H, Standard Specifications for Construction of Airports (2018).
- **Boundary and hold markers:** Suggested landing strip boundary markers as well as hold markings are recommended as part of turf runway construction. FAA guidance recommends the installation of low mass cones, frangible reflectors, or low intensity runway lights to mark the landing strip boundaries. High mass non-frangible items, such as tires or barrels, should not be used. The preferred distance between landing markers is 200 ft. Hold position markings on adjacent taxiways or aprons should also be provided to ensure runway clearance for holding aircraft and mitigate runway incursions.

Taxiways design criteria are based not only on the critical aircraft wingspan but also on the undercarriage dimensions of the aircraft. The Taxiway Design Group is based upon the aircraft undercarriage and standardizes taxiway/taxilane width and fillet standards, and in some instances, runway to taxiway and taxiway/taxilane separation requirements. As highlighted previously, it is appropriate for airfield facilities, especially taxiways and aprons,

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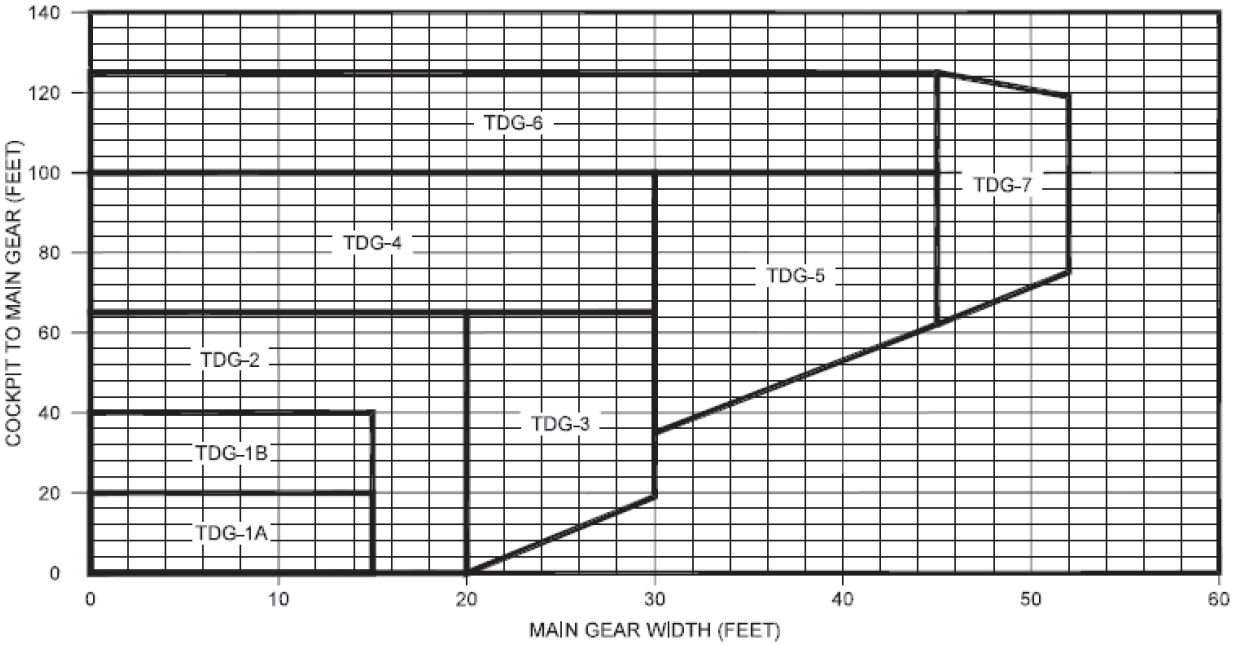


Figure 2. Sample taxiway design standards (FAA, 2014a).

to be built to different standards based upon expected use. Taxiway design standards are provided in Figure 2 and Table 6 and should be used for UAS airport infrastructure design.

Using various guidance, key infrastructure needed to support UAS and related operations may be identified. Again, suggested design guidance was based upon data obtained from the U.S. DOD, DOA, UAS manufacturers, UAS operators, and FAA design guidance.

5.9.1.1 Launch and Recovery Infrastructure

UAS use a variety of launch and recovery systems including pneumatic/hydraulic launch systems, vertical takeoff and landing, as well as horizontal takeoff and/or horizontal landing. The systems used are dependent upon the type of UAS as well as location of the launch and recovery systems.

Pneumatic/Hydraulic Launch. A pneumatic or hydraulic launcher is used to launch fixed wing drones. The system which uses gas or hydraulics, launches the UAS from a stable platform to achieve airspeed necessary for sustained flight. To achieve traditional horizontal

Table 6. Design standards based on Taxiway Design Group (FAA, 2014a).

ITEM	TDG							
	1A	1B	2	3	4	5	6	7
Taxiway Width	25 ft (7.5 m)	25 ft (7.5 m)	35 ft (10.5 m)	50 ft (15 m)	50 ft (15 m)	75 ft (23 m)	75 ft (23m)	82 ft (25 m)
Taxiway Edge Safety Margin	5 ft (1.5 m)	5 ft (1.5 m)	7.5 ft (2 m)	10 ft (3 m)	10 ft (3 m)	15 ft (4.6m)	15 ft (4.6m)	15 ft (4.6m)
Taxiway Shoulder Width	10 ft (3 m)	10 ft (3 m)	15 ft (3 m)	20 ft (6 m)	20 ft (6 m)	30 ft (9 m)	30 ft (9 m)	40 ft (12 m)
Notes: For specific taxiway separation or fillet design criteria, see Chapter 4 of FAA AC 150/5300-13A								

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launch, small UAS would require significantly more power and a long runway which is prohibitive to their effective operation. Transport of these launch vehicles is typically "hitch-mounted or trailer with weights ranging between 50 to 4,200 pounds" (Saddiqui, 2017; Davis, 2015).

Dimensional requirements for the mobile launch platform or pad are based upon the type of launch platform operational requirements as well as operator safety area. The mobile launch platform or stationary launch pad must be located within an open area free from ground and airspace obstructions.

A sample launch footprint based upon the Robonic OHTO Pneumatic Launch System (Robonic, n.d.) is:

- Stowed launcher dimension requirements (Length x Width x Height) = 5400 mm x 2100 mm x 1950 mm (17.7 ft x 6.9 ft x 6.4 ft)
- Deployed launcher requirements (Length x Width) = 16100 mm x 2100 mm (52.8 ft x 6.9 ft)
- Recommended launch pad/area dimensions (Length x Width) = 60 ft x 30 ft to allow safe distance (approximately 10 ft) between launcher and on-site operators.

The UAS launch pad should be designed to support the most demanding UAS and launch system as well as provide safety zones for UAS operators and launch personnel.

Unmanned aircraft retrieval systems associated with pneumatic or hydraulic launch vehicles may include a hardpacked surface with or without a net system, a skyhook system, or horizontal landing on runway or another similar surface. The preferred recovery system should cause minimal impact to the unmanned aircraft and be located near the UAS operations area. Recovery design specifications are dependent upon the operational requirements and size of the UAs likely to operate at the airport.

Sample recovery system dimensional criteria (Eastern Oregon Regional Airport, 2018) and considerations are as follows:

- Runway recovery:
  - Hardpacked, paved, gravel or dirt
  - Less than 1000 ft required
  - Belly landing may damage unmanned aircraft based upon runway condition and approach speed
- Skyhook recovery (an unmanned aircraft with a hook on its wing is caught on a taut cable attached to a vertical boom):
  - Stowed dimensions (Length x Width x Height): 19 ft x 7.2 ft x 6.25 ft
  - Deployed dimensions (Length x Width x Height): 28.75 ft x 17.5 ft x 58 ft
  - Operate off airport or segregated from other critical airfield facilities
  - Skyhook recovery is bulky and could damage vehicle
  - UAS vehicle could miss target, thus greater safety area required.
- Net recovery:
  - Off airport or segregated from other airport facilities
  - Net recovery is bulky and could damage the aircraft
  - Dimensions will depend upon type of vehicle supported. Typical size (Length x Height): 25 ft x 40 ft

Dimensions provided are based upon typical weight and wingspan of average unmanned aircraft likely to use this launch and recovery system. As unmanned aircraft and launch equipment become larger, the size of the launch pads must also increase in both size as well as pavement strength.

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Using these types of launch and recovery systems on airport property will require segregation of facilities from manned aircraft operations as well as coordination with users and ATC. Specific procedures should be instituted to limit interaction of unmanned aircraft launch and recovery with critical airport and aircraft activity, and safety areas should be established and marked for unmanned aircraft launch and recovery infrastructure. Further, the hydraulic and pneumatic launchers have machinery that implies a higher cost. It can be argued that this is an increase in the unmanned aircraft's operational cost and not isolated to the launch system.

Horizontal UAS Takeoff and Landing. Horizontal launch of small unmanned aircraft requires a longer runway than traditional manned aircraft due to power limitations needed to achieve required lift. This is one of several reasons why unmanned aircraft tend to have longer wingspans than typical aircraft as they maximize lift and fuel efficiency during flight. FAA Test Center and DOD data shows that conventional runways are likely to be used by fixed wing aircraft with maximum takeoff weights greater than 6,000 pounds. Unmanned vehicles at this weight and above have the engine power to reach critical airspeed needed for horizontal takeoff. Runway length, width, safety area, and separation criteria will all be impacted by critical unmanned aircraft design and visibility standards.

Unmanned aircraft larger than 55 lbs are still approved only for research and development. Therefore, in the near term, unmanned aircraft runway use will likely be limited to unmanned aircraft landings only. However, as the size and propulsion systems of unmanned aircraft become greater, use of conventional runways to support domestic and international operations is expected.

Vertical/Horizontal Takeoff and Horizontal Landing. Aircraft that can support vertical takeoff and horizontal takeoff and landing is referred to as a tiltrotor. A tiltrotor is an aircraft that is equipped with powered rotors mounted on rotating engine pods either at the ends of a fixed wing or mounted on the fuselage. This type of aircraft has the flexibility to operate in various environments where traditional horizontal takeoff and/or landing is not possible. In other words, a tiltrotor type aircraft/UAS can operate like a typical fixed wing aircraft or helicopter.

The tiltrotor drone platform consists of a combination of traditional fixed wing aircraft and helicopter designs.

For airports that support tiltrotor operations, existing runways and taxiway infrastructure are primarily used. However, runway length requirements are much shorter than fixed wing, horizontal takeoff aircraft/UAS only. Preferred runway use for tiltrotor UAS will depend upon wind conditions, runway capacity, aircraft operational specifications, runway traffic patterns, and operating requirements.

A sample tiltrotor aircraft with maximum takeoff weight of less than 12,500 lbs, a wingspan of 65 ft, and approach speed of less than 91 knots would likely fall under FAA AAC and ADG of A-II Small. Assuming visibility of not less than 1-mile, FAA AC 150/5300-13A runway design standards are provided in Table 7.

Since tiltrotor aircraft have the flexibility of both vertical takeoff and landing when necessary, another option is to design UAS helipad facilities to support both tiltrotor type UAS as well as single and multi-rotor vertical takeoff and landing (VTOL) UAS. U.S. DOD, DOA, and FAA do not provide specific guidance related to tiltrotor UAS dimension requirements.

Although FAA AC 150/5390-2C, Heliport Design, specifically states that dimensional criteria provided are not for multi-rotor aircraft, it is possible to determine likely dimensional needs by combining safety and separation criteria from multiple sources. Suggested tiltrotor takeoff and landing pad dimensions are provided as follows. Note, depending upon the type of

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Table 7. Runway design standards for A-II small aircraft design criteria not less than 1-mile visibility.

Parameter	Design Criteria
Runway Length	Depends upon Aircraft Operating Criteria
Runway Width	75 ft
Runway Shoulders	10 ft
Blast Pad Dimensions (Width x Length)	95 ft x 150 ft
Crosswind Component	13 knots*
Runway Safety Area (Width x Length)	150 ft x 300 ft
Runway Object Free Area (Width x Length)	500 ft x 300 ft
Approach and Departure Runway Protection Zone (Inner Width x Outer Width x Length)	250 ft x 450 ft x 1000 ft
*Dependent upon operating characteristics of aircraft	

UAS, propulsion requirements, and payload, additional safety separation standards may be required.

- Tiltrotor Operating Pad Proposed Dimensions:
  - Takeoff and Landing Area (TLOF)
    - Width = critical aircraft wingspan + 10 ft
    - Length = critical aircraft total length + 10 ft
  - Final Approach and Takeoff Area (FATO)
    - Width = 1.5 critical aircraft wingspan + 10 ft
    - Length = 1.5 critical aircraft total length + 10 ft
  - Safety Area Dimensions = Depending upon size and engine wake turbulence, at least 20 ft between FATO and safety area perimeters.

The siting of tiltrotor VTOL facilities must consider visibility, instrument operations, turbulence, communications, access, obstructions, and electromagnetic impacts. In the short-term, due to limited data on potential impacts, segregation of tiltrotor VTOL facilities can be useful.

VTOL. VTOL unmanned aircraft, such as quadcopters, are currently the most common commercial UAS on the market. These unmanned aircrafts are used for a variety of applications including survey, emergency management, traffic management, photography, and training due to the flexibility of their operations. Larger VTOL unmanned aircraft, currently being tested, resemble manned helicopters and are often referred to as drone helicopters. Suggested design and dimensional recommendations are provided by the FAA helicopter and DOD UAS helipad design criteria. Areas defined for unmanned vehicle VTOL operations should include a TLOF based upon design load and dimensional requirements of the unmanned aircraft as well as the FATO, safety areas, and parking positions.

Manned general aviation helipad dimensional requirements are outlined in Figure 2-2 of FAA AC 150/5390-2C (FAA, 2012), which includes detailed measurements (Figure 3).

The siting of a helipad whether for manned or unmanned aircraft must consider visibility, property requirements, obstructions, turbulence, access, communications, and electromagnetic effects.

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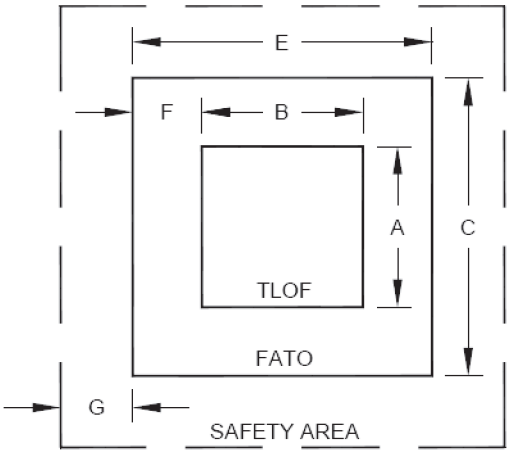


Figure 3. Manned general aviation helipad dimensions.

Depending upon use and design, dimensional protocols provided by the state-approved emergency management services may be used. A summary of approved protocols established by National Fire Protection Association and Emergency Management Services is provided as follows:

- Limited UAS helipad TLOF: 50 ft × 50 ft
- Limited UAS helipad FATO: 75 ft × 75 ft
- Standard VFR and IFR Helipad TLOF: 100 ft × 100 ft
- Standard VFR and IFR Helipad FATO: 150 ft × 150 ft

Note: Limited refers to the minimal size allowed to safely support emergency helicopter operations.

5.9.1.2 Aircraft Movement Areas

Other airfield facilities to consider for UAS integration include taxiways, parking, and movement aprons. Navigational aids, signage, and markings are critical for the safe movement of both manned and unmanned aircraft within the airport operating area. Further discussion of navigational aids is provided under support facilities. At airports that currently support UAS activities, operations are segregated from manned aircraft. Further, UAS operators must give right-of-way to larger manned aircraft. However, given testing and planned aviation industry improvements in communications, navigation, and aircraft design, segregation of operations in the long-term will no longer exist.

Taxiways. As noted, taxiway dimensional requirements are based upon the undercarriage and landing gear configuration of the most demanding aircraft regularly using the taxiway. Based on approved UAS weight and dimensions, the largest unmanned aircraft in the short-term that would likely use airport facilities would fall within Taxiway Design Groups 1A or 1B. Thus, according to dimensional criteria provided in this section and Table 7, taxiway width of 25 ft plus 10-ft shoulders would be needed. Further, assuming that most UAS that would regularly use taxiways have wingspans between 49 and 79 ft (ADG II), a separation of 66 ft between taxiway centerline and fixed or movable objects can be useful.

However, DOD recommends widths between 40 to 75 ft due to overall size, undercarriage, and UAS wingspans. If an airport is a joint-use airport which supports a variety of governmental and commercial research, taxiway widths should range between 35 and 75 ft

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depending upon critical UAS design requirements (e.g., landing gear configuration and wingspan) as well as potential manned aircraft use.

Civil/Commercial UAS Aprons. Apron requirements are based upon the size and mix of aircraft regularly using the apron. Aprons should be designed to support future expansion without major alteration of existing infrastructure and disruption of existing apron use. Aprons include aircraft movement areas, aircraft parking, as well as ground service roads. UAS aprons should also include service roads, launch and recovery, and potentially UAS Mobile Operating Vehicle parking facilities.

Apron parking needs are based upon peak hour transient and based aircraft demand, aircraft dimensions, as well as wingtip and service equipment (i.e., fueling and aircraft ground vehicle) clearances. A simplistic calculation for determining aircraft apron parking needs is ½ transient peak hour operational demand + peak hour 'based' aircraft multiplied by aircraft parking design and separation criteria. At airports where aprons support large, medium, and small aircraft, apron parking needs would further be refined to consider the peak hour fleet mix. This methodology may be used to determine apron parking requirements for anticipated UAS demand.

Military UAS Aprons and Facility Needs. If an airport is a joint-use facility, DOD apron requirements must also be considered. Although the U.S. Air Force and Army have slightly different apron parking requirements as outlined in UFC 3-260-01, an estimate of likely UAS parking needs may be developed assuming based and transient UAS/Drone activity. Based upon design information provided in UFC 3-260-01, additional apron area may be needed to accommodate various support facilities such as:

- Jet Blast Deflectors are used to reduce the impacts of jet blast on structures, equipment and personnel in addition to reducing noise and fumes related to jet engine operations. See UFC Airfield and Heliport Planning and Design Manual, Appendix B, Section 8 for more information.
- Line Vehicle Parking relates to the requirements for parking mobile station assigned and squadron assigned vehicles and equipment. See UFC Airfield and Heliport Planning and Design Manual, Appendix B, Section 12 for more information.
- Utilities including, but not limited to:
  - Storm water runoff collection system, including inlets, trench drains, manholes, and pipe
  - Deicing facilities and deicing runoff collection facilities
  - Apron illumination
  - Fire hydrants
  - Refueling facilities
  - Apron edge lighting (U.S. DOD, 2008).

The Army, Air Force, Navy, and Marine Corps all use "block" dimensions for aircraft parking. The parking blocks are roughly based upon the overall length of the aircraft, aircraft wingspan, or rotor-wing length. Service points and an interior taxiway are provided between the blocks as well as a perimeter or peripheral taxiway to support aircraft movements. Separation between parking blocks is double the parking block width, and the peripheral taxiway width is at a minimum the same width as most demanding aircraft wingspan or rotor-wing length.

A sample block parking orientation is provided in Figure 4.

In addition to aircraft apron facilities, other pavement areas include:

- Warm-up pad (holding apron)
- Unsuppressed power check pads
- Arm/disarm pad
- Compass calibration pad

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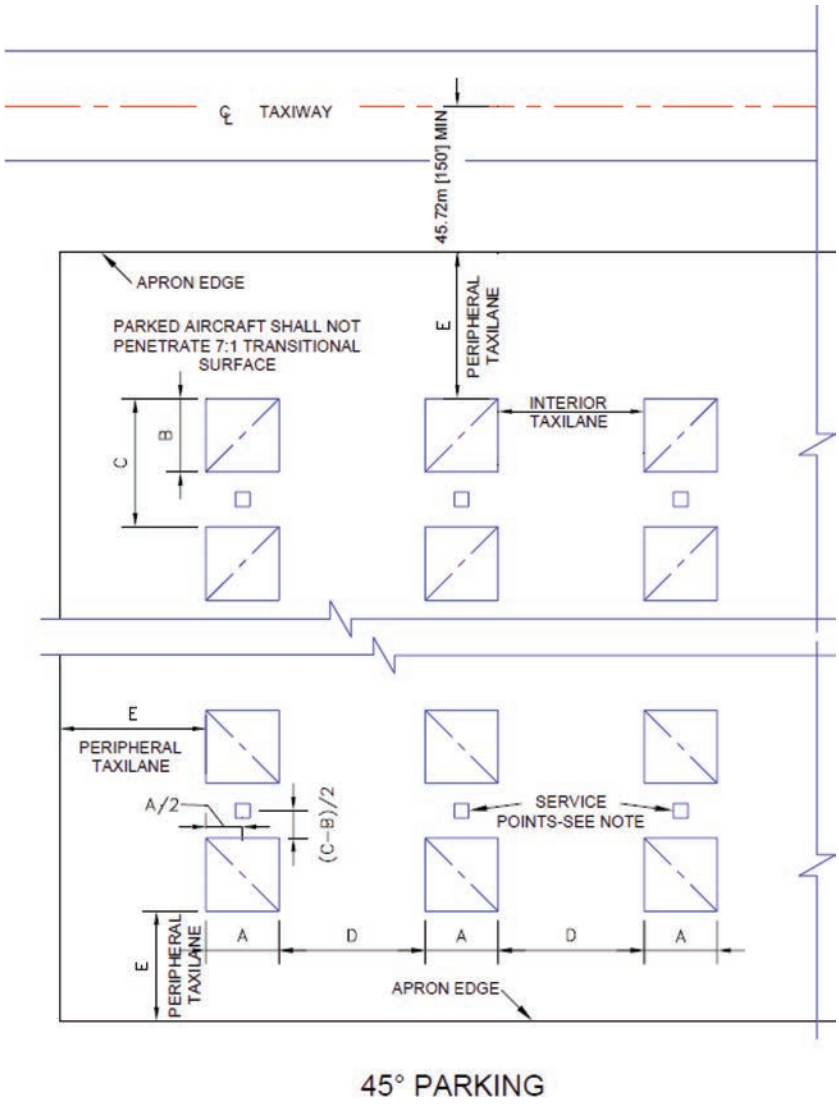


Figure 4. Sample Navy and Marine Corps 45 degree parking (U.S. DOD, 2008).

- Hazardous cargo pad
- Alert pad
- Aircraft wash rack (U.S. DOD, 2008).

Whether commercial, general aviation, or military, all apron expansion plans and movement areas must be designed to consider ATC line of sight criteria. Although FAA Air Traffic Division currently designates UAS like "birds," expanded use of UAS at airports both on the ground and in the terminal area airspace will ultimately require ATC to monitor UAS like a conventional aircraft. Thus, it is essential for all aircraft movement areas on the airport to be visible to the controllers in the ATC cab.

Also adding airfield markings and signage to identify areas where military and UAS activities will be operating should avoid accidental incursions and conflicts between manned and unmanned aircraft. Ultimately a well laid-out apron minimizes runway incursions and effectively expedites aircraft services. Additional guidance related to apron planning and design can be found in Appendix 5 of AC 150/5300-13A, AC 150/5360-9, and AC 150/5360-13.

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Holding Pads. Airport operators that support UAS activity indicated that some UAS require additional time on runways or taxiways to allow the onboard computer and operator to establish coordinates, navigation, and communication systems. As a result, UAS operations may negatively impact airfield capacity. Airport operators also suggest installation of UAS holding bays. Holding bays typically are used as standing space for aircraft awaiting clearance and are typically located adjacent to the taxiway serving the runway end. Holding bays must be located outside of the runway obstacle free zone, precision obstacle free zone, and runway safety areas to avoid interference with the instrument landing system or other navigational aids.

Holding bays can support multiple aircraft and should include clear entrance, aircraft movement, and parking as well as exit markings. Holding bays should be designed to provide adequate taxiway wingtip clearance as well as clearance between parked/standing UAs and those in route to the runway. Again, like taxiways, dimensional criteria should be based upon the most demanding unmanned aircraft likely to use the facility.

5.9.1.3 Airfield Pavement

Commercial UAS currently approved weigh less than 55 lbs whereas the largest military UAS currently in service has an operating weight of less than 40,000 lbs. As a result, most airport pavements, except for small airports catering to aircraft less than 12,500 lbs, will be able to support all current UAS activity.

Airport pavement strength must be designed to support the heaviest aircraft or family of aircraft that regularly use an airport's runways, taxiways, taxilanes and/or apron areas. In the case of UAS, depending upon the type of launch and recovery platform and support facilities, the pavement will need not only to support the unmanned aircraft but also the launch and recovery equipment, mobile operations center, and other facilities needed to support UAS operations. If UAS will use a turf or grass strip or compacted dirt apron area, then the dimensions of these areas must comply with turf runway design guidance outlined at the beginning of this section and in AC 150/5300-13A. These regulations are based on the requirements of manned aircraft and not on the safety margins unique to UAS. Until additional research is conducted on UAS-specific airfield requirements, construction guidance for turf runway and apron area should follow AC 150/5370-10H, Standard Specifications for Construction of Airports, to accommodate the unmanned aircraft, launch, recovery, operating, and support equipment, as well as airport emergency equipment.

5.9.1.4 Airfield Summary

Table 8 summarizes specific physical infrastructure based upon the type of UAS size and activity. The dimensions are based upon estimates of the type of UAS. The DOD UAS recommendations and the FAA airport dimensional standards are based on manned aircraft. Therefore, this table is designed only to provide a first step for future testing and research to establish UAS-specific design standards.

5.9.2 Airport Landside Facilities, UAS, and Autonomous Vehicles

The landside area is defined as the area of the airport that provides facilities necessary to process passengers, cargo, freight, and ground transportation vehicles. Landside facilities may also encompass facilities that may not require direct airside access such as an on-airport industrial park. Landside passenger and user access to the airport and associated parking is a major revenue stream at large general aviation and commercial airports. However, with the growth of autonomous vehicles both on the ground and in the air, demand for on-site parking is anticipated to decrease. Reuse of existing parking infrastructure to support other revenue development needs to be considered as part of not only UAS integration but overall long-term airport development.

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Table 8. Suggested UAS design standards.

	DOD UAS Recommendations	Small UAS	Ultralight/Light Sport/Small Aircraft Platform UAS	Medium and Large Unmanned Aircraft Platforms
Weight (lbs)		4.5 to 55	55 to 12,500	> 12,500
Length (ft)		< 15	15 < 60	> 60
Wingspan (ft)		< 9	9 < 79	> 79
Mission Speed (knots)		< 65	65 < 174	> 86
UAS ARC Code		A-I, A-II	C-II	D-IV
Visibility (mile)		> 1	> 1	> 3/4
Airport Infrastructure				
Launch and Recovery Requirements				
Pneumatic/Hydraulic Launch (ft)		52.8 x 6.9	NA	NA
Runway Recovery (ft)		< 1000	NA	NA
Net Capture (L x H) (ft)		25 x 40	NA	NA
Skyhook Recovery (L x W x H) (ft)		28.75 x 17.5 x 58	NA	NA
Launch Pad Dimensions (L x W) (ft)		60 x 30	200 x 100	400 x 200
Conventional Runway Launch and Recovery (ft)	Length based upon aircraft operating criteria adjusted for elevation, temperature and slope Width: 100	NA	Length based upon aircraft operating criteria adjusted for elevation, temperature and slope Width: 100	Length based upon aircraft operating criteria adjusted for elevation, temperature and slope Width: 150
VTOL Launch				
Fixed Wing (ft)	1600 x 100 (correct length for elevation and temperature)	1200 x 75 (correct length for elevation and temperature and width for landing gear)	1600 x 100 (correct length for elevation and temperature and width for landing gear)	6000 x 150 (correct length for elevation and temperature and width for landing gear)
Rotary Wing (TLOF) (ft)	50 x 50 to 100 x 100	50 x 50 (depending upon overall size of UAS and mission)	100 x 100 or critical UAS rotor diameter	100 x 100 or critical UAS rotor diameter
Launch and Recovery Safety Areas				
Approach Runway Protection Zone (IW x OW x L) (ft)	Clear Zone Range Width ranges from 200 to 3000	500 x 700 x 1000	500 x 1010 x 1700	1000 x 1510 x 1700
Departure Runway Protection Zone (ft)	Clear Zone Range Length ranges from 200 to 3,000	500 x 700 x 1000	500 x 1010 x 1700	500 x 1010 x 1700

(continued on next page)

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Table 8. (Continued).

	DOD UAS Recommendations	Small UAS	Ultralight/Light Sport/Small Aircraft Platform UAS	Medium and Large Unmanned Aircraft Platforms
Runway Safety Areas Prior to Threshold (W x L) (ft)	Runway width x up to 1000 length	120 x 240	500 x 600	500 x 600
Runway Safety Area beyond Departure End (ft)		120 x 240	500 x 1000	500 x 1000
Runway Object Free Area prior to threshold (ft)	NA	400 x 240	800 x 600	800 x 600
Runway Object Free Area beyond runway end (ft)	NA	400 x 240	800 x 1000	800 x 1000
Runway to Taxiway Separation (ft)	NA	225	250	400
Runway to Aircraft Parking Separation (ft)	Beyond primary surface (1000 to 2000 from runway centerline)	200	400	500
FATO (Rotary Wing Aircraft) (ft)	Varies based upon type and mission	75 x 75	150 x 150	150 x 150
Safety Area (Rotary Wing) (ft)	Varies based upon type and mission	95 x 95	170 x 170	180 x 180
Taxiway Requirements				
Width (ft)	40 to75	35	50	75
Safety Area (ft)	50 to 75	79	118	171
Object Free Area (ft)	NA	131	186	259
Centerline to Fixed or Movable Object (ft)	150 to 200	66	93	129.5
Apron Parking Recommendations				
Peak Hour GA UAS Apron Area (SY)	Based upon aircraft type and demand	~67	~1,133	~4,245
Peak Hour Military UAS Apron Area (SY)	Based upon aircraft type and demand	~195	~1,784	~5,960
Parking Wingtip Clearance (ft)	10 to 50	10	10	20
Fueling Clearance (ft)	25 to 50	25	25	50

Notes: FATO based upon 1.2 RD or TLOF

Taxiway criteria based upon taxiway design criteria outlined in AC 150/5300-13A. In the short-term, anticipate that most UAS may fall within Groups 1A and 1B. Proposed taxiway width includes taxiway width and taxiway shoulder width.

Sources: FAA AC 150/5300-13A; AC 150/5390-2C; ACRP Apron Planning and Design Book; U.S. DOD airfield and UAS design criteria; A3 LLC.

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UAS landside needs relate to access to UAS facilities in addition to traditional tenant parking and surface access. Long-term landside criteria should also consider the impacts of ground access improvements related to intermodal, autonomous vehicles, personal air taxis, as well as other technology. Based on discussions with FAA and other regulatory personnel, ground and air taxi vehicle automation integration is a low priority. However, based upon several industry models, automation and some types of self-driving technology is inevitable. Several car manufacturers including GM, Honda, Ford, Toyota and others anticipate some level of self-driving car technology to be in place by 2020 (Fagella, 2017). Uber, Google, Boeing, Airbus, and other UAS manufacturers and operators expect full automation and artificial intelligence to be implemented as early as 2030 based upon current published data (Wyman, 2018a). Although no specific infrastructure guidance is provided for UAS, ground automation, and other landside development, several international airports (e.g., Singapore's Changi International and London Heathrow) working with autonomous industry partners have already implemented or are implementing landside infrastructure to support autonomous ground vehicles and baggage handling systems (Park, 2018).

Landside demand related to UAS is anticipated to include tenant auto parking, access roads, as well as additional terminal parking and a pick-up and drop-off zone. It is expected that landside design criteria will mimic existing regulatory landside and parking design guidance adjusted to accommodate specific vehicle design needs. Initially, given current personnel requirements to operate a commercial UAS (typically 4 individuals: pilot, launch, recovery, observers (~2), and on-site data management/information technology personnel), parking should support at least one mid-size car, one heavy duty truck and trailer, and/or parking for large mobile transport vehicle.

Depending upon the size of the UAS and its mission along with anticipated changes to regulations, parking needs may increase. However, ground demand may also decrease once urban air mobility and UAS air taxi operations become more affordable and convenient.

5.9.3 Hangar, Administration, and Airport Terminal Facilities

Additional airport facilities likely needed to support UAS activity include hangars and adjacent apron, administrative space for business operations and remote pilot operating stations and UAS monitoring as well as terminal facilities to support future UAS air taxi and commercial passenger demand. In addition, facilities will be needed to support UAS operators permanently or temporarily based at the airport as well as transient UAS activity. Demand for transient UAS facilities is unlikely in the next 10+ years based upon existing technology, trends and demand. However, the airport sponsor should consider both based and transient UAS needs as part of long-term planning efforts. In addition, the airport sponsor before investing in UAS facility development and/or retrofitting existing infrastructure should carefully evaluate the financial impacts of proposed UAS development on existing operations and revenues, especially since limited funding is available, and evaluate the likely timing and return on investment.

5.9.3.1 Unmanned Aircraft/Vehicle Storage and Equipment Needs

Hangar and equipment storage requirements are based upon customer demand and aircraft design criteria. Conventional and corporate style UAS hangar facilities may also include areas for an administration area, service areas, parts storage as well as UAS storage. Since the needs of UAS operators may vary, airports can provide land leases and possible financial incentives to support UAS on-airport development. Airports may also want to evaluate opportunities for nearby land acquisition to support UAS growth if much of the airport property is either bound by a long-term lease or includes environmentally sensitive parcels.

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Another option is to develop or retrofit existing facilities to address UAS operator needs and mission. Initially, these facilities could be located near the general aviation apron or on-airport research facilities. Hangar development apart from the multi-use buildings should be constructed by the actual UAS tenant.

5.9.3.2 Office and Administrative Space

Office and administration space requirements are dependent upon the needs of the operator. Some operators may use a mobile operations center (MOC) while others may require a fixed administrative and operational facility.

Mobile Operations Center. Mobile facilities are typically used to support short-term research and testing or commercial needs. These mobile facilities could be located on or adjacent to an existing apron or on separate UAS pads constructed to segregate traffic while supporting activity. Like transient hangars, these UAS pads could be leased on a short-term basis.

According to companies such as NASC and Peak 3 LLC, MOC trailers are self-contained, field deployable unmanned aircraft command centers which include ground control station as well as unmanned aircraft transport, mobile workshop and ground support equipment. These mobile units are primarily designed to accommodate small UAS activity. According to user data, average MOC dimensions are 24 ft long x 8 ft wide and can be towed by a large pick-up truck. These vehicles are climate controlled and are equipped with the following facilities:

- Computer workstations;
- Other workstations;
- Ground control station and antenna suite;
- Heads up display;
- 25- to 30-ft telescopic antenna;
- Airfield communications;
- Unmanned aircraft transport;
- Mobile maintenance workshop;
- Internal communications system; and
- Weather station, WiFi, back-up power, and external lighting.

In addition to the MOC, small UAS users sometimes also rent administrative space on the airport typically located within the airport terminal. Average space requirements needed to support an average team's data processing, training, and equipment storage needs is approximately 160 to 180 square ft. However, customer demand will drive actual requirements.

Fixed Administration Facilities. The size of fixed operational and administrative facility needs is dependent upon the type of UAS platform, number of operations, and personnel requirements. The administrative footprint for UAS operations supporting medium to large aircraft platforms or small "swarm" type operations proposed by Uber and Amazon require personnel office space, aircraft operations, ground control and communications space, conference rooms, break-rooms and bathroom facilities. Separate hangar facilities would house unmanned aircraft maintenance, vehicle storage, and equipment. Depending upon needs, administrative and command control facilities could be housed above or within large hangar facilities or in an adjacent building.

5.9.3.3 General Aviation and Commercial Terminal Facilities

General aviation and commercial terminal facilities may be impacted by unmanned and autonomous short-haul, on-demand air transportation that is currently being tested by companies such as Uber, Boeing, Airbus, Kitty Hawk as well as countries including the United States,

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China, Dubai, Israel, New Zealand, and Norway. Approved commercial operations are expected to be in place by 2025 (Smart, 2018).

While there are concerns that on-demand air transportation may take traffic away from airports, it is more likely that airport facilities to support mid and long-haul operations will remain strong. However, ultimately existing landside and airside facilities will likely need to be reconfigured to support not only new self-driving land and air vehicles but also the growth of electric passenger aircraft and large UAS cargo operations.

Various airport management and governmental staff described expansion or construction of a larger general aviation terminal facility to support UAS air taxi and general aviation activity. With the Uber air taxi model, passengers would not be required to pass through security before using the service. Therefore, a process to separate secured and non-secured airport users is needed. Another option includes retrofitting existing landside facilities, such as a parking garage, to support expanded ground and air unmanned vehicles. However, control of aircraft on the landside without modifications to various ATC procedures needs to be evaluated. Therefore, airport sponsors can work with stakeholders including local authorities and FAA Flight Standards to address potential opportunities and issues related to UAS integration. For additional information on stakeholder engagement, refer to Volume 1 of this report, *Managing and Engaging Stakeholders on UAS in the Vicinity of Airports.*

5.9.4 Support Facilities

Support facilities at an airport encompass a broad range of functions to ensure the smooth, efficient, and safe operation of the airport. Support facilities typically assessed as part of the master plan update include airport rescue and firefighting facilities, airport maintenance, fuel storage, aircraft maintenance, deicing, and ground circulation, access and parking. These facilities support both manned and unmanned aviation demand and will automatically be addressed, if required, as part of an integrated master plan update. These facilities can support UAS demand with little to no modifications.

Nonetheless, due to the nature of UAS and supporting technology, additional infrastructure is needed. The need for support facilities will vary based on a number of factors including approach, departure and missed approach procedures, airport/ATC airspace and ground maneuvering, aircraft priority and communication procedures, demand, project justification, environmental review, and airport compliance related to infrastructure development as well as facilitating and maintaining airport security and funding. To identify needed support facilities, airport sponsors should coordinate plans with FAA personnel at the UAS Integration Office, Airport District Offices, and at Flight Standards as well as working with state DOT aviation representative and local DOD personnel if applicable.

5.9.4.1 Communications Infrastructure

Unmanned aircraft are tethered (virtually) to ground-based links which may be widely distributed geographically. These ground-based links are used for vehicle control, monitoring, and air traffic communications and are, to varying degrees, vulnerable to jamming, spoofing, and interference. To prevent lost link due to poor communication or intentional theft of UAS controls, a system of high-integrity, secure data links between the aircraft, the ground control stations, and air traffic facilities is needed. Securing mobile and wireless communication networks is an ongoing challenge for effective unmanned aircraft integration as well as other critical communication technologies. Therefore, the type and level of security applied to communications links will depend on the vehicle type, potential lethality (determined by size, speed, and proximity to manned aircraft and population centers),

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intended operations, and flight environment. The encryption integrity level will be defined in certification requirements (DeGarmo, 2004).

Due to these issues, the FAA Reauthorization Act of 2018 under Section 374, now requires the FAA, NTIA, and the Federal Communications Commission (FCC) to work together to provide recommendations to Congress regarding whether UAS operations should be permitted to use the current aviation spectrum and the establishment of other spectrum frequencies to support UAS (Kestleloo, 2018).

The current FAA policy requires that the UAS operator monitor the on-site ATC tower frequency during operations and call the ATC tower, on a landline, upon initiation and completion of operations. This allows the ATC tower to issue Notices to Airmen (NOTAMs) as well as provide information to manned operators regarding UAS activity in the area. If an airport is not equipped with an ATC tower, the UAS operator must monitor the Unicom Frequencies. According to FAA Airport Integration personnel, it is up to the UAS operator to look out for manned aircraft operations and defer airspace to them if there is a conflict (Williams, 2018).

UAS operators should only monitor rather than interact on the ATC or Unicom Frequencies because of concerns about "flooding the system." Therefore, in addition to monitoring the airport ground stations used by manned aircraft, UAS operators should use a separate frequency that doesn't interfere with manned or emergency management operations. The FAA Technical Center is currently leading the research on frequency overload and UAS communication needs for safe integration.

Thus, airport infrastructure will need secure communication/control and back-up power systems will be needed to accommodate UAS activities. At the time of this writing, several companies provided secure communication platforms for tracking, storing, sharing, and flight data management as well as other tools to support UAS operations. Some of these firms include: Skyward, Kitty Hawk, FreeWave and U-Team (Northeast UAS Airspace Integration Research Alliance, 2018). The costs of these services vary depending upon the type of equipment and facilities needed to support UAS operations. To support increased communications and power generation needs, larger conduits should be used to support additional cable communication needs in addition to back-up power and communication systems.

Although larger aircraft may be equipped with a collision avoidance system, small UAS are not and are still difficult to see especially in low light conditions. Ground-based airfield and airspace sensors could assist both manned and unmanned vehicles and operators in identifying and avoiding proximate traffic. However, such a solution needs to be sensitive to cost, power and siting requirements, and accuracy.

5.9.4.2 Navigational Aids, Lights, and Markings

Although commercial UAS operations may only be flown during visual flight rules (VFR) conditions and require a chase plane or ground spotter, beyond VLOS or instrument flight conditions are being tested at the various UAS test sites. As part of FAA's planned upgrades to the NAS, it is expected that UAs, like manned aircraft, will be equipped with Automatic Dependent Surveillance-Broadcast (ADS-B) systems along with onboard sensors to provide on-airfield situational awareness. The type of navigational aids required to support UAS activity will depend on whether the unmanned aircraft is manually operated or autonomous. Traditional visual NAVAIDs such as runway end identification lights, approach lighting systems, lighted signage and even wind cones may not be needed to support autonomous UAS activity since guidance will rely less on visual aids and more on radio frequencies and airport mapping (i.e., AGIS).

However, since regular use of autonomous UAS is not anticipated within the next 10 years, sponsors have time to obtain funding and install additional airfield lighting, signage, and

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markings along with meteorological and magnetic equipment that will support both manned and unmanned aircraft operators. Markings include the addition of specific UAS hold lines on the apron movement areas and taxiways to limit accidental runway incursions and wake turbulence impacts as well as provide for greater visibility. Also, if general aviation aprons support UAS commercial activity, the aprons should include markings, lights and signage to alert manned operators of UAS activity.

5.9.4.3 Meteorological Facilities

Unmanned aircraft are lighter, slower, and more fragile than their manned counterparts and consequently are more uniquely sensitive to certain meteorological events such as surface/terrain-induced (boundary layer) winds, turbulence, icing, extreme cold, and precipitation. Small unmanned aircraft and those having a light wing load are especially sensitive. Therefore, installation of additional wind cones, segmented circles, and magnetic wind rose near or adjacent to the UAS operating areas would also be needed to support safe operations. Because of limited sight distance associated with UAS activities and the need for sense and avoid technology to be added to these aircraft to support beyond VLOS operations, use of the magnetic wind rose will be key to allowing UAS navigational and other sensors to be accurately configured prior to operations.

5.9.4.4 Physical and Data Security Infrastructure

The design and operation of UAS present unique security challenges for users, ATC, and airports. The variety of UAS as well as their missions can make secure control of UAS flights challenging. In addition to airport physical security requirements to prevent unauthorized access to critical facilities, UAS introduce some additional requirements. Security requirements of the ground control station, data link infrastructure, vehicle, and even the data must be a fundamental consideration in system design and operational policies and procedures of UAS (DeGarmo, 2004). Along with UAS control facilities, redundant and alternate systems for communication infrastructure must also be considered. Thus, site development for fixed UAS facilities as well as pads for mobile operating centers should integrate additional security and safety systems to protect cable, power, and other critical utilities.

5.9.4.5 UAS Geofencing and Counter-UAS Technology

Geofencing and similar technology limit where UAS may fly based upon the installation of specific built-in software, firmware and global positioning system (GPS) tracking to avoid entry into controlled or protected airspace. This type of technology does not require on-site facilities to operate but could be supported by improved airport surveys and the availability of detailed GIS data.

As previously discussed, technology vendors are again contacting airports proposing to demonstrate or install their UAS detection and countermeasure systems. Since the FAA has not authorized any specific technology, and unauthorized UAS countermeasure systems can cause a variety of problems, airport sponsors should contact their local FAA Airport District Office before pursuing any agreements or testing.

5.9.4.6 Fuel Storage and Handling

Unmanned aircraft use a variety of fuel systems based upon the engine propulsion system, aircraft type (aircraft fixed wing, rotary, and lighter than air), mission, weight, heat management, operator, and flight range requirements and auxiliary power requirements. Current propulsion systems are based upon different types of internal combustion engines which may use gasoline, 100LL, JP-8, Jet A, and kerosene. Other potential energy sources include electricity, solar energy via photovoltaic cells, hydrogen, methanol and energy mechanics. Onboard fuel

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storage includes batteries, fuel tanks and capacitors. Coordination with UAS operators will be needed to determine the most efficient fueling systems.

Although existing UAS engines, especially larger commercial or military drones, are still driven by internal combustion engines and fossil fuels, alternative fuel systems are in development. These systems include alternative reciprocating engines fueled by biofuels, Jet A, or other clean fuel systems; electric engines which use batteries, electric fuel cell, or photovoltaic system, and hybrid systems which include a combination of both traditional fuel and battery back-up (Gonzalez, Leo, and Navarro, 2014).

Depending upon the type of fuel requirements, expansion of the existing fuel farm or construction of a new fuel farm may be needed to support UAS traffic as well as additional types of fuel including electrical, kerosene, and JP-8. In addition, special permitting will likely be required especially if UAS operators plan to provide their own fuel storage.

Growth in electric and hybrid unmanned aircraft engines will require charging stations to be located either adjacent to parking aprons or a designated fueling area could be developed to support manned and unmanned operations. This would allow all fueling facilities to be centralized and permitted at one location. If a centralized facility is warranted by manned aircraft operations, space should be set aside to accommodate future UAS fueling needs. In the next 5 to 10 years (short-term), a centralized charging station for ground vehicles as well as small UAS may be sufficient to accommodate UAS demand, assuming similar technology.

However, this will depend upon the type and level of UAS demand likely to be supported at that airport. For example, at Sebring Regional Airport which already supports UAS activity, its two electrical charging stations adjacent to the terminal facilities are currently sufficient to support ground vehicle electrical demand as well as UAS demand for the next 3 to 5 years according to forecast demands.

5.9.4.7 Emergency Planning

An Airport Emergency Plan is required for all airports certified under Title 14 CFR Part 139 § 325. Airport operators should reference the FAA's guidance under FAA AC 150/5200-31C, Airport Emergency Plan. For all other airports, even if an AEP is not required, a stand-alone UAS Emergency Plan would improve the overall safety of routine UAS operation at the airport and promote a safety culture. Refer to Volume 1 of this report, *Managing and Engaging Stakeholders on UAS in the Vicinity of Airports* for information on developing a UAS Emergency Plan.

Airports that serve scheduled and unscheduled air carrier/charter activity under 14 CFR Part 139 are required to provide on-site firefighting facilities and equipment based upon critical air carrier aircraft length and number of daily departures. If UAS military and/or civilian activity is anticipated, ARFF personnel should be trained on fire suppression and emergency management related to UAS activities. In addition, alternative fire suppression agents should also be available due to the varying types of fuel that could be used by UAS.

5.10 UAS and Airport Operational Guidance

The mix of unmanned and manned operations within an airport environment will require some specific operational requirements to maintain safe operations. Therefore, this section evaluates existing unmanned, manned, and ATC procedures such as segregation, and types of operational guidance needed to support the introduction of UAS into the uncontrolled and

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controlled airport environment. In addition, unmanned and manned fleet mix and types of activity should also be considered. Critical UAS operational concerns such as air traffic and ground control procedures (i.e., segregation of operations), right-of-way procedures, see and avoid procedures, communication procedures, data management needs, and security needs should be considered.

In addition to airport infrastructure, various operational studies and procedures should be coordinated/implemented to safely support UAS and manned aircraft operations at an airport. These include establishing specific airspace and ground operations procedures, communication requirements, safety and security criteria, minimum operating standards, and emergency safety procedures. Further, emergency plan requirements associated with UAS should be included in the Airport Emergency Plan (FAA AC 150/5200-31C) and training exercises. General aviation airports, who are not currently required to have an emergency plan, should develop an integrated emergency and operational manual to address UAS and manned aircraft needs. The operational studies and procedures should be coordinated with on and off airport stakeholders including law enforcement and medivac personnel, federal, state and local regulatory agencies, regional ATC, local community, and other interested parties. For more information, Appendix D identifies documentation and procedures based upon discussions with FAA, airport managers that support UAS operations, and UAS manufacturers and users.

5.11 Development of Alternatives

Based upon the findings of the forecast and facility requirements sections, alternatives are created to address identified needs. In addition to infrastructure needs, UAS alternatives should address operational, communication and safety/security requirements needed to safely integrate UAS with manned operations at a public use airport. The following provides suggested guidance on how to effectively develop alternatives related to UAS integration:

- For an airport master plan, UAS launch and recovery infrastructure needs could be integrated into the airfield alternatives. UAS launch and recovery, apron, and support facilities should be considered as part of the airport's long-term development plan. Evaluating both manned and unmanned needs simultaneously will allow for more efficient and cost-effective options to be considered. Although UAS operations are currently segregated from manned aircraft, ultimately manned and unmanned aircraft will use the same airport airfield facilities.
- Alternative development within a UAS planning document will focus on development needed to support UAS forecast operations. However, review of existing and planned airport development either through discussions with airport staff and/or review of previous planning studies is needed to create options that are reasonable and flexible to support continued airport growth.
- The timing of development within each alternative option should be based upon operational triggers provided by the forecast and facility needs assessment. Due to a variety of limiting factors (i.e., communications, size and speed of commercial UAS, altitude limitations, and facility needs), segregation of manned and unmanned facilities and operating procedures will likely be the norm for the foreseeable future.
- However, as technology evolves and the dissimilarities between manned and unmanned aircraft decrease, airport infrastructure will be designed to accommodate both types of operations. This assumption is supported by FAA, U.S. DOT, and the UAS industry. Thus, airport alternative options should be developed to consider dual use of potential infrastructure to support manned and unmanned activity.

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- Given that the UAS industry is rapidly evolving, alternatives that preserve future flexibility are preferable. This includes evaluating potential reuse options for existing facilities as well as designing infrastructure and other airport facilities that could serve multiple aeronautical uses.
- UAS alternatives analyses should account for potential environmental impacts (e.g., noise), land use and zoning implications, grant assurances and funding issues, stakeholder and public input, compliance with community comprehensive planning efforts, as well as other factors that may be particularly unique to the airport. If new approach and departure procedures are associated with a proposed UAS development alternative, noise and land use impacts should also be considered.

5.12 Airport Compliance

Public use airports must remain compliant with both federal and state requirements to remain eligible for funding and maintain safety. Federal airport compliance requirements and policies are outlined in FAA Order 5190.6B. Although UAS related capital improvements funding is not currently available, public use airports must comply with both federal and state grant assurances to remain eligible for federal and state grants and loans needed to finance traditional airport planning, environmental and infrastructure improvements.

Given recent changes in the FAA Reauthorization Act of 2018 and industry growth, it is expected that some level of federal and state funding may become available to support UAS airport integration within the next 5 to 10 years. Therefore, airports can consider the potential impacts of UAS activity on the following airport compliance requirements to ensure that proposed integration does not negatively impact future airport operations and funding.

5.12.1 Airport Grant Assurances

As part of the UAS analysis, planners should present recommended development to the public and airport stakeholders for input. It is also critical, as part of a planner's due diligence, to evaluate proposed UAS infrastructure and operational improvements as they relate to specific federal and state grant assurances. Public airport owners, sponsors, other airport representatives or organizations are obligated because of accepting federal or state funds to maintain and operate their facilities to comply with specific conditions (i.e., grant assurances). The duration of these obligations is dependent upon the life of the project or airport facility, the type of recipient and other conditions as outlined within the grant assurances.

As it relates to UAS, several federal grant assurances provided in FAA Order 5190.6B at a minimum must be reviewed (i.e., Grant Assurances 19, Operations and Maintenance; 22a, Economic Non-discrimination; 22h, Airport Safety and Efficiency; 23, Exclusive Rights; 24, Airport Fee and Rental Structure; 25, Airport Revenues; 27, Use by Government Aircraft; and 29, Airport Layout Plan) in addition to state and local assurances. Compliance with grant assurances, especially at small airports which rely on federal and state funding for airport maintenance and infrastructure improvements, is critical to the long-term longevity of an airport. Thus, planners and other representatives and advisors to the airport should carefully consider the impacts of proposed development as it relates to existing airport state and federal obligations. Planners should also review current state and local statutes and ordinances as they relate to UAS to avoid incompatibility with local and regional requirements.

A further description of federal grant assurance requirements as they relate to UAS is provided in Section 6.5.3 *Funding and Grant Assurances,* and Appendix A in *PDT Master Plan Capital Funding Sources and Programs.*

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5.12.2 Environmental Compliance

Typically, when considering environmental factors as part of an airport master plan, the planner and environmental specialist should use the guidance provided in FAA Order 5050.4, NEPA Implementing Instructions for Airport Projects, and the associated Desk Reference. As part of any planned improvements at an airport, including UAS, environmental due diligence and evaluation is required to determine potential impacts, consider alternatives and identify minimization and mitigation strategies.

UAS activities may introduce additional environmental impacts particularly noise. Proposed launch and recovery infrastructure may require new approach and departure procedures which in turn could result in new aviation noise exposure. Also, depending on UAS power sources, new waste streams (for example used batteries) may be generated. These issues should be considered when planning UAS facilities.

Finally, even if federal funding is not used to fund proposed UAS infrastructure, NEPA review will be required for federally obligated airports. Federally obligated airports are required to update their ALPs to show proposed infrastructure and updated ALPs must be submitted to the FAA for review. Approval of the ALP is a federal action and therefore compliance with NEPA is required.

There are three levels of NEPA review: categorical exclusion (CATEX), environmental assessment (EA), and environmental impact statement (EIS). Airport sponsors coordinate with the FAA to determine the appropriate level of review in accordance with the current versions of FAA Orders 5050.4 and 1050.1F, Environmental Impacts: Policies and Procedures. Many of the proposed UAS projects may qualify for a CATEX. Chapter 5 of FAA Order 1050.1F lists the types of projects for which a CATEX may be appropriate provided the project would not involve extraordinary circumstances.

For example, a proposed UAS hangar may be eligible for a CATEX according to Categorical Exclusions for Facility Siting, Construction, and Maintenance of FAA Order 1050.1F.

"f. Federal financial assistance, licensing, ALP approval, or FAA construction or limited expansion of accessory on-site structures, including storage buildings, garages, hangars, t-hangars, small parking areas, signs, fences, and other essentially similar minor development items."

5.12.3 Land Use and Zoning Compliance

Land use on and adjacent to airports is governed by federal and state legislative codes and grant assurance requirements, which apply to UAS. According to FAA Grant Assurance 21 and Title 49 U.S.C. § 47107 (a) (10), the airport sponsor and local municipality should adopt zoning requirements that "restrict the use of land adjacent to or in the immediate vicinity of the airport to activities and purposes compatible with normal airport operations, including landing and takeoff of aircraft."

Concerns about privacy and safety have limited UAS flights over people and populated areas. For this reason, UAS operations over unpopulated areas (i.e., agricultural and pasture) are preferred. Operations over the industrial areas may also be allowed. However, discussions with industrial users prior to initiating UAS activity should be done to mitigate any issues as well as establish operating and emergency management procedures in case of lost link. Further, with development of personal air vehicles and air taxi services, airport zoning requirements are expected to expand to incorporate urban air mobility approach and departure requirements as well as mitigate obstructions to air navigation.

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5.13 Facilities Implementation Plan

The facilities implementation plan shows how the airport sponsor will implement the planning recommendations, where manned examples serve as a valuable framework for UAS planning and integration. The plan may be complex (e.g., implementation plan for a large-scale master plan update) or simple (e.g., implementation plan for a limited UAS planning study). As with any proposed airport facilities implementation plan, UAS improvements can be scheduled based upon planning activity levels/demand triggers rather than specific years. Although the activity forecasts show demand based upon key years, development is not based upon a specific year but rather airport activity and capacity. Years are used merely for statistical purposes and to classify forecast demand into short-term (0 to 5 years), mid-term (5 to 10 years) and long-term (10 to 20 years) timeframes for planning purposes.

Airport infrastructure and capital improvements are driven by critical aircraft operating requirements, aircraft operational demand, technology, and safety and regulatory requirements. Activity based triggers ensure that an airport can support forecast demand. These demand triggers are used to identify the timing and justification for various airport capital improvement projects. Traditional timing for airport capacity and activity-based development based upon the National Plan of Integrated Airport Systems (NPIAS) guidance are as follows:

- 60 percent airport capacity triggers planning and environmental projects
- 80 percent airport capacity triggers project design and permitting
- 90 percent airport capacity triggers project construction

These activity levels may be applied unless a substantial change in the type or level of activity warrants immediate action. For example, a general aviation airport becomes certified under 14 CFR Part 139 to support scheduled or unscheduled commercial traffic. Due to socio-economic and technological factors, activity level projections become less reliable beyond the first 5 to 8 years.

UAS improvements may also be triggered by changes in regulations. Since airport UAS regulatory guidance is currently limited and is expected to remain so for the near future, near term scheduling could be focused upon forecast demand and projects to safely integrate UAS activity.

Projects needed to support UAS airport integration include obtaining a COA and development of a UAS operating and procedural manual. These steps ensure that operations remain coordinated and documented for airport specific procedures. The COA is typically renewed every 5 years unless a significant change in UAS or manned activity occurs or there is a substantial change in the airport's infrastructure or airspace. The UAS operating and procedure manual can also be revised upon renewal to reflect current conditions.

A COA can be pursued for both airports with and without a tower that intend to support substantial UAS activity including training associated with a college or university, use by law enforcement, and emergency management services as well as to attract manufacturing, maintenance and other UAS related businesses to the airport. Several general aviation and non-hub regional airports view UAS operations and businesses as an opportunity to improve revenue and expand economic and educational opportunities in their communities. However, it is important to note that it can take up to 2 years for an airport to obtain a COA; therefore, it is important to be aware of alternative methods (including agreements, coordination, and LAANC) in the meantime.

Other tasks that would facilitate safe integration of UAS operations within the airport environment include, but are not limited to:

- Conduct a safety risk assessment and create a safety management system for UAS;
- Update airport minimum standards;

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- Create secure communication documentation related to UAS including lost link procedures and apply for frequency request from the FCC;
- Coordinate frequency acquisition and use with an ATC;
- Perform airport geographic information survey (AGIS) of airport property, create electronic airport layout plan (eALP) and upload data to FAA database to assist with airspace and navigational support as well as obstruction identification; and
- Conduct airspace obstruction studies and remove obstructions in conjunction with new operational procedures and infrastructure development.

Refer to Section 5.6 for more information on these projects.

Since the UAS industry is still in its infancy, airport management should focus on near term (2 to 3 years) and short-term (5 years) suggested improvements. Proposed future development should be evaluated given operator demand, new technology, safety, funding, liability, and anticipated sponsor's return on investment.

A sample airport implementation plan based upon UAS civil, commercial and military demand is illustrated in the following sections. Note, that the implementation plan for each airport will vary based upon the type and levels of manned and unmanned demand in addition to existing infrastructure and long-term airport goals.

Phase I (year 0 to year 5)

Proposed short-term airport strategies associated with UAS could include:

- Obtain COA including development of UAS operating and procedure manual;
- Initiate safety risk assessment study and implement safety management system for UAS integration;
- Prepare secure communications documentation and frequency request;
- Perform AGIS Survey and either update existing ALP to show UAS infrastructure needs or create an eALP along with needed support documents;
- Complete a Part 77 airspace obstruction study and mitigation plan (7460 process if needed);
- Conduct a UAS and airport business and economic study;
- Environmental documentation associated with any new UAS facilities;
- Add pavement markings, lighting, and signage to designate areas where UAS activities, including launch and recovery, on the airport will occur; and
- Work with ATCT to establish procedures for coordination between manned and unmanned operations.

Phase II (year 5 to year 10)

As the UAS industry continues to grow and technology matures, the impacts and needs related to supporting UAS activity are anticipated to be more transparent. To support continued development, the following projects could be considered for Phase II.

- Renew COA and UAS operating and procedure manual, as needed;
- Prepare UAS development site and expand utilities including secure data lines to support new UAS infrastructure;
- Upgrade entire airport communications system and data network;
- Install back-up power systems and expand existing electrical vault(s) to support expanded power needs;
- Implement design plan to retrofit or design new landside access and parking plans;
- Design multi-use holding pads to support both UAS and manned aircraft operations; and
- Install electric and liquid UAS fuel systems.

Phase III (year 10 to year 20)

Proposed airport long-term development needs are difficult to forecast because of the predicted high growth of UAS, leaps in technology, unknown regulatory requirements as

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well as other unknown factors. Therefore, the capital improvements included in Phase III are a "best guess" of likely needs considering the current regulatory environment and UAS industry development.

Proposed development may include:

- Expand and improve vehicle access;
- Renew COA and UAS operating and procedure manual, as needed;
- Prepare site and construct additional UAS hangars and administration/operations facilities;
- Design and construct new general aviation terminal building and apron facilities or retrofit existing commercial terminal and relocate passenger terminal facilities;
- Expand airport fuel facilities and upgrades;
- Construct access road and parking improvements; and
- Add air and ground vehicle charging stations.

5.14 Financial Feasibility Analysis

A financial feasibility analysis is conducted to demonstrate if an airport can fund the proposed projects. Funding for UAS projects is currently limited. At the time of this writing, capital improvement funding for UAS related airport projects is not available nor can UAS operations and demand be used in conjunction with manned aircraft to support other airport infrastructure improvements (e.g., runway extension, taxiway or apron construction, and navigational aids).

Discussions with FAA personnel revealed that UAS activity and infrastructure are not eligible for AIP funding. Internationally, neither ICAO nor EASA have offered any guidance on UAS infrastructure types and funding that countries should consider applying to UAS and airport integration. Likewise, it is assumed that Passenger Facility Charges (PFC) at commercial service airports may not be used to fund UAS infrastructure. UAS infrastructure is also not eligible for states' funding via their designated aviation funding systems. However, given FAA's support of UAS and aerospace development, approval of AIP and PFC federal funding for UAS related airport capital improvements may occur within the next 5 or more years based upon guidance highlighted in the Reauthorization Act of 2018 as well as various Congressional requests. In the meantime, alternative sources of funding must be considered. For example, Cape May recently was awarded a U.S. Commerce Grant to support UAS development within Cape May County.

Also, certain short-term projects including AGIS and airport obstruction analysis, based upon previous discussions with FAA, may be eligible for some level of federal funding even if related to UAS in some way. Other documentation including a safety risk assessment and environmental documentation, although potentially eligible for FAA funding, is questionable.

Other potential funding sources for UAS infrastructure include:

- Federal and state economic funding to support aviation related development (U.S. Commerce Grant);
- Academic or FAA Technology Center research grants;
- Industry participation (public-private partnerships);
- U.S. DOD funds; and
- Local government investments.

Sample capital improvement programs associated with UAS integration into a general aviation, small commercial and international airport are provided in Appendix E while there are also some examples of the use of infrastructure funding in Appendix F.

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5.15 Airport Layout Plan

The ALP set is a graphical representation of planned airport development. Although FAA and state DOT grant funding for UAS planning and infrastructure is not currently available, proposed UAS development can be incorporated into the ALP set. Since an ALP update is done as part of the airport master plan process, the cost of adding data and additional sheets to the set will be minimal.

The following ALP sheets should include proposed UAS infrastructure:

- ALP,
- Data Sheet,
- Area Plans,
- Airport Airspace Drawing –14 CFR Part 77,
- Inner Portion of the Approach Surface Drawing (if a UAS runway is proposed), and
- Airport Land Use Plan.

However, if a UAS planning study is done independently, the cost of updating the airport ALP to show primarily proposed UAS development may be cost prohibitive and not very useful in the long-term. Rather, as highlighted in the following section, updates to the ALP including UAS development should be included as part of an ALP update study, airport master plan update or as part of AGIS and eALP development.

5.16 Airport Readiness Steps

Upon completion of the initial UAS planning study, whether as part of an integrated master plan or as a stand-alone study, the airport sponsor may need to take additional actions to support UAS integration. The following is a list of next steps. The timing of these steps is dependent again on need and available funds. Airport sponsors should individually evaluate potential UAS use case scenarios to identify potential costs, benefits, barriers to operations, liabilities, and airport stakeholder and community acceptance.

5.16.1 Obtain UAS COA

As a public entity, an airport sponsor can apply for a COA from the FAA to support UAS activity at the airport. A COA is issued by FAA Air Traffic Control to a public operator for specific UAS activity. The public entity must submit a completed application. A sample of a completed application is available on the FAA's website at https://www.faa.gov/about/office_org/headquarters_offices/ato/service_units/systemops/aaim/organizations/uas/coa/.

Once the application is filed, FAA performs a comprehensive review and imposes any provisions as part of the approval for the safe operation of UAS within the airspace. According to the FAA's website, approval is usually obtained within 60 days. Once the airport sponsor obtains a COA, it must renew the certification at least every 2 to 5 years unless major changes in operations are required.

As part of obtaining a COA, a UAS Operating and Procedure Manual is developed that includes information outlined in the application process (e.g., contacts, operational description, performance and procedural characteristics, and flight crew requirements) as well as emergency operations and accident response, security procedures, employee training and use, and memorandums of understanding.

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5.16.2 Conduct a Safety Risk Assessment and Develop Safety Management System

FAA encourages 14 CFR Part 139 certified and non-certified airports to implement a safety management system (SMS) program which includes safety risk assessment and management plan. According to the FAA Airport's SMS Desk Reference, safety risk management (SRM) consists of "standard set of processes to identify and document hazards, analyze and assess potential risks and develop appropriate mitigation strategies." SRM includes conducting safety risk assessments for triggering events. Adding or introducing UAS activity could be construed as a triggering event. Therefore, airports may wish to consider conducting a Safety Risk Assessment to facilitate integration of UAS. Refer to Volume 1 of this report, *Managing and Engaging Stakeholders on UAS in the Vicinity of Airports* for guidance on integrating UAS safety within a SMS program and *ACRP Report 131: A Guidebook for Safety Risk Management for Airports*, for guidance on SRM and safety risk assessments.

The LAANC system, according to FAA, is merely a temporary measure as part of integrating UAS into the NAS. Based on the experience of Golden Triangle Regional Airport, airports which complete an independent safety risk assessment and create a safety risk management plan that addresses UAS and manned operations will not be required to participate in the LAANC system. At Golden Triangle Regional Airport, developing and creating a safety risk management plan specific to the airport has enhanced overall safety and allowed for greater flexibility in handling manned and unmanned operations at the airport. An additional benefit is that the airport has and continues to attract manned and unmanned businesses to the airport.

5.16.3 Prepare Secure Communications Documentation and Frequency Request

Currently, FAA does not recommend that UAS pilots use the same communication frequency as manned aircraft pilots and the FAA ATC. Its concerns include UAS pilot limited training on radio communication procedures, radio static and confusion, over-capacity of the system, as well as the "lack of sufficient frequency spectrum necessary for the UAS control and air traffic control (ATC) communications links compounded by the need for protected frequencies" (Henricksen, 2008, p. iii).

Generally, commercial small UAS "use unlicensed bandwidths of 2.4 GHz, 5.8 GHz, unlicensed 900 MHz, and UHF [ultra-high frequency] bands for communication. But, UAS operators must comply with national regulations applicable for operation of other technologies in these bands" (Radio Division, TEC, p. 7). Radio communication is key to safe integration into the NAS. UAS use radio communications for ATC, UAS command and control including sending commands (uplink) and downloading data (downlink), and to support the sense and avoid function (Radio Division, TEC, p. 6) which will become even more critical when UAS operate beyond line of sight.

The FCC under 47 CFR Part 87 is responsible for aviation radio services which include aircraft and ground radio stations. Ground radio stations include two types: (1) the aeronautical and fixed service are stations used for ground-to-air communications associated with aviation safety, navigation, or preparation for flight and (2) the radio navigation service consists of stations used for navigation, obstruction warning, instrument landing and measurement of altitude and range (FCC, 2017a).

To support UAS activity some airport operators use mobile communication systems on the airport. These airports obtained specific frequencies from the FCC for the mobile communication systems to be used as ground communication systems. Other airports have added

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additional frequencies to their existing Automated Weather Observing Systems or Automated Surface Observing Systems by applying to the FCC. Lastly, others require UAS operators to coordinate with on-airport air traffic control and monitor the manned frequencies. These operators typically already have obtained specific operating frequencies to support their UAS operations to avoid impacting existing airport ground stations and aircraft station frequencies.

Thus, next steps could include a task to prepare FCC Form 605, "Quick-Form Application for Authorization in the Ship, Aircraft, Amateur, Restricted and Commercial Operator, and General Mobile Radio Services" (FCC, 2017b).

5.16.4 Complete Airport Geographic Information Survey, Update Existing ALP, or Develop Electronic Airport Layout Plan

Other steps include updating the existing ALP and/or completing an AGIS and an eALP to facilitate integration of UAS at an airport.

With the integration of NextGen, the FAA is moving toward widespread use of AGIS which involves the collection of various spatial data (i.e., survey, GPS, sensors, computer aided drafting and design data and imagery) for specific airport and off airport facilities, referred to as attributes, in order to create an electronic database of airport and contiguous natural and manmade objects (e.g. railroad lines, rivers, runway coordinates, roads, and antennas). This survey data, which is outlined in FAA AC 150/5300-16A, 17C, and 18B, is used to develop satellite-based approach and departure procedures as well as manage the NAS. AGIS data will assist airports, FAA, and state DOT in identifying potential obstructions to air navigation and assist in the development of airport airspace procedures for both manned and remotely piloted aircraft. In addition, several researchers and aviation/aerospace universities are evaluating using AGIS data to improve UAS situational awareness within the airport environment, especially given remote aircraft limited visibility and increased use of automation and sensors for "sense and avoid" maneuvers both in the air and on the ground. AGIS data could be used in conjunction with UAS equipped ADS-B or other system to provide UAS operator and airport ground control UAS location and situational awareness. Academic research associated with the FAA Test Centers and with the FAA ASSURE Team on using AGIS data to support semi and full-autonomous UAS activity within the airport environment is currently ongoing.

An eALP converts the ALP paper files and data to an electronic format that is uploaded to the FAA's AGIS system. Benefits of an eALP are that it provides one source for multiple ALP data, offers engineering level accuracy, assists in identifying existing and potential obstructions to air navigation, provides more detailed airport and nearby facility data, and allows the FAA and other governmental agencies to respond more quickly to airport issues and opportunities. An eALP system, although initial costs are higher than a traditional paper ALP, is in the long-term cheaper and data can easily be updated. Although AGIS and eALPs have been mostly limited to large and medium hub commercial airports, the FAA ultimately plans to have all ALPs be transitioned to eALPs.

The benefits to airport sponsors and UAS commercial, military, and general aviation users are substantial since it will allow airports to provide critical data needed to support UAS operational changes either under Part 107 or the airport's COA. It allows the airport sponsor and regulatory agencies to identify obstructions to air navigation, potential conflicts in operations between manned and remotely piloted aircraft, as well as helping to identify the highest and best use of on-airport property. These are just some of the existing advantages of both AGIS and eALPs. Other opportunities will develop as technology continues to grow and evolve.

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5.16.5 Conduct an Airspace Obstruction Study

As part of the UAS integration process, UAS approach, departure, launch and recovery airspace procedures must be developed in conjunction with the FAA Airport District Office and Flight Standards. This will be an important area of future research as standards equivalent to 14 CFR Part 77 have yet to be established for UAS. Eventually, proposed approach and departure surfaces must be illustrated in the ALP set. It is likely, given the low altitude of UAS operations, that obstructions to air navigation will be identified as part of the ALP and AGIS process. Proposed UAS airspace procedures including launch and recovery must be coordinated with FAA Flight Standards and the Airport District Office as well as local DOT personnel. These agencies will evaluate the proposed air traffic patterns, and, if approved, will publish this data to support both UAS and manned operations. As part of this analysis, in accordance with 14 CFR Part 77, existing and potential obstructions to air navigation will be identified and recommendations will be made on how to address the obstruction.

All proposed development on an airport as well as off airport which may exceed 200 ft above ground level are subject to an obstruction evaluation and airspace analysis regardless of federal funding. The FAA Regional Airports Division is responsible for initiating non-rulemaking aeronautical studies, which include evaluating the effects of construction or alteration on existing and proposed operating procedures; determining potential hazards associated with the proposed construction on air navigation; and identifying mitigating measures to enhance safe air navigation.

The first step in this process is to file documentation with the FAA's Obstruction Evaluation Group through the FAA OE/AAA portal at https://oeaaa.faa.gov/oeaaa/external/portal.jsp. This document will then be sent to one of the FAA's Obstruction Evaluation Specialists for review. This specialist will either issue a favorable determination or a notice of proposed hazard to air navigation. If development is determined to be a hazard, then adjustments should be made or additional review by the FAA may be requested. Ultimately, the goal of the obstruction analysis is to ensure the safety of aircraft, whether manned or unmanned, and the efficient use of airport facilities and airspace.

5.17 UAS and Airport Facility and Operations Checklist

The following is an updated checklist for highlighting specific infrastructure and operations needs necessary to safely and efficiently integrate UAS operations into a public use airport environment. The checklist is based upon a combination of guidance in *ACRP Report 144*, FAA airport advisory circulars, orders and guidance, input from FAA Airports and UAS Integration Office, UAS operators and airport managers. A baseline checklist is provided but should be adjusted to comply with the UAS type and mission as well as airport facilities, capacity and level of service (Table 9).

5.18 Summary

A variety of planning and design documents were considered, and various airport stakeholders were consulted to prepare the planning guidance in this chapter. Legislation and guidance are ever evolving as is UAS and aviation technology. Therefore, similar to *ACRP Report 144*, which was completed in 2015, some suggestions and guidance may become dated and even obsolete. Thus, airport sponsors and their representatives should continue to not only follow but participate in developing new rules and regulations related to UAS and airport integration.

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Table 9. Baseline UAS integration checklist.

UAS Design and Performance Criteria			
MTOW (lbs.)			
Wingspan/Rotor Length (in Ft.)			
Body Length (Ft.)			
Landing Gear Type (Ft.)			
Main Gear Width (Ft.)			
Approach Speed (Knots)			
Airport Design Category			
Visibility Minima			
Primary Use/Mission			
Takeoff Method (Pneumatic Launch, Runway, Vertical Takeoff, other)			
Recovery Method (Runway, Net, Skyhook, Vertical landing, etc.)			
Unmanned or Autonomous Operating System			
Facilities	**Description**	**Infrastructure or Operational Need(s)**	**Grant Eligible**
Communication Requirements			
FCC Frequency			
ATC Coordination			
Other			
Data Storage Requirements			
UAS Storage (e.g. hangar, stand-alone building)			
Apron space			
Ground-based control station			
Fuel Type			
Fuel Storage			
Maintenance Requirements			
Material Storage			
Waste Management			
Special Payload Accommodations			
Support Services Needed			
ARFF and Emergency Response Needs			
Security and Safety Requirements			
Potential Environmental Impacts			
Land Use and Zoning Needs			
Other Needs			

Sources: *ACRP Report 144,* FAA Office of Airports, UAS Integration Office, FAA Technical Center, Booz Allen Hamilton, and Astrid Aviation and Aerospace.

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CHAPTER 6

Anticipated Future Conditions

Industry innovation has outpaced both U.S. regulations and ICAO standards and practices until recently. FAA, ICAO, and the European Union Aviation Safety Agency have or are in the process of establishing UAS operational guidelines and airspace integration programs. The U.S. Congress, as part of the FAA Reauthorization Act of 2018, specifically enacted new legislation and guidance for FAA oversight and UAS integration into NAS.

In addition to the initial seven UAS test sites established in 2013, the FAA in May 2018 implemented the UAS Integration Pilot Program (IPP) to help local, state, and tribal governments partner with the private sector to accelerate safe integration of UAS into the NAS. These participants are testing UAS applications in various sectors (i.e., delivery of time-sensitive medical equipment and controlling mosquito populations). In addition, some IPP test sites, like the initial seven test sites, are evaluating beyond VLOS operations and technology while others are focused on night operations and sense and avoid technologies. For example, as part of the IPP, the North Dakota DOT is testing large commercial UAS, beyond VLOS, and UAS sensor technology at Grand Sky, which is part of the Grand Forks Air Force Base. The testing of beyond VLOS is crucial to developing an appropriate regulatory framework that will not hinder innovation (Wyman, 2018b). In addition, due to several natural disasters, in October 2018 the FAA granted two waivers (beyond VLOS and beyond VLOS above people) to State Farm Insurance Company due to the devastation caused by Hurricane Florence (AUVSI, 2018).

The IPP program currently provides an avenue for FAA to issue waivers to current regulations and is anticipated, like the EASA "Basic Regulations" framework (European Union Aviation Safety Agency, 2017), to help inform new enabling rules for low-altitude UAS operations and integration into national, and ultimately international, airspace. The FAA, ICAO and EASA have all re-established drone advisory committees or working groups comprised of industry leaders in UAS manufacturing and commercial operation. The overall purpose of these committees is to provide independent advice and recommendations for establishing standardized UAS operator, operating, and safety criteria for integration into the national and international airspace system.

As stated, federal regulations today are limited and a framework for widescale commercial UAS operations does not currently exist. This is believed to be due to lack of regulation and technological hurdles (e.g., beyond VLOS, sense and avoid, aerial communications and current radio spectrum limitations). Further, FAA's tendency to focus on UAS risks rather than potential benefits may have unintentionally helped foster a hesitant culture surrounding UAS, potentially affecting public perception of UAS operations (Wyman, 2018b; NASEM, 2018). Thus, future UAS regulations, particularly related to large commercial UAS operations, will need to promote safety and security while encouraging innovation. Integration of these

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new regulations and practices will impact airport operations and infrastructure. Therefore, airports must consider these new provisions and requirements as part of their planning, design, and operations efforts, whether through new provisions in the airport master planning process and/or additive measures as part of an airport's Airport Certification Manual and general Part 139 operations, minimum standards, and operational procedures related to airport safety and security.

Given current data and discussions with airport sponsors and government officials, UAS operating on or near an airport does impact airport air traffic procedures. However, the type and level of operational and infrastructure impacts is still unclear due to limited guidance and regulations. What is clear is that the type of UAS and application will shape the regulatory environment in the years to come. Thus, the focus of this chapter is on potential UAS development and airport opportunities and impacts anticipated to occur beyond 2028. Although the future of UAS continues to be debated, internal government and external industry forecasts both agree that a shift in demand is likely to occur in the late 2020s because of various factors including public acceptance, economic/market demand, technology, regulations and infrastructure needs. Since it is anticipated that UAS, especially large UAS, will replace aircraft in mature aviation market sectors such as agriculture, cargo, and passenger travel in addition to creating new aviation opportunities, airports at large need to remain flexible to adapt to changes that UAS technology is driving. The findings of this chapter highlight likely long-term impacts of UAS on airport infrastructure, operations, and airspace using some of the recommendations highlighted in *ACRP Report 144* as well as insight obtained from stakeholder interviews, available data, and anticipated market demand.

6.1 Public Acceptance

Public acceptance of UAS technology is considered the most vital for investment, favorable UAS regulation, and overall integration into the transportation system. If safety concerns are effectively addressed in conjunction with a public perception shift regarding automation and artificial intelligence, then UAS growth is expected to increase, especially in the large UAS market, exponentially. Public acceptance of unmanned and autonomous aircraft is anticipated but depends on how society embraces other daily technology that regularly makes decisions without significant human input.

Several studies were performed to determine public perception of UAS specifically regarding safety compared to manned aircraft, government and news descriptions of UAS technology, and other concerns. Both the Queensland Australian Study completed in 2015 (Clothier et al., 2015) and a U.S. study completed by Embry-Riddle Aeronautical University in 2013 (Ison, Liu, and Vincenzi, 2013) determined that public perception of the technology was neutral, which was the result of limited knowledge of UAS technology. However, less than half the respondents of the U.S.-based survey would be uncomfortable with UAS except in a firefighting and weather monitoring capacity.

Both studies revealed that privacy was the primary concern followed by safety, military use, and misuse (e.g., terrorism). Both the 2013 and 2015 studies revealed that the public was not yet ready to accept widespread use of such technology. However, if privacy and safety issues can be overcome through improved coordination and transparency with the public, then expansion of UAS technology beyond its current uses is likely.

Improvements in technology, especially regarding aircraft autonomy, are expected to cause a paradigm shift as people become more comfortable with the technology. As noted in Unmanned Systems Integrated Roadmap (2017–2042), advances in autonomy are expected to

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"greatly increase the efficiency and effectiveness of both manned and unmanned systems" (U.S. DOD, 2018). Autonomous systems allow the system to independently develop and select different courses of action based upon mission goals as well as knowledge of the environment and situation. In the future, it is anticipated that both military and commercial UAS will draw from past aircraft and air crew data when making decisions, which is expected to instill confidence in air traffic controllers, operators and the public regarding the overall safety of autonomous aircraft operations. Since the size of UAS will need to increase to meet data and mission requirements (e.g., passenger and cargo transport), acceptance of large UAS by consumers is critical. As shown by the Uber Elevate and other UAS passenger models, an optionally piloted aircraft may be useful in mitigating concerns in addition to liability and insurance concerns.

6.2 Market Demand

Although the DOD has used UAS since World War II, regulation and mass adoption of commercial and civil UAS are still in the early stages. *ACRP Report 144* was published in 2015 to assist airports and stakeholders in understanding how UAS can impact airports. Although much of the information found in that report remains accurate, additional steps should be taken by airport sponsors and regulators to address the demands of this quickly evolving industry.

Economic drivers will continue to push and shape demand by determining UAS applications with a viable customer base. UAS is currently being used on a small scale for the movement of objects as well as surveillance and photography with commercial UAS delivery services expected to be implemented within the next 5 to 10 years (Cohn et al., 2017). However, other potential uses such as large air cargo, personal and commercial air travel, law enforcement, and medical evacuation will require a longer development and approval timeline (Cohn et al., 2017).

Global commercial pilot demand forecasts for the next 20 years anticipate a substantial shortfall in available pilots, by approximately 600,000, because of attrition including retirements, increased number of aircraft in service, decrease in military trained pilots, high cost versus salaries, and less vocational interest. Due to this shortage, industry and DOD are looking to technology to help fill the gap. With enough automation built in, only a 'safety pilot' would be needed in case something unexpected happens. Automation continues to be integrated into passenger aircraft, improving safety and aircraft performance. In the 1980s, commercial airlines decreased the flight crew from three (flight engineer, co-pilot and pilot) to two (co-pilot and pilot).

UAS integration will be impacted not only by public acceptance, technology, and operating infrastructure, but by market timing and safety concerns as well. Future aviation is anticipated to consist of a combination of automation, new aircraft design, new propulsion systems (i.e., electric and/or hybrid) as well as introduction of subsonic, supersonic and long-haul hypersonic/suborbital flights (Crosby-Close and Ros, 2018). Based upon existing data and discussions with FAA personnel, U.S. and international industry, and airport stakeholders, early adopters of new UAS technology will be U.S. DOD, research and development agencies and organizations, including the FAA Technology Center, and various UAS test sites, in addition to early adopter commercial operators such as urban air mobility, agriculture, survey, cargo shipping, telecommunications as well as firefighting/emergency management. These early adopters are expected to expand their fleets to include large vehicles, autonomous but with pilot back-up, either on board or remotely, resulting from anticipated improvements in endurance, sensor payloads, and onboard data processing. Note that medivac patient transfer is not expected within this first stage of market integration.

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During this first stage of market integration, major infrastructure improvements within the airport environment except for additional storage facilities, additional markings and visual navigational equipment, communications, and back-up power supply, may not be warranted. However, since the face of transportation has not changed substantially since the 1950s and large infrastructure projects can take more than a decade to complete, airport infrastructure planning and environmental efforts should begin to address the anticipated next stage of market demand including integration of large passenger and cargo UAS, including manned and unmanned regional/short-haul electric aircraft and "last mile" electric air taxi operations.

The next stage, which will be based upon technology and regulatory improvements, will likely involve some level of UAS substitution of aircraft within mature market segments, but still likely requiring a 'safety' pilot, including large air cargo, corporate/business, air taxi and air carrier passenger, emergency management and medivac, civil and military transport and fighter aircraft as well as various aviation related training programs (ATC, aircraft maintenance and some flight training).

Improvements associated with substitution of some manned aircraft operations with UAS are also anticipated to create new uses for UAS applications including, but not limited to, use of UAS for non-aviation construction as well as for the transfer of products from inland ports to large marine vessels. Development of UAS along with other transportation technology improvements is anticipated to completely change transportation worldwide.

6.3 Improvements in Technology

As technology improves, it will enable new and expanded UAS applications. Ongoing technology improvements include autonomous flight, battery performance, detect and avoid technology, and development of electrical propulsion systems for passenger aircraft. Recently, NASA successfully flew a large UAS in the NAS without the use of a chase or safety aircraft (Northon, 2018). The aircraft was able to take off from Edwards Air Force Base and land successfully at Southern California Logistics Airport. This example is indicative of a statement cited in the previous *ACRP Report 144*: namely, large UAS may require facilities similar to those of manned aircraft and operate in like manner (Neubauer et al., 2015).

UAS military operations have and continue to occur at commercial and joint-use airports (e.g., Syracuse Hancock International Airport) with few issues since these vehicles, for the most part, operated like manned aircraft. Conversely, other smaller rotorcraft UAS for personal or recreational use are not likely to impact airport infrastructure as current regulation bans UAS operations in a 5-mile radius at airports (Matthews, Frisbie, and Cistone, 2017). It may be in airports' best interest to attract large UAS as opposed to smaller, commercial ones (Wanner, 2018) unless operational limitations and infrastructure (i.e., Air Traffic procedures, designated apron areas for landing and takeoff, and on-site coordination with stakeholders) can safely and effectively be implemented. This includes airport operations staff use of small UAS for on-site survey and wildlife management.

Large UAS (greater than 55 lbs), which are anticipated to be the primary UAS users of an airport, are likely to operate like traditional manned aircraft. Businesses utilizing large UAS would be treated like manned aircraft tenants by paying traditional airport operating fees to contribute to an airport's financial sustainability. This could also include fuel services associated with electrical charging stations or more traditional fuels like JP8, 100LL, Jet A, and gasoline. This would allow airports to generate additional revenue from UAS operators including fees obtained from charging stations as well as potential advertising revenue.

Another consideration related to larger UAS operations is the potential for the previously mentioned new on-demand services (e.g., Lyft and Uber business models) that can now be built

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in new, constrained areas within cities much akin to heliport or short takeoff and landing airport (STOLport) operations. A STOLport is an airport designed to accommodate short aircraft takeoff and landing operations and is limited to certain types of aircraft. STOLports typically include short runways (<2000 ft) and have limited infrastructure compared to traditional airports.

New concepts include Uber aerial vehicle services located over existing highways for Uber vehicle transfer to aerial vehicles. Such new vertical type "airports" would require their own rules, regulations, planning/design/construction requirements. Nearby airports would need to coordinate with these types of facilities based on air traffic routing, Uber split services into an airport versus a STOLport, and overall traffic avoidance through ATC. As a result of these new operations and consumer demand, significant changes in infrastructure both at the airport and within the intracity environment will be needed.

While public acceptance of a fully-automated airliner is not envisaged within the next 20 years, it would allow the industry to more effectively handle the pilot shortfall. In addition, the airline industry could save as much as $30 billion if it replaces pilots by adopting autonomous flight technology (Zhang, 2017). According to a member of the Executive Board of the German Aerospace Center, the greatest challenge to aerospace development is "not technological, but financial or operational" (Crosby-Close and Ros, 2018). The planned use of electric propulsion for short-haul flying as well as tiltrotor technology for intracity air taxi operations will transform air travel. In addition, some tasks that have historically occurred within the airport environment, such as luggage check-in and identity verification via biometrics could be completed as part of the UAS air taxi service, which may fix "one of the major pain points in the air travel experience" (Crosby-Close and Ros, 2018).

Today unmanned aircraft are remotely controlled or perform pre-programmed tasks, but over time, aircraft autonomous functions will expand eventually allowing a fully independent unmanned system that can operate with little to no human input. Growth of autonomous technology will also push improvements in communications, cybersecurity, data management, avionics and sensors, all of which will support both the growth of UAS and the transmutation of aviation itself.

6.4 Regulations

Regulations continue to determine the viability of different UAS applications and typically cover three specific areas: UAS operations, operators, and vehicles. Operations focuses on beyond VLOS, autonomous operations, altitude restrictions, flights over people, and airspace integration. Federal regulations will continue to govern operator certification as well as training requirements, whereas regulations governing UAS vehicles are likely to be a combination of federal and state regulations. Vehicle regulations have and will likely include UAS identification, aircraft propulsion systems, airworthiness, use (e.g., passenger, cargo, surveillance, and military), and weight restrictions.

Unmanned aerial systems cover a large spectrum of vehicles including the smallest of nano drones to the largest military aircraft. To date, UAS operations are used for a variety of missions including surveillance and safety purposes, fire and rescue, security, law enforcement, wildlife management, environmental monitoring, etc. (Matthews, Frisbie, and Cistone, 2017). Additionally, more recent operations include providing cellular service to areas affected by disaster for emergency response purposes (Margaritoff, 2018), the use of the current telecommunication network LTE for wide-area connectivity to allow UAS-to-UAS communication (Asplund, et al., 2018) and even to deploy flotation devices to save swimmers caught in rough tides (Kwai, 2018). These examples are by no means a

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comprehensive list of UAS applications. There will certainly be additional uses in the future as technological advances occur, some of which have not yet been conceived. However, this begs the following questions: how well equipped is the current regulatory environment to promote the safe use of UAS in these emerging applications? How do these UAS operations impact current airport facilities or the NAS at large?

The regulations that have been addressed to this point have been primarily at a national level. However, state and local government will also play a role in shaping the UAS regulatory environment. Two years ago, Uber Elevate published a white paper called *Fast-Forwarding to a Future of On-Demand Urban Air Transportation*. The document highlights a case for urban mobility using eVTOL aircraft. The paper describes using existing infrastructure such as 'repurposed tops of parking garages, existing helipads, and unused land surrounding highway interchanges' as potential operation sites for the aircraft. The document is interesting for two reasons. First, a large corporation is pursuing large-scale aircraft operations that would not primarily occur at airports. Second, Uber Elevate has partnered with local governments in Dallas, Texas, and Los Angeles to make on-demand aerial ridesharing (Uber Air) a reality. This development raises questions pertaining to the role of state regulation compared to federal regulation. The FAA regulates the NAS, however, it is less clear the extent of that jurisdiction (Donohue, 2018). Some academic literature suggests that states and local government may be better equipped to regulate UAS operations as opposed to the federal government (Donohue, 2018). On the local level, this would require set coordination protocols between local airports and the communities surrounding them to ascertain the type of UAS that will impact the area and NAS.

Based on the current market and academic literature, it is apparent that the impact of UAS at airports will differ depending on the type of UAS and application. For widescale adoption of UAS into the NAS, it will be necessary for UAS to operate autonomously and beyond VLOS (AIA and Avascent, 2018). It is also likely that UAS will need to grow in weight and size (beyond 55 lbs) to provide value on a large scale. Sense and avoid technologies for UAS have been improving as technological innovations push the boundaries of autonomy. Yet, the regulatory framework does not yet exist to support beyond VLOS operations for personal or commercial UAS use. "Absent the 5-miles restriction, such limiting operating parameters might suffice for some on-airport uses of UAS (e.g., surveying or wildlife monitoring), but ultimately more sophisticated capabilities will be integral to most practical business and commercial operations" (Matthews, Frisbie, and Cistone, 2017). Future regulation for UAS will require thoughtful consideration of UAS applications as well as integration with other aviation activities.

6.5 Infrastructure Needs

ACRP Report 144 poses several questions that airport operators should consider when evaluating the potential UAS infrastructure changes (Neubauer et al., 2015). If there are future infrastructure changes that need to be made at airports, the following list provides several areas to consider.

- Does the UAS need a runway for takeoff, landing, or both? If so, what runway length and width is required?
- Can the UAS taxi to/from the runway and follow ATC commands and other voice commands?
- Does the UAS need hangar space when not flying?
- Does the UAS need ramp space prior to or after flight?
- What sort of control station is required (truck, trailer, office space [or area for UAS pilots to operate their aircraft])?

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- Does the UAS need launch and recovery space (in lieu of a runway)? If so, how close to the airport does this space need to be?
- What sort of communications infrastructure is needed? Does the UAS operator need special towers of antennas to ensure communications are established and maintained with the UAS?
- Will the communication frequencies needed create conflicts? Will they interfere with existing frequencies used by airport staff, the FAA, tenants, airlines, fixed base operators, or others?
- Will the UAS need special emergency standby equipment? Is it available at the airport or does it need to be brought in from an outside source? As an example, a large general aviation airport might need to bring in a local fire department truck to standby for UAS operations as a matter of protocol.
- What type of fuel facilities are needed and where do they need to be located?
- Do operators need a place to dispose of batteries and other UAS aircraft operational waste management?

UAS applications and market demand ultimately will dictate whether operations will need to occur at an airport or at an offsite location. Infrastructure needed to support UAS activity will include, at a minimum:

- Charging stations;
- Landing facilities and other assets;
- Air traffic management facilities;
- Vertiports;
- UAS service centers where UAS Air Taxi and other transport vehicles can be stored, inspected, and repaired;
- Distribution hubs to load and receive goods from UAS; and
- Receiving stations.

As UAS become more sophisticated and the missions expand, additional infrastructure will be required.

Although current market conditions and academic literature indicate that substantial improvements to airport infrastructure are not warranted, use of large UAS, whether commercial or military, will have a substantial impact. According to both the AIA and Avascent, "through 2036 large unmanned aircraft are expected to drive nearly $150 billion in total spending and sustain up to 60,000 R&D, manufacturing and services jobs annually" (AIA and Avascent, 2018).

It is anticipated that all UAS, depending upon mission and need, may use airport infrastructure at some level. However, UAS aircraft with weights greater than 10,000 lbs will need airport infrastructure or similar landing and takeoff infrastructure to effectively operate.

The Corgan CONNECT Mega Skyport concept supports Uber Air eVTOL urban air operations as well as provides a facility that is not only a transportation hub but provides an urban location that meets the needs of the community. According to the Corgan Blog (Corgan, n.d.), their vision uses the space above major highways to provide locations for development of commercial and social environments that support and connect communities.

It is apparent that the types of UAS and UAS applications will continue to evolve, and development will depend upon need. This in turn will drive regulatory changes specific to the UAS application. At present, airports would benefit from considering the types of facilities that would attract large UAS operators, among these could be available apron, hangar, and office space (Neubauer et al., 2015). Large UAS operators, when treated as an airport tenant, have the

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potential to generate additional revenue if procedures are implemented to allow for safe coordination with manned operations. The potential economic impact of UAS at airports is already documented in several cases where airports and local government have partnered to pursue such opportunities. For example, the Economic Development Administration has recently awarded a $3 million grant to help develop a 20,000-square ft facility in Cape May County Airport, New Jersey, specifically to promote innovation in the UAS industry (EDA Public Affairs Department, 2018). In Oregon, the Pendleton City Council has approved a $600,000 task order to extend utilities to a new industrial park, north of the airport, meant to support UAS businesses (Sierra, 2018). Airports along with government, academia, and industry can collaborate using governmental grants (e.g., U.S. Department of Commerce, U.S. DOT, FAA, and U.S. DOD), public-private partnerships, public-public partnerships, and academic development to enhance opportunities for UAS development while creating an environment for UAS operations to innovate.

General aviation, reliever airports, and commercial facilities with available capacity and infrastructure, such as UAS-specific runways, STOLports and/or vertical takeoff and landing pads, could provide an incubator for UAS growth and aviation technology evolution. UAS runways or STOLports would allow segregation of traffic such as commercial, cargo or flight training activities as well as support additional aviation related development and efficient use of available airport property.

6.5.1 Environmental Changes/Needs

As UAS technology improves, it is anticipated that more electric means of propulsion will become more common. It is not unreasonable that airports and UAS operators will require a means of properly disposing of used batteries. Although this does present an environmental concern, at the time of writing this report, little information was readily available beyond speculation. It is feasible that airports could begin considering used batteries in a waste/recycling program (Hodgman, 2018). Additionally, noise levels are different for UAS which should be taken into consideration from the public perspective (Neubauer et al., 2015). Beyond considerations of battery disposal and potential noise concerns, no other environmental challenges were noted based on a review of the literature and none are expected based on the current trends related to UAS.

6.5.2 Communication and Operational Needs

Airports should take measures to ensure safety by offering redundancies for UAS operations. Most commercial and civil UAS, unlike their military counterparts (e.g., Global Hawk) require additional time on the runway or taxiway to calibrate their systems, thus impacting airfield capacity. UAS operations may also require additional protocols, communication, and safety measures compared to manned aircraft. Thus, due to these current technological and operating limitations, manned and unmanned aircraft operations generally remain separate (Neubauer et al., 2015).

The communication challenges will be overcome as technological advances and innovations occur to sensor, GPS, spectrum and communication technology. Although it is possible that UAS communication technology may require the expansion of remote towers, which have the potential to affect airport infrastructure/operations, the previous *ACRP Report 144* noted that "no specific challenges encountered by airports with UAS operations were discovered." This finding is supported by current developments at the IPP test sites which show that the FAA has and is allowing UAS operations in more densely populated areas.

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6.5.3 Funding and Grant Assurances

If UAS operate like manned aircraft at airports, it is reasonable to assume that UAS operators should be treated like manned aircraft operators. *ACRP Report 144* notes that some airport operators believe that UAS facility development will ultimately become AIP eligible (Neubauer et al., 2015). Additionally, to avoid potential conflicts, airports must charge fair market rates comparable to their manned counterparts to comply with existing federal and state grant assurances. However, concerns have been raised about how UAS operators will contribute financially to maintain infrastructure if operations do not require runway facilities (Wanner, 2018).

Airport infrastructure costs will depend on the type and mission of the UAS. Funding will also depend upon anticipated infrastructure use: public or private. As UAS operations become more of a 'public good,' airport infrastructure is more likely to be eligible for federal funding. Ultimately, a combination of governmental funding and public-private partnership funding is expected to support UAS infrastructure at airports, as well as other infrastructure available for public use such as those used by emergency responders. Yet, if federal, state, or local governments do not establish a stake in UAS infrastructure, businesses could create a monopoly through a closed system of vertiports or other related infrastructure (AIA and Avascent, 2018). Therefore, like private airports, governments, both federal and state, should take a role in at least regulating private UAS infrastructure and requiring public access as needed for aviation safety and security. What is clear is that the UAS industry has been growing as more money is poured into research and development. Beyond these general speculations, funding concerns surrounding UAS will continue to remain uncertain until UAS operations become more typical.

6.6 Findings and Anticipated Future Conditions

Based on current industry news, informed speculation, and emerging UAS technologies, airport sponsors should be mindful of changes that need to occur. Because the industry is making significant investments in UAS technology, research, and development, the outcome of these efforts will invariably require changes to both regulations and airport infrastructure. The type of UAS operations will determine, in part, the impact at and to airports. Other factors affecting airports will include the regulatory environment, available funding and scale of UAS deployment. Figure 5 outlines the different types of UAS and some of their anticipated effects on airports in the future.

Commercial UAS activity is likely to impact airports, depending on the scale of the activity and UAS. It is anticipated that large-scale delivery of packages by UAS would require airport infrastructure. For example, for the carriage of property, one potential scenario could be the distribution of packages by UAS from the cargo area of an airport. Boeing CEO has said, "the full vision of self-flying cars ferrying people through busy urban areas will take longer than five years to realize, . . . but vehicles that start with more simple functions like cargo aren't far away" (Levy, 2018). As the cargo aircraft arrive at an airport, UAS could be used to deliver and distribute the payload directly from the cargo facilities, replacing other equipment and/or humans that are conducting some of this today. For smaller commercial operations such as aerial photography or inspection, UAS may not require the use of an airport due to their smaller size and minimal/nonexistent payload requirements. Such examples include UAS operations by insurance companies or real estate agencies. However, these types of UAS operations can also occur on airport property such as the inspection of pavement conditions or security perimeters. Simply put, the effect UAS technology will have on airports depends greatly on the application and type of UAS.

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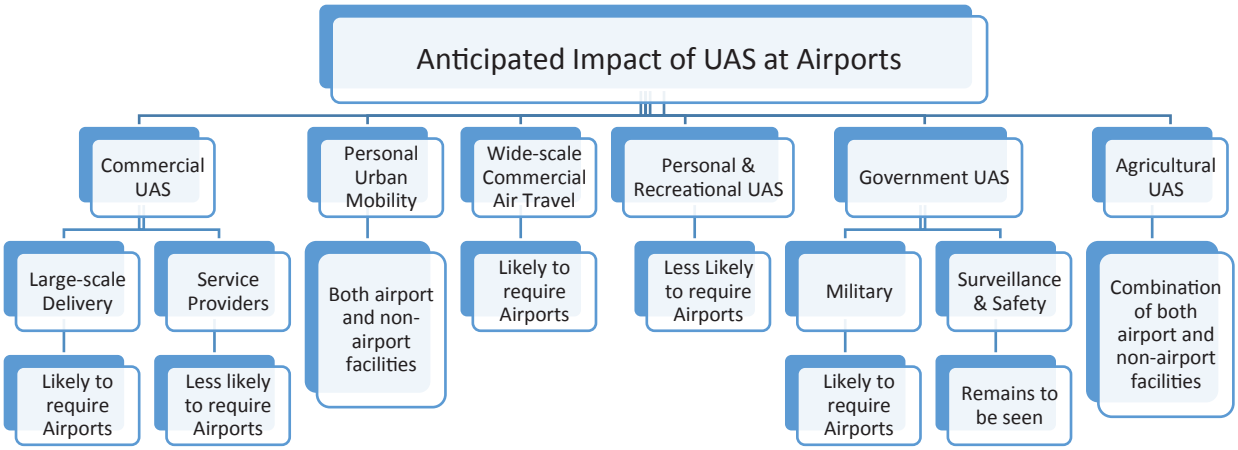


Figure 5. Anticipated impact factors of UAS at airports.

Urban mobility—as demonstrated by the efforts of Uber, Boeing, and Airbus—has the potential to affect facilities both on and off airport property. The industry is already seeing evidence of this. Like Uber Elevate, Vahana announced their first successful flight earlier this year with their eVTOL designed for urban mobility (Lovering, 2018). The air taxis that Uber envisions could "siphon off a chunk of shorter flights [at airports] that are 500 miles or less" (Bachman, 2017). This scenario envisions UAS replacing short, commuter flights. However, battery technology currently doesn't support the sort of range that would be required for this change. These types of UAS operations, if realized at airports, may trend toward uncongested smaller general aviation or small commercial airports for several reasons. First, testing UAS operations at an airport may be easier in terms of proving emerging technology, especially to the FAA. Secondly, smaller and less busy airports have existing infrastructure such as runways, helipads, and taxiways, and potentially terminal facilities and hangars that can be utilized to operate and accommodate UAS activity, thus sparing capital investment in new infrastructure. As an example, eVTOLs could utilize existing helipads for operational activity. However, as the technology becomes more commonplace, these UAS would require other areas for maintenance and storage. This may require vertical infrastructure that would require proper fueling stations (electricity or otherwise) and access much like a general aviation terminal. While automobile parking may be a near term need to accommodate travelers who are being transported, it may not be needed in the future depending on the evolution of automated and/or shared vehicles. Either way, the ability of passengers to access the "terminal" or locations where the UAS are operating will be required so ground transportation and access need to be considered. Most existing airports typically have these facilities already in place.

Airspace conditions would need to change accordingly to accommodate UAS activity, whether passenger or freight. This is another advantage of using existing airports since the airspace is already reserved and generally protected for existing manned activity, compared to activity that may take place at new locations or in metropolitan areas that do not today have flight corridors.

Conversely, should regulation and conditions not be conducive to on-airport UAS development, it is likewise feasible that private investment for UAS infrastructure could occur elsewhere, off airport property. The case of Uber Elevate highlights the use of existing infrastructure for the operation of eVTOLs on vacant land, existing helicopter pads, hospitals, hotels, and other buildings for takeoff and landing. Should the industry trend this way, UAS urban mobility operations may require changes to regulation, more than physical changes at airports due to the potential prevalence of low-altitude UAS operations. It is feasible that a private facility could be developed for storage, maintenance and deployment

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of on-demand eVTOLs. Airspace conditions are likely to be less stringent at a private facility as compared to at an active airport with existing conditions. Availability of funding also affects where investment in UAS development occurs. Regardless, UAS for urban mobility has the potential to significantly affect how our cities and airports operate, opening an entirely new dimension of invisible highways in the sky. Airport sponsors should be aware of how the market is trending to anticipate the changes that may or may not be required at their airports. Should airport sponsors feel the need, these impacts can be prepared for by preserving space for future UAS activities, both airside and landside, on ALPs while also considering addressing potential UAS and other new technology infrastructure needs as part of future infrastructure design.

As it relates to widescale commercial travel, current aircraft can travel great distances on auto-pilot. Logically, the next step is full autonomy for commercial aircraft and passenger travel. Although airlines will continue to utilize airports, aircraft may become more and more autonomous, eventually allowing ATC to monitor and control aircraft operations. The explosion of artificial intelligence in recent years promises advances that could potentially lead to a machine-directed NAS. Steps towards this are evidenced at Fort Lauderdale–Hollywood International Airport which employs artificial intelligence developed by Searidge Technology to enhance ATC and airport efficiency (Searidge Technology, 2017). Public acceptance of artificial intelligence in ATC and autonomous commercial flights is another issue to be considered.

It is anticipated that personal and recreational UAS operations will largely remain separate from the airport environment, although there are low-activity airports looking to accommodate this activity. Several low-activity airports see the potential to be used for personal and recreational operations like model aircraft that have been around for many years. These airports could serve as hubs for this type of activity, especially if their traditional manned counterparts are not operating at the facilities. Whether individuals want to operate at an existing low-activity airport will be driven by factors such as cost, distance, and ability to conduct the types of operations they are interested in performing.

Government sponsored UAS operations will likely continue to occur at airports. Military UAS operate like manned aircraft with subtle differences. Other state-sponsored services such as firefighting and search and rescue operations may also require airport infrastructure akin to the current manned operations. This is primarily because larger UAS may require a runway for takeoff and landing. It is not anticipated that military or other government type UAS operations will significantly alter the way airports currently operate.

Similarly, agricultural UAS may not have as significant an impact to the average individual as urban mobility UAS. Agricultural UAS can operate from both airport and non-airport facilities. Larger UAS that carry heavy payloads for crop spraying would likely require airport infrastructure while smaller UAS used for inspection and monitoring could be deployed in proximity to the activity which may or may not be close to an airport.

Informed speculation is meant to capture a glimpse of what the future of UAS could look like. It is important to consider that informed speculation is merely that and that the future of UAS is a world of new possibilities. However, what is clear is that the challenge for UAS will be both technical and regulatory. As battery technology, autonomy and sense and avoid systems improve, legislation must also keep pace. ". . . Overly restrictive rulesets curtail exports and risk spoiling U.S. leadership in this emerging global market [of UAS]" (AIA and Avascent, 2018). It is anticipated that the FAA will administer more waivers for organizations to test UAS applications. These tests, which will occur both at the IPP test sites and elsewhere, provide the foundation for new and appropriate regulation. UAS is a disruptive technology which has the potential to not only impact airports, but how we communicate, interact, provide service, deliver, travel and much more.

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6.7 Final Thoughts

Based on interviews with professionals in the UAS industry, academic literature, and industry news, it is anticipated that UAS will affect airports though to what extent, is still uncertain. Airport sponsors can take a proactive approach to pursuing such opportunities. *ACRP Report 144* provided practical actions that airport sponsors could take to be prepared for future UAS conditions (Neubauer et al., 2015). Table 10 from that report outlines which steps may lead to action items and plans that, to the extent possible, should be captured by airport sponsors in their master plans.

Several challenges face UAS integration into the NAS. Technological innovations are rapidly occurring. Some of the biggest barriers to UAS integration include the "status quo mindset" of regulatory agencies toward UAS and infrastructure needs, inflexible rulemaking and/or guidance that does not consider potential technological innovations, and export definitions that currently designate UAS as a cruise missile, hampering commercial development (AIA and Avascent, 2018).

Multiple test sites are evaluating the application of commercial UAS operations in several applications. It is anticipated that the FAA will incrementally allow more complex UAS operations as technology such as beyond VLOS and sense and detect/avoid improves.

Aviation continues to evolve as new technologies are introduced and as the population and U.S. economy grows. Thus, according to the U.S. DOT, the following policy options can support aviation growth:

- "Ensuring that sufficient revenue is available to support the operating and capital needs of our national airspace system.
- Balancing the system's multiple and sometimes conflicting needs for modernization, maintenance, access, efficiency, capacity, environmental sustainability, and services.

Table 10. Airport action items and resulting benefits to the airport.

Airport Action	Benefits to the Airport
Engage with a UAS National Test Site	Test sites have available segregated airspace; COAs in place; potential research requirements for airports.
Engage with Area Universities	Multiple universities offer UAS related courses; multiple universities conduct UAS research; universities are partnered with national UAS test sites and Center of Excellence proposal teams.
Contact State Government	Departments of Aviation; Commerce, Agriculture, and Forestry; Mines, Minerals, and Energy; state police may be potential advocates for UAS businesses at airports.
Attend UAS Conferences and Seminars	Conferences and seminars on aspects of the UAS industry are conducted regularly to network and become informed on upcoming technologies.
Investigate Complementary UAS Businesses	Research UAS businesses that could be supported by the airport or by the local economy.
Determine UAS Facility/Infrastructure Requirements	Inventory airport facilities and infrastructure that could be used by UAS operators for marketing purposes.
Contact the FAA	FAA Office of Airports and FAA UAS Integration Office (AFS-80) can inform and offer direction to interested airports.

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- Enabling the safe integration of commercial space flights and unmanned aircraft systems into the NAS while minimizing risk to other users of the system.
- Shifting to a more collaborative, data-informed and risk-based safety management approach to proactively address emerging safety risks.
- Improving surface access to airports for passengers and freight" (U.S. DOT, 2018).

In short, "a combination of technological advances and growing consumer comfort—enabled by sound policy decisions and a supportive regulatory environment—will ultimately fuel economic growth and job creation [in the UAS industry]" (AIA and Avascent, 2018).

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Acronyms

AAGR	Associated Annual Growth Rate
AC	Advisory Circular
ACIP	Airport Capital Improvement Program
ACY	Atlantic City International
ADO	Airports District Offices
ADS-B	Automatic Dependent Surveillance-Broadcast
AGIS	Airport Geographic Information Survey
AI	Artificial Intelligence
AIA	Aerospace Industries Association
AIP	Airport Improvement Program
ALP	Airport Layout Plan
ANPRM	Advanced Notice of Proposed Rulemaking
APO	Office of Aviation Policy and Plans
ARC	Airport Reference Code
ARFF	Aircraft Rescue and Fire Fighting
ARP	Office of Airports
ASAP	Aviation System Action Program
ASOS	Airport Surface Observation System
ASV	Annual Service Volume
ATC	Air Traffic Control
ATCT	Air Traffic Control Tower
ATL	Hartsfield-Jackson Atlanta International Airport
AUVSI	Association for Unmanned Vehicle Systems International
AWOS	Automated Weather Observation System
BVLOS	Beyond Visual Line of Sight
CAGR	Cumulative Annual Growth Rate
CATEX	Categorical Exclusion
CFR	Code of Federal Regulations
COA	Certificate of Authorization
DHS	Department of Homeland Security
DLR	German Aerospace Center
DOD	Department of Defense
DROTAM	Drone Notices to Airmen
EA	Environmental Assessment
eALP	Electronic Airport Layout Plan
EASA	European Aviation Safety Agency
EIS	Environmental Impact Statement
EMS	Emergency Medical Services

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EPIC	Electronic Privacy Information Center
ERAU	Embry-Riddle Aeronautical University
ETL	Engineering Technical Letter
FATO	Final Approach and Takeoff
FCC	Federal Communication Commission
FTZ	Foreign Trade Zone
GAO	Governmental Accounting Office
GBAS	Ground-Based Augmentation System
GBSAA	Ground-Based Sense and Avoid
GTR	Golden Triangle Regional Airport
ICAO	International Civil Aviation Organization
IFR	Instrument Flight Rules
IPP	Integration Pilot Program
LAANC	Low Altitude Authorization and Notification Capability
LRE	Launch and Recovery Element
MCE	Mission Control Element
MCN	Middle Georgia Regional
MKT	Mankato Regional Airport
MOA	Memorandums of Agreement
MOC	Mobile Operations Center
MRO	Maintenance Repair and Overhaul
MSL	Mean Sea Level
MTOW	Maximum Takeoff Weight
NAS	National Airspace System
NASA	National Aeronautics and Space Administration
NBAA	National Business Aviation Association
NCSL	National Conference of State Legislators
NEPA	National Environmental Policy Act
NOTAM	Notice to Airmen
NPRM	Notice of Proposed Rulemaking
NTIA	National Telecommunications and Information Administration
OANG	Oregon Army National Guard
OIRA	Office of Information and Regulatory Affairs
OMB	Office of Management and Budget
PALS	Planning Activity Levels
PANS	Procedures for Air Navigation
PDT	Eastern Oregon Regional Airport
PFC	Passenger Facility Charges
PUR	Pendleton UAS Range
RAFB	Robins Air Force Base
RD	Rotor Diameter
RPAS	Remotely Piloted Aircraft Systems
RPASP	Remotely Piloted Aircraft Systems Panel
RPZ	Runway Protection Zone
RSW	Southwest Florida International Airport
RTCA	Radio Technical Commission for Aeronautics
SARPS	Standards and Recommended Practices
SEF	Sebring Regional Airport
SJTA	South Jersey Transportation Authority
SMS	Safety Management System
SOP	Standard Operating Procedure

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SRM	Safety Risk Management
STOLport	Short Takeoff and Landing Airport
SWOT	Strengths, Weaknesses, Opportunities, and Threats
TAF	Terminal Aerodrome Forecast
TALS	Tactical Automated Landing System
TALS-TS	Tactical Automated Landing System—Tracking System
TLOF	Touchdown and Liftoff
UAM	Urban Air Mobility
UAS	Unmanned Aircraft Systems
UAS-AG	Unmanned Aircraft Systems Advisory Group
UAVSA	UAV Systems Association
UFC	United Facilities Criteria
UHF	Ultra-High Frequency
USC	United States Code
UTM	Unmanned Traffic Management System
VFR	Visual Flight Rules
VHF	Very High Frequency
VLOS	Visual Line of Sight
VTOL	Vertical Takeoff and Landing

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APPENDIX A

PDT Airport Master Plan

PDT is a commercial service, non-primary airport that also facilitates UAS landings and takeoffs for the Oregon Army National Guard (OANG) and the Pendleton UAS Range (PUR). The addition of UAS activity associated with PUR was one of the issues the Master Plan update addressed. The PDT Master Plan update was funded with an AIP grant and local funds (Century West Engineering, 2018). UAS operations were considered in many components of the PDT Master Plan including:

- Activity Forecast;
- Existing Conditions;
- UAS Infrastructure/Airspace Needs;
- Alternatives Analysis;
- ALP;
- Airport Capital Improvement Program; and
- Capital Funding Sources and Programs and Cash Flow Analysis.

The following sections describe how UAS planning was included in each of these components.

PDT Master Plan Activity Forecast

Two UAS forecasts, Baseline and Growth, were prepared for the Master Plan Update due to the uncertainty in future civilian UAS activity. Both the Baseline and Growth UAS forecasts included moderate growth (5 percent annually) for military UAS operations since military UAS operations were already well established at PDT and would not be impacted by future FAA civilian UAS regulations. It is important to note that PDT is part of PUR. This UAS range is "part of the Pan-Pacific UAS Test Range Complex, one of six FAA designated Test Sites. . . . (whose purpose) is to provide the FAA with testing data to assist them in the development of regulations for integration of Manned and Unmanned Aircraft into the NAS" (Century West Engineering, 2018).

Civilian (i.e., all UAS activity not operated by the DOD) UAS activity is especially difficult to forecast. The Master Plan noted that a few customers alone could potentially generate hundreds of annual operations. Also, future FAA UAS regulations could positively or negatively impact UAS civilian operations. Regardless, a baseline was needed to develop a UAS forecast. Thus, a baseline of 500 civilian UAS operations per year, mostly revolving around operator training, research, and flight testing was assumed. The Baseline UAS forecast included 500 civilian UAS operations annually throughout the 20-year planning period. The Growth UAS forecast included 500 civilian UAS operations annually for the first 5 years after which operations were forecast to increase 10 percent annually for the remaining 15 years. The 10 percent growth rate accounted for the potential civilian UAS market growth and the activity expected due to

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Table A-1. UAS operations forecast for Eastern Oregon Regional Airport.

ACTIVITY	2014	2020	2025	2030	2035
BASELINE PROJECTIONS					
Civilian	500	500	500	500	500
Military	280	380	480	610	780
Total	780	880	980	1,110	1,280
GROWTH PROJECTIONS					
Civilian	500	500	800	1,300	2,100
Military	280	380	480	610	780
Total	780	880	1,280	1,910	2,880

the PUR (Century West Engineering, 2018). Table A-1 shows the two UAS forecasts prepared for the Master Plan.

PDT Existing Conditions

Existing UAS infrastructure was identified as part of the inventory of existing conditions. Existing UAS infrastructure at PDT includes 15 50 ft by 50 ft UAS compacted gravel operation pads. These pads featured potable water, electric, and fiber internet access. The OANG used one of these pads for operations. The Master Plan also noted that the "OANG uses a catapult launcher located southeast of the Taxiway Golf and Foxtrot intersection, and typically recovers the UAS on Taxiway Foxtrot. The use of Taxiway F for UAS recovery requires the taxiway to be temporarily closed by a Notice to Airmen" (Century West Engineering, 2018). The documentation of existing conditions also noted the advantage of being the PUR's designated test site.

PDT UAS Infrastructure/Airspace Needs

A separate chapter of the PDT Master Plan was developed to address UAS planning, infrastructure needs, airspace requirements, and to provide a phased plan for UAS integration. The Master Plan noted that the need for UAS infrastructure, equipment, and support service varies and is dependent upon the size and type of UAS. Therefore, a description of infrastructure needs for various categories of UAS based on the informal DOD UAS Groups was included. Table A-2. summarizes infrastructure requirements for each UAS Group.

The PDT Master Plan described then current and future UAS airspace requirements. Current requirements were based on the existing OANG Shadow (RQ-7) UAS operations. In discussing current airspace requirements, the Master Plan stressed the importance of the ATC tower. The ATC tower had a visual display provided by the DOA that controllers used to track the shadow while in flight. Under current airspace conditions, manned and unmanned operations were being segregated to avoid traffic conflicts and mitigate risk. The Master Plan noted that mixing of manned and unmanned traffic within Class D airspace during launch and recovery UAS operations was prohibited except for a chase aircraft. Lost link procedures were also described. The PDT Master Plan stated that the current procedures could accommodate UAS testing and Oregon ANG training operations for the next

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Table A-2. PDT airport infrastructure requirements.

	Runway Requirements	General Services	Facilities	Administrative/ Office Space
Group 1	None.	Mobile Operations Center (MOC), radio communications equipment, crew shelter, data processing space.	None.	Data processing, training, secure storage.
Group 2 and Group 3	Wide range of requirements for Unmanned Aircraft platforms and associated launch, recovery and control mechanisms ranging from pneumatic launchers, skyhook recovery, to runway and net system recovery.	Fuel, UAS pad maintenance, utility support (e.g. internet, power, trash, and sewer), transportation, security and labor associated with safety, compliance, and administration support. Memorandums of Agreement (MOA) will be required with the ATC tower for airfield movement and airspace coordination/approval.	Many Group 2 systems utilize an MOC to support operations in the field. The UAS pads located on the airport can accommodate a wide range of trailers to meet the needs of current and future UAS customers.	Data processing, training, secure storage.
Group 4 and Group 5	As a general rule, Group 4 & 5 UAS operate very similarly to manned aircraft aviation and require very similar infrastructure and equipment support.	Large UAS will require airfield services such as towing, refueling/ de-fueling, deicing, power, security, and hangar space. MOA's will be required with the ATCT for airfield movement and airspace coordination/approval.	Depending on the owner/operator, Group 4 and Group 5 UAS platforms utilize command and control stations that may be building-based or housed within mobile ground stations. The DOD developed mobile ground stations to support overseas locations and separated the Mission Control Element (MCE) and Launch and Recovery Element (LRE) functions. These stations are typically housed in commercially available trailers outfitted with Ultra High Frequency (UHF) and Very High Frequency (VHF) radio links, a C-band line of sight data link, and KU-band (12-18 GHz) communication satellite data links. Other users, such as National Aeronautics and Space Administration (NASA), utilize a building-based operations center where ground, support, and communications equipment are permanently installed.	The administrative footprints for large platforms are significant with personnel office space ranging from 10-20 offices with a conference room, break-room, and bathrooms. Space located above a large hangar or a small-detached building would meet the needs of required administrative personnel.

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5 to 10 years unless significant changes occurred to the ANG training operations (Century West Engineering, 2018).

The PDT Master Plan then discussed the three phased plans for the PUR. The Phase I plan had already been implemented and the Phase II and Phase III plans informed the future UAS infrastructure requirements outlined in the Master Plan. These plans were purposefully flexible as they were designed to accommodate both manned and unmanned aviation until the UAS market was more mature (Century West Engineering, 2018).

As part of Phase I, a dedicated 2,800-foot UAS strip and a full service UAS operating area was provided. Also, the fifteen 50-foot by 50-foot compacted gravel pads previously noted were installed adjacent to the UAS strip. Phase II included construction of several multi-purpose hangars and Phase III included long-term development of an industrial park with road access; various buildings and hangars for UAS use; and a new UAS launch and recovery runway (Century West Engineering, 2018). Phase II and Phase III proposed developments were carried forward to the alternatives' analysis.

PDT Master Plan Alternative Analysis

The analysis of airport development alternatives included consideration of UAS infrastructure requirements. As part of the introduction to the alternatives, the PDT Master Plan acknowledged that "[t]here are no FAA design standards specifically developed for UAS airside facility planning. For this planning process, existing FAA design standards for comparably-sized conventional aircraft (Airplane Design Group I and II) will be used to define operating areas (runways, taxiways, etc.)" (Century West Engineering, 2018).

All the preliminary airport development alternatives included a proposed UAS-only runway. "The proposed UAS runway is intended to allow improved separation between conventional aircraft and UAS equipment. The proposed UAS-only runway (U7L-U25R) is located 700 feet north, and parallel to existing Runway 7–25, which meets the FAA standards for accommodating simultaneous operations during visual flight rules (VFR) conditions" (Century West Engineering, 2018). The North UAS Development Alternative included UAS airside and landside development. The airside development consisted of an 1,800-foot long (optional 2,800-foot) by 60-foot wide UAS-only runway and a parallel taxiway. Support facilities included a UAS apron, UAS launch pads, UAS tower, hangars, commercial buildings, internal access roads, parking and an extended vehicle access road (Century West Engineering, 2018).

PDT Master Plan Airport Layout Plan

The PDT ALP also incorporated planned UAS facilities. The North UAS Development Alternative including the proposed UAS-only runway was shown on the future ALP, Airport Land Use Plan, Airport Airspace Plan (CFR Part 77), and Airport Data Sheet. Also, proposed UAS facilities were included on one of the On-Airport Individual Area Plans Drawing, the UAS Development Area Plan. The ALP set also included a Runway Protection Zone/Inner Approach Plan and Profile sheet for the proposed UAS Runway U7L-U25R.

PDT Master Plan Airport Capital Improvement Program

Proposed UAS facilities were also included in the 20-year Airport Capital Improvement Program (ACIP). The intermediate term projects included:

- "UAS Runway (2,800' x 60') [National Environmental Policy Act Documentation];
- North Access Road (to UAS Development);

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- UAS Runway (2,800' x 60');
- UAS Parallel Taxiway; and
- UAS Apron." (Century West Engineering, 2018)

Cost estimates for these facilities were included in the ACIP and the Master Plan indicated that these projects would be eligible for AIP funding.

PDT Master Plan Capital Funding Sources and Programs

Finally, the PDT Master Plan identified capital funding sources and programs and provided a cash flow analysis. In addition to AIP funding, two potential state funding sources/programs were noted, namely, Connect Oregon and the Aviation System Action Program (ASAP) Fund. "Connect Oregon is an initiative to invest in air, rail, marine, and bicycle/pedestrian infrastructure to ensure Oregon's transportation system is strong, diverse, and efficient" (Oregon DOT, n.d.). "Connect Oregon projects are eligible for grants that cover up to 70 percent of project costs" (Oregon DOT, n.d.). The Oregon State Legislature established the ASAP Fund in 2015 which is distributed through three programs: Critical Oregon Airport Relief Program, Rural Oregon Aviation Relief Program and State Owned Airports Reserve Program (Oregon Department of Aviation, n.d.). The PDT Master Plan noted that the City of Pendleton applied for a Connect Oregon grant in 2016 for PUR but was not successful. The PDT Master Plan did not indicate whether these programs would be applicable to UAS development. Like state airport infrastructure investments, these happen on a case-by-case basis and differ per state based on funding availability.

Beyond the Master Plan

Several interesting UAS developments including new airport users and additional funding have occurred at PDT since the initial publication of the Draft Master Plan in December 2016. Airport users now include Airbus A^3 that is testing the Vahana, PAE ISR that is testing new vertical and landing reconnaissance unmanned aircraft, and Cubic Corporation that is flying the Ares surveillance drone (Banse, 2018). The airport has been successful in obtaining state economic incentive program grants. For example, in late 2018, the airport received a $301,122 state grant for a new mobile command center, and the mission control and innovation center that houses the PUR administrative offices and UAS training, operations and manufacturing. The grant also provides funding for new equipment to enhance 3-dimensional printing and metal fabrication. A U.S. Economic Development Administration grant is also being pursued to help fund construction of new hangars to meet drone company demand (Sierra, 2018).

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APPENDIX B

General Airport Issues and Challenges with UAS

Airspace and UAS Traffic Management

UAS integration with the NAS and associated traffic management with manned aircraft is a top priority of the FAA and aviation industry. Based upon meetings with FAA Airports, ATC and UAS Integration personnel, UAS operating within the airport airspace, including transitioning to and from the NAS, must follow established airport terminal airspace and ground operations procedures in addition to FAR Part 107 operational requirements. These include monitoring aircraft and ATC communications, contacting airport management and ATCT personnel (if available) regarding planned operations, monitoring for manned aircraft, giving "right-of-way" to manned aircraft operations. Thus, depending upon airport airspace class, additional UAS operating rules and air traffic management procedures will be required to support UAS airport activities. As part of this effort in March, 2019 the FAA proposed requiring UAS operators to use the LAANC system within 5 miles of airports.

Airport Airspace and UAS

The airport operating area includes controlled airspace, which is defined by the level and type of traffic. Depending upon mission and equipment, UAs operate at different altitudes. Commercial small UAS remain limited to altitudes of less than 400 feet, but military UAS can operate at higher altitudes. Figure B-1 illustrates the NAS operating levels for different UAS missions.

Airspace surrounding an airport, depending upon type and level of activity and ATC, is defined as Class A, B, C, D or E. Class A is airspace from 18,000 feet to 60,000 feet above mean sea level where commercial UAS are not allowed to operate. Table B-1 describes FAA designated airspace. Each class of airspace requires different equipment and pilot ratings to allow legal entry. To be fully integrated into the civil airspace system, all aircraft, including UAS must comply with these standards for safety, especially with respect to collision avoidance.

Airports with airspace classifications of B, C and D are equipped with an on-site ATCT, so a UAS pilot must obtain permission from ATC to operate within this airspace. For UAS operations to be allowed in Class B and C airspace, an FAA waiver and special operating procedures are required. Class B airspace surrounds airports that support high volume airport operations or passenger enplanements (e.g., John F. Kennedy International Airport). Class C airports are also equipped with an ATCT and regularly support commercial passenger service. These airports are similar to Class B airports but on a smaller scale (e.g., Atlantic City International Airport).

UAS transitioning from the NAS to the airport airspace must follow operating guidance provided by ATC, if available, or the airport sponsor. This includes specific operating and noise

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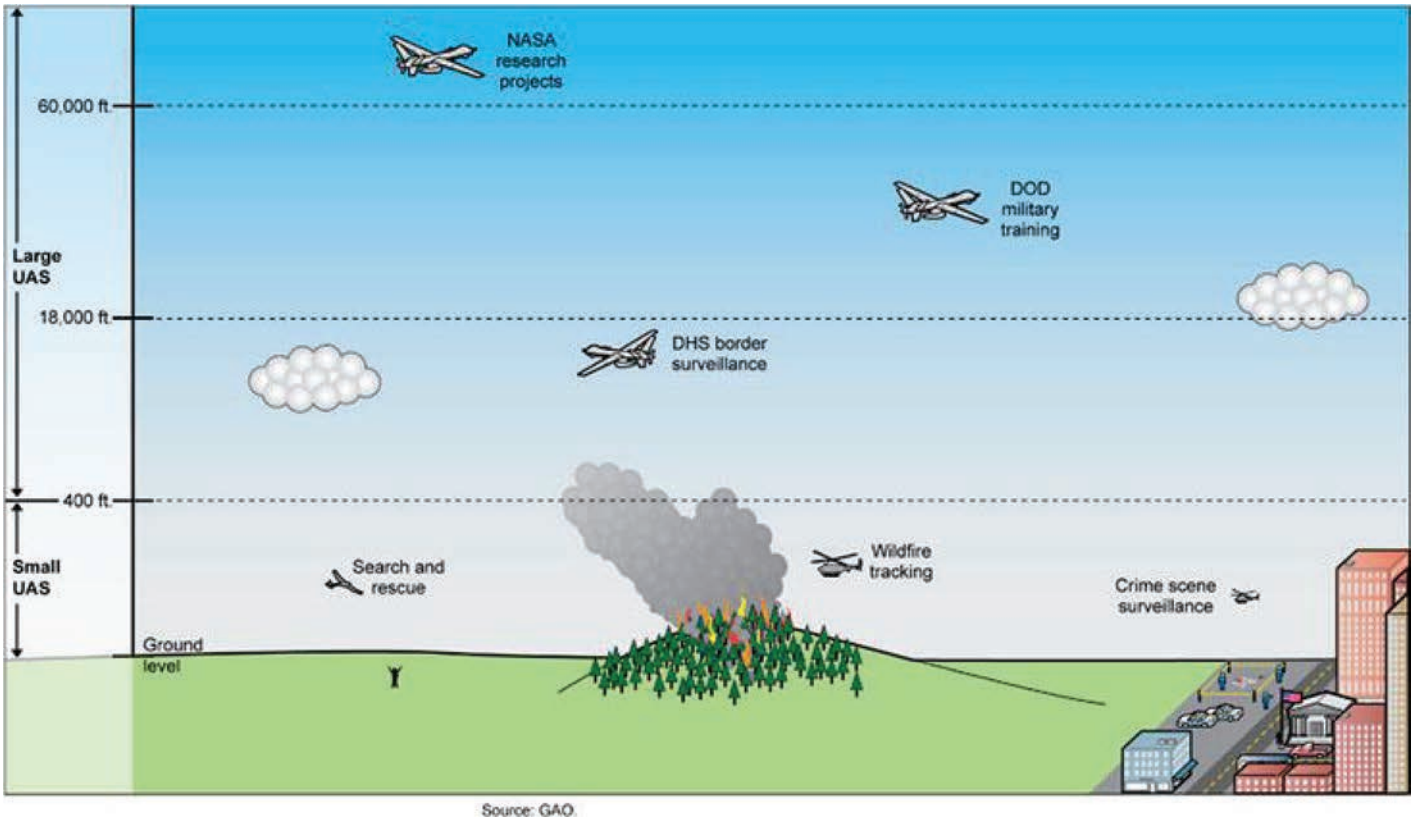


Figure B-1. Examples of current uses for drones and their altitudes of operation (GAO, n.d.).

Table B-1. Classification of air space and entry requirements.

Class	Coverage	Entry Requirements
A	All airspace that is at or above 18,000 ft MSL (mean sea level) and below 60,000 ft MSL	IFR clearance and two-way communication with ATC.
B	Usually has a radius of 15 miles and a height of 10,000 ft MSL	A Mode C transponder within 30 (nautical miles) NM of the airport. Two-way communication with ATC and at least a Private Pilot Certificate.
C	Usually a 5-mile radius from the surface to 4,000 ft above ground level (AGL), and an additional 10-mile radius from 1,000 ft AGL to 4,000 ft AGL	Mode C transponder to fly in or above the airspace and two-way communication with ATC.
D	5-mile radius from the airport, including the surface to 2,500 ft AGL	Two-way communication with ATC
E	From the surface, sometimes 700 ft AGL and most often 1200 ft AGL or 14,500 MSL and below class A at 18,000 MSL	Controlled airspace, IFR flights must have two-way communication while VFR must follow weather minimums
F	Ranges from the surface but below 14,500 MSL	Uncontrolled Airspace does not require communication or special equipment

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mitigation procedures. With the growth of UAS operations including the viability of air taxi, package delivery, and personal air transport, the airspace environment surrounding the airport will likely become larger to support additional approach and departure paths. Expansion of airport airspace, low airspace operations and the transition of UAS from the NAS to the airport environment will necessitate development of new airspace procedures and zoning requirements, which are addressed in later sections of this analysis.

Air Traffic Management

FAA and NASA are developing a UTM to support civilian low-altitude airspace and UAS operations. The purpose of the UTM is to avoid accidents between UAS and other aircraft. UTM keeps civilian UAS operations to below 400 feet thus limiting interactions with manned aircraft. However, effective coordination is required to address low-flying manned aircraft such as helicopters, crop dusting aircraft and other low-flying vehicles, as well as operations within controlled airspace.

NASA's UTM research platform is designed to provide an initial system for safely integrating UAS into the NAS while obtaining critical data needed to safely incorporate UAS activity into higher altitudes and existing air traffic routes. Under UTM, UAS and manned aircraft would communicate in real time to determine which operation has priority and an alternate route would be provided if needed. Commercial and recreational UAS operators are limited to this lower altitude. However, some UAS research vehicles operated through the FAA UAS Test Centers in addition to DOD have permission to operate at higher altitudes.

The impacts to air traffic management at the airport level involve coordination and transition of aircraft to and from the NAS for approach and departure. To address UAS operations in controlled airspace while UTM is being developed, the FAA is implementing the Low Altitude Authorization and Notification Capability (LAANC) system. The system provides some specific ATC and operational requirements for low altitude (400 feet or less) UAS operations. It has been recently proposed that FAA would require UAS operators to use the LAANC system within 5 miles of an airport so they can receive near "real-time" airspace authorizations to allow for better flight planning and issuance of NOTAMs and other guidance to manned and unmanned operators (FAA, 2019). The FAA states that LAANC uses airspace data obtained through "temporary flight restrictions, NOTAMS and UAS facility maps (https://www.faa.gov/uas/request_waiver/uas_facility_maps/) showing the maximum altitude ceiling around airports where the FAA may authorize operations under Part 107" (Lillian, 2018). It is important to note that the LAANC system also includes information for recreational, commercial and public UAS users operating outside airport terminal airspace according to Air Traffic Division Order N JO 7210.909.

Other information regarding LAANC, COAs, and Part 107 waivers is highlighted below. For additional information, refer to the FAA UAS Resources Webinar (https://www.faa.gov/uas/resources/webinar/faq/).

- "If an operator already has an existing 107.29 waiver, they will still need to request authorization through the FAA's Drone Zone to operate since LAANC does not currently recognize waivers.
- Under LAANC, when a part 107 operation is approved via LAANC, there is no requirement to contact local ATC, as the approved flight will be at/below the pre-determined altitudes.
- What do we do if our local controlled airport isn't listed as participating in LAANC? If you do not see your airport on the list of facilities, participating in LAANC, you will need to apply

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airspace authorizations through DroneZone (https:FAAdronezone.faa.gov). LAANC is only available at FAA-owned and operated airports (e.g. DFW, LAX, BWI, and BOS). If your airport does not show, it means your airport is either a Federal Contract Tower or is owned and operated by the DOD.

- Are cities and local governments required to have a waiver? Yes, City/county/state/federal governments need an authorization to fly. Those that qualify under the public statutes 49 USC 40102(a)(41) and 49 USC 40125, can fly some qualified missions as a public aircraft operator and need a COA to fly. To fly under the civil rules (Part 107) they need Part 107 certificated pilots and may need waivers to certain portions of Part 107. City/county/state governments must fully comply with either their COA or under Part 107 if flying under Part 107. If they want a waiver to fly at night under Part 107, for example, they must obtain a waiver to Part 107.29 just like any other Part 107 pilot" (FAA, n.d.).

The LAANC system is a temporary measure to support safe UAS operations within the airport airspace and NAS. In discussions with Airport Operators and FAA, airport sponsors have two options. If the airport does not currently have procedures in place to address UAS, including FAR Part 107, COA, or other guidance, then LAANC guidance should be applied. However, as is the case for GTR and Cape May County Airport, UAS air traffic management procedures are in place, thus UAS operators must follow the airport sponsor's approved air traffic procedures. For either option, UAS operators must coordinate with ATC personnel before entering controlled airspace and give priority to manned aircraft operations. However, new regulations currently being drafted may impact the requirement to contact ATC, and require only that a proper flight plan is filed through the LAANC system (FAA, 2019).

With the passing of the FAA Reauthorization Act of 2018, Congress directed the FAA to prepare a full plan for UTM services and applicable standards. The Act allows the FAA to authorize private UTM service providers on an interim basis, and publish all data associated with public use UAS COAs. In addition, the GAO was tasked to study the potential impacts of allowing local control of low-level airspace.

Local control of low-level airspace may set a bad precedent since it would impact not only commercial UAS activity but low-flying manned operations as well. Safe use of various aeronautical technology requires standardized national requirements. UAS local restrictions vary throughout the country and sometimes conflict with federal law. If such a transition in airspace control occurs, airport operators will need to work with the FAA, other airports, industry and local governmental personnel to develop simple and standardized operational and communication procedures.

UAS Communications

Communication infrastructure and procedures are needed to support UAS activity within a controlled and uncontrolled airport environment.

Radio Communication—Pilot and Unmanned Aircraft

UAS consists of an aircraft, a remote operating platform and a communication, command and control system (C3), which links the UAS and operating platform. Since pilot control is provided using a radio data link from a remote location, communication is critical for safe operations. The majority of small UAS operations fly within visual line of sight and are controlled using unlicensed radio frequencies or spectrums. However, given planned expansion of UAS activities BVLOS, access to spectrum and infrastructure to transmit command signals to the unmanned aircraft is needed.

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Existing aeronautical radio navigation spectrum between 700 megahertz (MHz) and 5 gigahertz (GHz) are typically used to support UAS command communication links. However, use of these spectrum options along with a lack of licensing requirements is impacting UAS operator use. To address the issue of limited spectrum, the FAA Reauthorization Act of 2018 directs the FAA, the National Telecommunications and Information Administration (NTIA) and the Federal Communications Commission (FCC) to submit a report to Congress on whether UAS operations should be permitted to operate on the same spectrum as manned aviation use.

To support UAS growth, a fair and efficient system for administering and assigning frequencies to government and commercial users is needed. In addition, the UAS industry must standardize the way command signals are delivered. Some international protocols allow for the use of commercial fixed satellite systems. However, it is anticipated that new ground-based facilities will also be needed to provide sufficient bandwidth and coverage.

Use of the existing commercial wireless network and infrastructure could support a substantial portion of low-altitude BVLOS missions. However, it is unlikely that the current commercial wireless system can accommodate operations in remote areas or higher altitude UAS operations. As a result, some communications infrastructure will need to be constructed at an airport to support UAS operations. The FAA and FCC are evaluating whether the system could "piggy-back" off existing airport systems or if new ground-based infrastructure would be required. Currently, UAS operators use mobile communication systems and existing wireless radio frequencies to support the command and control process. But as the spectrum becomes more congested, regulatory action will be needed since the FCC will have to change its rules to allow UAS to use some of the commercial spectrum bands (Turner, 2016).

Sense and Avoid Radar Communications

FAA regulations require manned and unmanned aircraft operators to apply "see and avoid" to evade other aircraft and terrain. This requires UAS operations to remain within visual line of sight. However, as growth continues, BVLOS UAS operations will become the norm. Therefore, an alternative means of compliance is needed.

One means has been to employ onboard sensors to provide the location of aircraft that might pose a collision threat. Given the size of most commercial UAS to date, the additional equipment would negatively impact operations by increasing the aircraft weight and takeoff needs as well as limiting range and time in the air. To address this issue, several U.S. and international companies have developed ground-based sense and avoid radar systems for UAS. These systems have and are being used by the U.S. military to allow UAS to fly within U.S. or international civil airspace. Currently, ground-based sense and avoid (GBSAA) is the only sense and avoid system certified by the FAA and other international regulatory organizations for routine use in civil airspace.

According to one manufacturer, SRC Inc., its GBSAA System called LSTAR™ can be used to support terminal area operations; lateral transit operations and vertical transit operations from different classes of airspace (SRC, Inc., n.d.). There are multiple options for installation including tripod or pedestal, rooftop or vehicle mount. Ultimately as UAS use expands, GBSAA antenna will become a permanent fixture at the airport. Current information on range and antenna safety area is proprietary. However, applying current ground-based augmentation system (GBAS) guidance provided by FAA, an antenna safety area with a radius of 300 to 500 feet is likely with an antenna height of at least 33 feet. Airport development near the antenna must not interfere with the system signal. Therefore, buildings within 500 feet of the antenna can be no more than one-half the height of the antenna array. Building heights

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and development beyond the GBSAA safety area need to be designed to avoid limiting or blocking the signal.

Radio Communication and Aircraft Management

Radio communication between manned aircraft, UAS, and ATC continues be a concern. Since many UAS Part 107 operators are not certified as manned commercial or instrumented rated pilots, they are not familiar with existing radio communication procedures. Although UAS operators must contact local ATC, if available, or the sponsor before operating within airport airspace, ongoing radio communication between air traffic, manned aircraft and UAS operator is not required. Rather, FAA suggests that unless operating under special conditions, UAS operators must merely monitor the radio frequency and provide right-of-way to manned aircraft. Discussions with FAA personnel revealed concerns about radio communication congestion associated with too many operators using the same frequency. The FAA Technical Center is currently studying communication procedures and spectrum congestion.

Manned aircraft can enter Class B airspace only after obtaining ATC permission, whereas entry into Class C, D, E and G requires prior two-way communications if there is an ATCT present. FAA working with local ATC has issued individual waivers to allow some limited UAS operations within Class B, C and D airspace. However, as UAS operations increase, the LAANC system will no longer be feasible. UAS, ATC and manned aircraft need a communication system which allows for the efficient and safe movement of all aeronautical activity operating within and transitioning to and from airport airspace.

Airport Charting

Another issue highlighted by UAS users and airport operators involved airport charts and identification of UAS operations. With the growth of UAS and urban air mobility (UAM), alternative vertical takeoff and landing (VTOL) sites are forecast to be developed inside and outside the airport environment. Therefore, in addition to development of new flight routes and procedures, aeronautical charts need to be updated to identify UAS and UAM operating areas.

FAA is considering revamping the existing sectional charts to enhance flight planning and readability. SKYVector Aeronautical Charts, provided at https://skyvector.com, identify drone activity areas or drone notices to airmen defining UAS operating areas (DROTAMs™). Hovering over the DROTAMs, pilots can obtain information about the UAS operating area designation, operating days and times, and operating area. A sample DROTAM for Southwest Florida International Airport (RSW) is shown in Figure B-2.

An additional challenge posed by NOTAMs, particular to UAS operations, is determining the area that is being affected. New technologies are being developed by the FAA to help ensure airspace awareness and a safely integrated NAS. Tools that disseminate information rapidly for both manned and unmanned operations include the Low Altitude Authorization and Notification Capability (LAANC) or UAS Facility Maps. Airport operators can take

NOTAM UAS Operating Area RSW_07/026
DEFINED AS 3NM RADIUS OF RSW262022.4 (19.0NM WSW FMY)
SFC-420FT (SFC-400FT AGL) THU FRI TUE WED 1400-2100 1807121400-1812062100.

Figure B-2. Sample DROTAM.

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advantage of these tools to distribute cartographic representations when communicating with airport users.

As noted in a discussion with Jonathan Daniels, CEO of Praxis Aerospace Concepts, specific charting and identification information needs to be provided to highlight airports that support UAS activity, airports specifically designed to support military and/or commercial UAS (drone) operations (i.e., Droneports) as well as VTOL facilities in urban areas that support UAS and UAM activities. As UAS and UAM operations continue to expand, additional documentation and data need to be provided as part of flight planning and to promote safe operations in and around the airport airspace.

Counter-UAS Technology and Airport Infrastructure

UAS technology raises concerns relating to safety and security risks such as unauthorized access to critical facilities, terrorism, or other malicious purposes. In 2019 the FAA published Order 8900.504, Expanded UAS Oversight as a recognition of the risks posed by UAS (FAA, 2019). For these reasons, several counter-UAS systems which use radio waves to detect and restrict UAS operations were tested at a few civil airports as part of an FAA pilot program.

There are two types of counter-UAS technology. The first type identifies and detects when the UAS enters controlled airspace. This type of technology may include radar, audio and camera systems, but does nothing to deter the UAS from entering the airspace. This type of counter-UAS system involves installing several antennas near the perimeter of the airport property or airport operating area. The ground-based radar system's initial detection area varies depending upon terrain and other obstructions, but preferred detection radius is approximately 2 kilometers (6400 feet).

The second type of system involves true countermeasure technology allowing the airport to use various methods to take control, capture or destroy an unauthorized UAS entering the controlled airspace. Airports are just one of several industries interested in applying counter-drone technology.

In 2016, the FAA UAS Integration Office, along with Department of Homeland Security (DHS), DOD and other federal agencies, initiated a pilot program with UAS detection manufacturers to evaluate UAS detection and identification capabilities on and near airports. According to FAA guidance provided in a July 19, 2018 letter to airport sponsors, "a number of significant safety implications and practical issues, as well as legal restrictions, exist" (Dermody, 2019). For these reasons, the FAA has not authorized any UAS detection assessments at any federally obligated airports which had not previously participated in the FAA's UAS detection pilot program. New guidance provided in a May 9, 2019 letter to airport sponsors stressed that any "entity considering installing a UAS detection system may wish to seek systems specific and site-specific guidance from its legal counsel and/or the appropriate authorities" (Dermody, 2019). Therefore, any airport considering this technology should evaluate potential impacts on grant assurances as well as how new systems may require updating airport certification manuals and Airport Layout Plan documents. The installation of UAS detection systems may lead to FAA aeronautical study, while national guidance continues to be developed.

Further, because of the concerns raised by airport sponsors and elected officials, the FAA Office of Airports and Safety Standards released additional guidance and information about the pilot program findings. This information can be found at https:/www.FAA.gov/uas. The pilot program highlights several safety, legal and practical issues associated with airports deploying UAS detection or other counter-UAS capabilities on or near airports (Dermody, 2019).

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A brief list of FAA's findings is as follows:

- Due to substantial high radio spectrum congestion within the airport environment, UAS detection was difficult and, in some cases, impossible.
- Certain operational states also limited detection and require a high level of manpower to operate equipment and discern false positives.
- Current costs of the system are high and prohibitive. The current system requires several layers of redundant coverage and is likely to become obsolete as UAS technology continues to mature.
- The system requires the installation of a permanent system with several sensors to provide adequate airspace coverage. In addition to the sensors themselves which could result in an obstruction to air navigation, the safety areas and coverage distances for these detection technologies varies and would likely negatively impact airport development.

Lastly, based upon discussions with FAA Office of Airports and documentation (Dermody, 2019), the FAA reiterated that it does not endorse or support the use of countermeasures in the airport environment since it may introduce greater hazards to the NAS, cause legal issues associated with unauthorized "taking" of another's property (i.e., conversion), and violate several provisions of federal criminal law in Title 18 U.S.C. Further, the detection technology has not yet advanced to the point where it provides reliable and immediate information regarding security threats.

The FAA Reauthorization Act of 2018, in addition to UAS funding and guidance, includes a variety of new regulations related to UAS operations and security. One such measure is the *Preventing Emerging Threats Act of 2018* which authorizes the Department of Justice and Department of Homeland Security to 'take actions' to 'mitigate a credible threat' (defined by the Secretary or Attorney General in consult with the Secretary of Transportation) that an unmanned aircraft system poses a safety or security threat of a covered facility or asset. This mitigation would not require a warrant or judicial review or oversight and could include physically disabling the drone, taking it over, intercepting communications or seizing the drone itself (Greenwood, 2018). As a result of this legislation, the Act requires FAA to deploy counter-UAS systems at five airports within 1 year.

Also, under Sections 364 and 365 of the FAA Reauthorization Act of 2018 (2018 Act), the FAA must

- coordinate with government agencies authorized to operate counter-unmanned aircraft systems (i.e. Department of Justice, Department of Homeland Security and other federal law enforcement personnel);
- review interagency coordination and standards for use of these systems;
- establish a program to test existing remote detection and identification technologies; and
- "establish and publicize a mechanism for the public and Federal, State and local law enforcement to report suspected unlawful operations of UAS. . . . and adds a civil penalty regime" (Kestleloo, 2018)

Finally, according to the 2018 Act, the DOT must consult with the DOD related to "identification and defensive" counter-UAS technology within the NAS.

Recent events have increased international interest in applying counter-UAS technology. London Gatwick Airport was brought to a standstill due to drone sightings for 36 hours on December 19, and again for an hour on December 21, 2018 (Christian, 2019). More than 1,000 flights and 140,000 passengers were affected (Christian, 2019). Heathrow Airport's departures runway was closed for more than an hour on January 9, 2019, because of reported drone activity (White & Grafton-Green, 2019). Both airports have announced they will make significant investments in counter-UAS technology (Morrison, 2019).

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Urban Air Mobility and Autonomous Vehicles

UAS technology has also spurred the development of urban air transport/urban air mobility (UAM). The UAM system involves an on-demand network of small, electric aircraft that takeoff and land vertically to allow for rapid passenger and cargo transportation between suburbs and cities and within cities. UAM systems would have the flexibility of having an onboard pilot, being remotely piloted or being fully automated. Uber Air, the company's planned urban aviation ridesharing program, plans initially include an onboard pilot before transitioning to a remote pilot system and ultimately to a fully autonomous system. According to the company, operations are expected to start as soon as 2023 (Uber Elevate, 2016). In addition to Uber Air, over 75 UAM developers are anticipated to commence operations within the next 5 to 10 years. The UAM community includes FAA, NASA, DOT, General Aviation Manufacturers Association, transportation researchers, civil aviation authorities, aircraft manufacturers and startups. In New Zealand, China and Dubai, entrepreneurs are already testing and refining different prototype personal air taxi aircraft with governmental support.

UAM uses low-altitude (500 to 5,000 feet AGL) airspace to shuttle one to five passengers or cargo to destinations between 5 and 50 miles. Since proposed UAM systems launch and land vertically, the system is anticipated to provide substantial cost benefits compared with traditional transportation infrastructure. Given the short-haul nature of UAM and existing landside and airside infrastructure at traditional airports, the two systems provide users a symbiotic platform for short and long-haul transportation needs. Many commercial airports are already equipped with large parking garages and ground parking facilities that could be repurposed to create VTOL hubs consisting of multiple takeoff and landing platforms.

General aviation airport infrastructure could also be repurposed to support vertiport hubs as well as personal, business and academic manned aircraft activity. Many general aviation airports are located within 10 to 20 miles outside the commercial city center making airports ideal locations for cargo and passenger transportation hubs. By supporting UAM infrastructure demands on an airport's landside property and infrastructure, conflicts with manned and potentially unmanned large commercial aircraft would be limited.

Self-driving cars and other ground automation technology is already being used at airports internationally. Although parking currently represents a significant portion of commercial airport revenues, a decrease in ground parking facilities would allow airport property to be developed based upon highest and best use while also decreasing airport operating and maintenance costs. A multimodal network and airport city concept will allow airports to maximize their property, reduce environmental impacts, enhance economic growth, while also providing social benefits to its employees, users and the local community.

Challenges currently impacting UAM integration include:

- Vehicle certification and regulatory operating requirements;
- Battery technology;
- Vehicle efficiency, performance, and reliability;
- Cost and affordability;
- Safety and ATC;
- Potential noise, emissions, and waste management impacts;
- Vertiport infrastructure (Uber Elevate, 2016); and
- Pilot, mechanic, and ATC shortages.

Although it is anticipated that pilot augmentation technology and automation will reduce the need for pilots and ATC personnel, mechanics to support UAM, UAS and manned aircraft will continue to remain in demand for the foreseeable future. Thus, on-airport facilities to support aircraft maintenance, manufacturing and avionics would continue to remain in demand.

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Operational Challenges and Airport Land Use Zoning

Integration of UAS operating in and around the airport environment is likely to impact airport operational procedures and airport zoning. Concerns identified by various stakeholders are highlighted in the following sections.

Operational Considerations and Challenges

Small UAS are difficult to see and be seen by manned aircraft operators and air traffic controllers since they typically lack the onboard avionics used by larger manned aircraft. Since small UAS are operating outside ATC, several operational challenges were identified in relation to airports.

- "How will airborne surveillance of small UAS be assured? Should all UAS operating in and around airports be required to have transponders?
- What surveillance is required for airport surface operations?
- What airport infrastructure (i.e., information and communication systems and ATC tools) will be required to support anti-collision and airport efficiency?
- What type of control procedures need to be in place when airspace, runways, landing areas or parking is limited or capacity constrained?
- What is the minimum performance capability required of UAS to operate within the Airport terminal environment?
- To what extent would segregation of airspace and surface operations be required?
- What failure contingencies must be established to address UAS failures (e.g. lost ATC communications, navigation, surveillance, and emergency)?
- How can airports assess the operational and financial impacts of UAS operations on their current business, including liability?" (Matthews, Frisbie, and Cistone; 2017)

Other operational concerns identified in discussions with FAA Airports and UAS Integration Office personnel as well as representatives from Praxis Aerospace, Golden Triangle Airport, TBI Airports, and others contacted as part of this study include:

- UAS use of airfield facilities may negatively impact capacity. UAS have slower ground speeds and typically require more time than manned aircraft to establish communication and navigation links.
- Current UAS, like manned small aircraft, are impacted by wake turbulence, wind and visibility conditions. Thus, larger separation requirements will be required between manned and unmanned aircraft to support safe operations. Since airfield capacity will likely be impacted, can UAS operations be evaluated as part of the airport capacity analysis?
- UAS operators must contact ATC and airport sponsors prior to flying in and around the airport environment, so NOTAMs may be issued. Discussions with airport operators revealed that coordination, especially at general aviation airports, is lacking. An electronic notification system warning manned aircraft operators of UAS activity within the area is needed.

Operational challenges associated with integration of UAS are numerous and varied. However, several airports such as GTR, Cape May Municipal, Sebring Regional, and Syracuse Hancock International Airport have all safely integrated some level of UAS operations within the airport environment. Each of these airport operators have established specific procedures and communication protocols, which promotes a safe operating environment. The management of these airports also work with the local community, government, manned aircraft tenants as well as UAS users to identify any issues, concerns and mitigation strategies.

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Operational protocols must be flexible to allow for the growth of the UAS market and changes in technology. Coordination and communication according to operators is the key to providing a safe and efficient airport environment.

Land Use Zoning

Airport zoning regulations dictate land use adjacent to or in the immediate vicinity of an airport. Zoning ordinances in and around airports fall under the discretion of each state and are usually enforced by the state's department of transportation. Airport zoning guidance is provided by FAA as it relates to airport and airspace obstructions and land use compatibility. Discussions with UAS subject matter experts suggest that with improved technology and the rise of automation that current approach and departure zoning criteria may shrink. While others suggest that with the introduction of UAM and redevelopment of the landside areas to support these operations, airport airspace and zoning may in fact increase.

To effectively address airport zoning and operational impacts, airports can update their ALP. The ALP depicts both existing and planned future development including existing and future airport property, existing and future airport infrastructure, safety zones and approach and departure profiles, existing and proposed land use on and adjacent to the airport as well as existing and planned non-aviation areas.

Airport sponsors asked if UAS development including airspace and surface operating areas should be included on the airport layout plan documentation. Since the ALP is used by the FAA and DOT to identify future needs and zoning requirements, any existing or planned UAS development can be incorporated as part of an ALP update.

Airport Compliance and Funding

Airport sponsors continue to raise concerns about compliance and funding. At a recent meeting with airport managers in Florida[1], several sponsors have and continue to be contacted by businesses requesting to establish UAS passenger air taxi, cargo services and other types of UAS activity at their airport. The FAA has officially designated UAS as an aeronautical activity. "UAS are aircraft under the law and therefore fit the definition of "aeronautical user." (49 U.S.C. §40102(a)(6), 14 CFR §1.1). . . . The FAA Office of Airports is in the process of updating the definition of "aeronautical activity" – as it pertains to airport access – to include certain unmanned aircraft systems" (Price, Bonset, and FAA; 2018). However, under current Airport Improvement Program (FAA Order 5100.38) and Critical Aircraft and Regular Use Determination (FAA AC 150/5000-17), UAS facilities are not currently eligible for AIP funds. Although UAS infrastructure is not eligible, federally obligated airports must still comply with federal and state grant assurances to remain in compliance and eligible for traditional airport capital improvement funding.

Airport Grant Assurances and Compliance

All federally obligated airports are required under FAA Order 5190.6B to comply with standard airport sponsor assurances to remain eligible for federal funds or conveyance of federal property for airport purposes.

[1] East Central CFASPP Meeting, October 22, 2018, discussion with FDOT Aviation representatives and Airport representatives from Flagler Executive, Orlando-Sanford International, Kissimmee Gateway, Orlando International, Valkaria, Ormond Beach, Deland Municipal and Melbourne International Airports.

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Airport sponsor grant assurances can be broken into two categories: general requirements for airport operations, finance and management, and requirements related to airport planning and AIP-funded projects. Since UAS activity cannot be used to support AIP-funded projects, only general airport sponsor grant requirements are currently relevant. Overall, the airport sponsor under Grant Assurance 22 must 'make reasonable and not unjustly discriminatory rules, regulations and minimum standards for the safety and efficiency of the airport' and maintain safe operations and conditions in accordance with Grant Assurance 19.

Grant assurances that relate to UAS airport integration are as follows:

Grant Assurance 19, Operations and Maintenance: Some key aspects of Grant Assurance 19 related to UAS activity include requirements for the airport sponsor to notify pilots through the FAA NOTAM system of any closures or conditions that affect airport use. Currently, airports that provide certificates of authorization for UAS use are not required to issue NOTAMs since the COA is already identified on the airport charts. However, all other airports that allow UAS activity within the 5-mile airport radius should issue NOTAMs reflecting the location of UAS activity, day and time of operations and operating altitude and airspace envelope. An ongoing issue impacting airport operators are "rogue" UAS operators who do not coordinate with ATC or the airport before initiating operations. Without this coordination, the airport is not able to issue a NOTAM prior to the UAS operator initiating operations. The question of whether the airport is "in compliance" with Grant Assurance 19 in this case is yet to be addressed.

Grant Assurance 22a, Economic Non-Discrimination: This grant assurance requires sponsors allow aeronautical users access to the airport on reasonable terms without unjust discrimination. Since UAS are defined by law as an aircraft, the sponsor should permit operators to use the airport unless there is a reasonable justification to prohibit uses based upon safety or civilian aviation needs of the public. Sponsors may not limit or discourage UAS activity by denying use of the airport without reasonable justification, charging excessive or discriminatory fees or denying airport space without reasonable justification. Given the lack of standardized regulations and communications associated with UAS, many airport sponsors do not support UAS activity because of safety concerns. Airport sponsors can coordinate with their local airport district and flight standards offices to determine if refusal to allow UAS activity is a violation of airport grant assurances.

Grant Assurance 22h, Airport Safety and Efficiency: An airport sponsor may make reasonable and not unjustly discriminatory rules, regulations and minimum standards for the safety and efficiency of the airport. Sponsors must apply the rules, regulations and minimum standards consistently to all similarly situated aeronautical users. Therefore, UAS operations and business activity should be treated as any other similar aeronautical activity at the airport. Minimum standards and operating procedures can be implemented to support safe and efficient airport activity. There is no requirement under 22h for an airport to adopt rules, regulations or minimum standards, but failure to do so could violate other assurances (i.e., Grant Assurance 5, 19 and 22).

Grant Assurance 23, Exclusive Rights: A sponsor may not grant an exclusive right to provide aeronautical services or conduct aeronautical activities. Sponsors must avoid imposing conditions or restrictions that grant an exclusive right to conduct any aeronautical activity. Given the mission and operating requirements of commercial UAS, development of segregated UAS-only infrastructure may be viable to support safe operations. This does not violate the Exclusive Rights grant assurance since development would not be limited to only one user but rather to a specific type of aeronautical activity.

Grant Assurance 24, Airport Fee and Rental Structure: Airport sponsors are required to have a fee and rental system that is designed to make the airport as financially self-sufficient

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as possible. However, the fee structure must be reasonable and not discriminatory. Airport sponsors can charge like fees for similarly situated users. Applicability to UAS operations depends upon whether they are based at the airport and require specific infrastructure or are transient users of the airport facilities. Rental fees may include transient or based aircraft hangar rental, building rental and user fees (e.g. waste management, electricity, and water), and apron parking fees. Sponsors can establish an agreement with users based upon current FAA Rates and Charges Policies and market information, if available. The FAA Airport District Office and state DOT aviation division personnel should be consulted regarding fair rates and charges. Note, the airport sponsor must allow federal government aircraft and thus UAS to use the airfield at no charge unless they represent a substantial use (See Grant Assurance 27).

Grant Assurance 25, Airport Revenues: Sponsor must use airport revenue and aviation fuel taxes for airport capital and operating costs or other facilities directly and substantially related to the actual air transportation of passengers and property. Although UAS facilities are currently not eligible for federal capital improvement funding, the airport sponsor is not precluded from using airport revenues to support UAS infrastructure and operational activity. If the airport sponsor is complying with all other grant assurance requirements, airport revenues may be used in concert with other potential funding sources to support UAS development.

Grant Assurance 27, Use by Government Aircraft: As noted in Grant Assurance 24, the airport sponsor must make available all facilities developed with Federal financial assistance as well as landing and takeoff of aircraft to be used by U.S. Governmental aircraft without charge unless use is substantial. Substantial by Government aircraft is defined, unless agreed to otherwise, to exist when Government operations would unduly interfere with the use of the airport landing areas by other authorized aircraft, or during any calendar month where: "Five (5) or more Government aircraft are regularly based at the airport or on land adjacent thereto; or b. The total number of movements (counting each landing as a movement) of Government aircraft is 300 or more, or the gross accumulative weight of Government aircraft using the airport (the total movement of Government aircraft multiplied by gross weights of such aircraft) is in excess of five million pounds" (FAA, 2014). If the governmental aircraft activity is determined substantial, then the airport sponsor may charge the government for use of the facilities. This charge is based upon the proportionality of the facility use and is to be used for the cost of operating and maintaining the facilities used.

Grant Assurance 29, Airport Layout Plan: Public airports which receive federal funding are required to keep up-to-date at all times an airport layout plan showing the following: "boundaries of the airport and all proposed additions thereto, together with the

- boundaries of all offsite areas owned or controlled by the sponsor for airport purposes and proposed additions thereto;
- the location and nature of all existing and proposed airport facilities and structures (such as runways, taxiways, aprons, terminal buildings, hangars and roads), including all proposed extensions and reductions of existing airport facilities;
- the location of all existing and proposed non-aviation areas and of all existing improvements thereon;
- and all proposed and existing access points used to taxi aircraft across the airport's property boundary" (FAA, 2014).

ALPs are subject to conditional approval by the FAA, and the sponsor is required to not make or permit any changes at the airport or its facilities that are not in conformity with the approved ALP or that would "adversely affect the safety, utility or efficiency of the airport"

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(FAA, 2014). If the changes are found to adversely impact the airport, the airport sponsor or operator must:

- Eliminate the adverse effect; or
- Bear all costs of relocating the airport facilities to a new site as well as the costs of providing the facilities necessary to meet the level of safety, utility, efficiency and operations that existed before the unapproved changes were made.

As this relates to UAS integration, UAS cannot as an aeronautical activity be prevented from operating at an airport if they comply with all operational and safety requirements. However, infrastructure used to support UAS operations must be approved by the FAA and state DOT, even though no funding is currently available. UAS facilities as well as proposed approach and departure procedures can be incorporated into an airport's ALP set if the airport plans to support regular UAS operations and related activity. Thus, if an airport or associated public entity obtains either a Part 107 waiver or COA to support UAS activities at the airport, this data should be incorporated into the airport's layout plan set to maintain conformity with airport sponsor grant assurances.

Local and State Restrictions

Historically, the authority to regulate aircraft and flight operations is controlled by the FAA. However, given the proliferation of UAS and concerns about privacy, law enforcement and safety, an inconsistent patchwork of local and state regulations are being created. This lack of uniformity is having a negative impact on UAS development but will also impact airport operations and development.

Several airport sponsors and operators have had conflicts associated with local zoning ordinances and land uses within existing and planned approach and departure paths which have delayed or negated future airport development and capacity. Restrictions on aeronautical activity by local and state law may also cause airports to be in direct conflict with their federally obligated grant assurances. Conflicting regulations impacts the overall safety of aircraft and airport operations.

Further, according to the FAA Reauthorization Act of 2018, Section 373, the U.S. Government Accountability Office is tasked with evaluating federal control of low altitude airspace and viability of enhanced state and local control. State control of low altitude airspace would impact not only UAS operations but also low altitude manned operations (e.g. medivac and emergency operations, crop dusting, and aerial survey). State and local regulations along with proposed changes in airspace management would undermine development and implementation of uniform flight rules thus, negatively impacting the safety of airport operations.

UAS Revenues and Airport Infrastructure Funding

Recent legislation, combined with advice from governmental entities and drone/UAS advisory committees, highlights some of the potential sources for funding UAS integration and development into the NAS and airport arenas. Some funding and fee changes related to UAS provided in the 2018 FAA Reauthorization Act requires:

- "[The] Comptroller General of the United States to identify the appropriate fee mechanisms to recover the costs of "the regulation and safety oversight of unmanned aircraft and unmanned aircraft systems" and "the provision of air navigation services to unmanned aircraft and unmanned aircraft systems" (Rupprecht Law P.A., 2018).

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- FAA to update 107.205, flying a drone from moving vehicle, and 107.25, beyond line of sight, to allow for carriage of another person's property for compensation or hire, and
- FAA to establish a small UAS air carrier certificate for the transportation of property for compensation or hire (Rupprecht Law P.A., 2018).

Financial UAS and UAM forecasts predict a billion-dollar global industry supporting cargo, passenger, civil and military transportation needs as well as robotics and automation. A December 2017 McKinsey & Company study (Cohn, Green, Langstaff, & Roller, 2017) estimates that by 2026 "commercial drones – both corporate and consumer applications – will have an annual impact of $31 billion to $46 billion on the country's Gross Domestic Product." Since traditionally, airport infrastructure development can take up to 10 years from planning to construction, airports need to evaluate how UAS will impact business activity while identifying potential funding sources and return on investment.

The advantage of an existing airport compared to other transportation systems is that most infrastructure to support UAS activity, both large and small, is already in place. Therefore, likely capital costs associated with UAS integration, depending upon the location and type of airport, could involve retrofitting or modifications of existing facilities as well as installation of additional radio communications and navigational aids. In March 2018, the final Drone Integration Funding Report was published by the RTCA (Radio Technical Commission for Aeronautics) (Drone Advisory Committee, 2018). Potential funding mechanisms include user fees, public-private partnerships, and lease of airport facilities, properties and/or airspace. With the growth of UAM, airports, including general aviation airports, should also consider not only lease and user fees but also landing and parking fees as well to offset any lost ground parking revenues.

Ultimately, airports should work with their local and regional FAA airport personnel to identify opportunities and documentation (i.e., market studies and benefit cost analysis) to determine potential funding sources.

Regulatory Environment—Domestic and International

The FAA's primary focus has and continues to be on UAS integration into the national and international airspace systems. Still, operations by larger UAS for cargo, personal travel and potential other commercial and civil activity is being tested as is beyond VLOS operations. To date, some foreign governments, including Israel, Dubai, Saudi Arabia, United Kingdom, France, Germany, as well as the Netherlands, are supporting growth and development of UAS technology.

Discussions with UAS operators and manufacturers revealed that there are significant operational and regulatory differences among the United States, European Union and other ICAO member countries. Currently, UAS operators must follow at least two sets of operating rules and technical requirements if they want to operate both domestically and internationally. These operational and regulatory differences impact an airport sponsor's ability to attract users and tenants from outside the United States, thus limiting revenue development. Several airports throughout the United States are equipped with a foreign trade zone (FTZ) since they may be operated by both public and private entities. An FTZ is a geographic area within or adjacent to a U.S. Port of Entry where domestic and foreign commercial merchandise receive the Customs treatment as if it were outside the commerce of the United States. Firms use FTZs to defer the payment of duties and taxes. FTZs also allow the re-export of cargo while avoiding applicable duties and taxes since merchandise never entered the U.S. consumption area (Hawaii Foreign Trade Zone, n.d.). Thus, given

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foreign investment and development of UAS and UAM technology worldwide in addition to the global nature of aviation, standardized UAS operating, manufacturing and design criteria will benefit all parties.

To this end, the 2018 FAA Reauthorization Act provides the following support to UAS operators:

- "Requires the FAA to assist U.S. companies experiencing delays in foreign authorizations;
- Requires within 1-year a foreign engagement action plan;
- Requires FAA to coordinate with ICAO to ensure that any new international standards for aircraft tracking and flight data recovery is consistent with a performance-based approach and is implemented in a globally harmonized manner" (Dombroff & McKinnon, 2018).

Airport Tenant Privacy Concerns

In the case of Electronic Privacy Information Center (EPIC) v Federal Aviation Administration (June 19, 2018), the FAA argued that privacy concerns are beyond the scope of its rulemaking authority. Since the Court dismissed the case, the FAA will stay out of regulating privacy issues associated with UAS activity leaving it to other federal agencies (FCC) and state and local governments (Makarious, 2018).

Privacy continues to be an issue not only for the public but also for on-airport tenants. Airport tenants have raised concerns regarding on-airport UAS activity and intellectual property, operations as well as client and personnel privacy issues. For instance, Piper aircraft has requested that the City of Vero Beach and airport management not allow UAS activity near their manufacturing site located adjacent to airport property.

Tenant concerns about privacy are not new. One method to address these concerns is to develop "no UAS fly zones" to eliminate operations over highly sensitive airport and tenant facilities. Under the 2018 Reauthorization Act, the Federal Trade Commission has the authority to pursue violations of privacy policies involving a UAS as an unfair trade practice. Thus, as part of the airport's minimum operating procedures and tenant lease agreements, restrictions may be put into place to limit overflight and recording of tenant facilities without prior permission. Also, concerns may be mitigated through stakeholder outreach and education seminars on UAS for airport tenants.

UAS Rogue Operators

An ongoing issue plaguing the FAA and airport operators is "rogue operators." "The question of how to deal with rogue operators remains unsettled but is perhaps one of the most vexing problems facing the industry" (Turner, 2016). Rogue operations include a substantial number of commercial flights conducted without FAA authorization in addition to recreational users not following the rules regarding where they can or cannot fly. As discussed in *Counter-UAS Technology and Airport Infrastructure*, rogue UAS operators can and have caused substantial delays and safety hazards at major metropolitan airports such as Gatwick International, Charles De Gaulle, Ronald Reagan, Miami International as well as at other airports and critical infrastructure (e.g. nuclear powerplants and dams). Although the FAA does not currently support counter-UAS technology either for UAS identification or UAS capture or destruction, there are several businesses working with the DOD and other organizations (e.g. powerplants, electrical grids, and state and federal prisons) to implement counter-UAS technology. A sampling of businesses providing these services include:

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- SRC, Inc (www.srcinc.com)
- SENSOFUSION (www.sensofusion.com)
- DCAA (Dubai Civil Aviation Authority) & Sanad Academy (www.dcaa.gov.ae, www.sanadacademy.ae)
- ANTIDRONE (www.anti-drone.eu)
- Battelle (www.battelle.com)
- Blighter Surveillance Systems (Carlini, 2017)

Unregulated UAS threaten not only air transportation operations and safety but have a negative impact on commerce and national security. Over 30 anti-drone companies worldwide are already in business to deal with these threats.

During ongoing discussions with airport operators, recurring scenarios associated with unauthorized UAS activity within controlled airspace highlighted several operational issues:

- Airport tenants flying their UAS next to an active runway or taxiway without permission.
- Commercial operators not contacting the sponsor or ATC to get permission to enter airport airspace.
- Operators asserting they obtained permission to operate on the airport, but did not or were refused due to safety concerns.
- UAS operators acting with disdain or aggressively with Airport and Air Traffic Personnel when requesting permission to operate within the airport airspace.

Most of these operators do not intend to jeopardize public safety but are merely trying to "shortcut" the regulatory process. Still the risks posed by these operators can be significant. First is the potential to cause injury to people on the ground or in the air. Since most UAS are less than 55 lbs. and are small, they are hard to see in the air. Further manned aircraft operators during approach and departure procedures are focused on adjusting aircraft speed and trim as they transition to and from the ground to airspace rather than looking out for UAs.

Because of these concerns, the FAA Reauthorization Act of 2018 now includes the following penalties:

- Intentional interference with a manned aircraft or airport is a felony with 1 year in prison,
- Causing injury is 10 years in prison, and
- In furtherance of another crime is life in prison.

Further, if the incident occurs within the airport airspace or property, the sponsor may be found liable for not providing a safe and secure environment. Thus, making the sponsor a possible party to a potential lawsuit and impacting compliance with federal grant assurances and associated funding.

Another indirect impact of rogue operations is the risk that state and local regulators will take actions that may negatively impact activity by rogue operators but also by "law-abiding" commercial and recreational users as well. Although airports and airspace both are under federal jurisdiction, local and state regulatory actions that limit UAS activity may have a negative impact on airport operations and revenue development. Further, given Section 373 of the Reauthorization Act of 2018 suggesting the possibility of shifting low altitude airspace from the federal jurisdiction to state and local jurisdiction, this may raise a variety of issues including: non-standard procedures, differing operational requirements, and communication and coordination issues.

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Summary

The increase in the number and type of UAS operations within the airport setting will provide airports several revenue enhancement and operational opportunities, but stakeholders must also consider the challenges of integrating this new aeronautical activity. Given forecast demand, integrating UAS into routine airport operations will impact all airport stakeholders by affecting airport capacity. Airports could attempt to segregate operations, but given lack of funding and compliance requirements, construction of separate and potentially redundant UAS-specific facilities is unlikely.

Thus, given the various uncertainties and challenges facing integration of UAS operations at an airport, it is suggested that the airport sponsor consider only one type of UAS use or business case initially to evaluate quantifiable costs, revenues, potential regulatory barriers, liability, community acceptance and other impacts. Sponsor actions can include to:

- Determine if proposed operations can be performed under existing FAR Part 107 or if additional waivers may be required;
- Evaluate whether the airport has the facilities, capacity, services and insurance to support planned UAS operations while maintaining the overall safety of airport operations;
- Determine if the UAS operator mission is authorized and approved by FAA, and that the operator can perform planned operations safely both on and beyond the airport airspace;
- Assess whether tenants, users and other airport stakeholders will object to proposed UAS operations, and whether those concerns can be mitigated; and
- Assess whether the community has issues with proposed operations, and how these concerns can be addressed or mitigated (Matthews, Frisbie, and Cistone; 2017).

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APPENDIX C

UAS Forecast Process

Step 1: Gather Current UAS Forecast Data

Several industry and governmental forecasts related to military, non-model, civil and commercial UAS were published by Teal Group, U.S. DOT/VOLPE, FAA, Business Insider, as well as other industry and academic organizations. UAS growth ranged from 3 to 51 percent during a 20+ year average forecast period. Given differences in how data was reported and variations in baseline information, the U.S. DOT and FAA Aerospace UAS forecasts provided the most consistent and realistic estimates of likely UAS demand in the United States during the next 10+ years. Tables C-1 and C-2. highlight the U.S. DOT and FAA UAS forecasts.

This data is used to extract current trends regarding UAS operations, which then may be modified when applied to each individual airport considering various operational (e.g. commercial hub, non-hub, general aviation airport), infrastructure (on-site ATCT, available apron, hangar and land, and type and condition of runways), and other socioeconomic factors (e.g. support of community and stakeholders).

Step 2: Gather Airport Terminal Area Forecast or Approved Master Plan Forecasts

Obtain the most current FAA TAF for the airport and most recent approved aviation activity forecasts (e.g. master plan and state aviation system plan). This data can be used to establish likely UAS based aircraft, the type of UAS operations that may occur at the airport on a regular basis (e.g. military, commercial, and general aviation) as well as the likely breakdown between itinerant and local operations given UAS industry data and trends.

Step 3: Determine Historical UAS Operations by Type

Historical UAS operational data may not be available. No historical UAS operational data was available although the airports had supported some UAS activity. Therefore, if possible, UAS activity should be collected and recorded by the airport. If no UAS activity currently occurs at the airport, using national UAS industry demand and the type of aviation activities currently and likely to occur, airport planners, as they did in the PDT Master Plan, must make a "best guess" estimate of the number, type and timing of when UAS operations may occur at the airport. This information will provide the baseline for future operational and infrastructure assessments. It is important to note that forecasts merely represent likely scenarios of future demand, so estimated baseline data in the forecasts can be shifted forward or back to represent actual demand.

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Table C-1. U.S. DOT forecasts for UAS (commercial, public, and DOD) (USAF, 2013; DOD, 2005).

Year	Total UAS Activity	Commercial Only UAS	All Other Public Agencies including U.S. DOD	US DOD Only – Unmanned and Pilot Optional
2015	11,000	1,000	10,000	9,005
2021	20,125	5,125	15,000	10,005
2026	50,000	20,000	30,000	12,211
2031	160,000	100,000	60,000	14,813
2035	250,000	175,000	75,000	19,000
AAGR 2015-2026	15%	31%	7%	3%
AAGR 2015-2035	17%	29%	11%	4%

Table C-2. FAA forecasts for UAS (FAA, 2018).

	Total Non-Model Fleet	
Year	**Base**	**High**
2017	110,604	110,604
Forecast		
2018	158,900	168,339
2019	229,400	268,937
2020	312,100	410,862
2021	407,400	604,550
2022	451,800	717,895
CAGR 2017-2022	33%	45%
CAGR 2021-2022	11%	19%
2023	501,039	852,491
2028	840,422	2,012,951
2033	1,409,689	4,753,097
2038	2,364,555	11,223,290
CAGR 2023-2038	11%	19%

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Step 4: Create Range Forecasts, Apply Forecast Methods, and Evaluate Results

FAA personnel (Conference Call with FAA Forecasting Branch, 2018) within the Office of Aviation Policy and Plans, Statistics, and Forecast Branch (APO-110) anticipate that UAS activity will merely replace general aviation, military and even some commercial operations as outlined in the TAF rather than provide additional operations above those already anticipated. However, UAS users, published information from DOT and DOD, and other U.S. and international organizations and agencies anticipate that UAS will not only replace some manned aircraft but will also add aircraft into the worldwide fleet. As a result, a range forecast consisting of three models (low, mid and high) can evaluate potential demand given various scenarios.

Low Forecasts: The low forecast assumes that UAS based "aircraft" and operations are already included in the airport's TAF forecast. This forecast assumes that UAS operations will replace manned aircraft operations on a one-to-one basis.

Mid-Forecasts: The mid-forecast assumes that UAS activity will not only add to forecast demand, but in the long-run also replace some manned activity. The likely percentage of aircraft and operations replaced may be determined considering current and anticipated activity at the airport, operating missions, as well as type and size of likely UAS activity.

High Forecasts: The high forecasts anticipate that a limited number of manned aircraft, if any, will be replaced by UAS given the types of activity currently occurring and anticipated to occur at the airport. Thus, most of the forecast demand is added to forecast activity in the TAF.

The range forecasts may be developed using either a recently approved airport aviation activity forecast or the most recent FAA TAF. If the TAF forecast base year differs substantially from data provided by the airport in its Airport Master Record, the TAF forecast base year may be updated to reflect actual airport activity. Applying growth rates provided from the U.S. DOT 2015–2035, FAA Aerospace Forecasts, 2018–2038, UAS and manned aircraft demand forecasts can be created.

Step 5: Compare to TAF

Since the FAA does not currently review or approve UAS-specific forecasts, this step is optional. However, given the unique nature of UAS forecasting and FAA's interest in evaluating the impacts of UAS on the aviation system, a comparison between the forecasts could highlight some needed changes in the governmental forecasting methodologies especially at small general aviation airports. According to FAA Order 5090.3C, Field Formulation of National Plan of Integrated Airport Systems, which is currently being updated, forecasts traditionally should not vary significantly (more than 10 percent) from the FAA's forecast to be approved and incorporated into the TAF and NPIAS. However, even if proposed operational forecasts exceed 10 percent, due to the unique nature of UAS and aviation industry trends, the FAA may ultimately incorporate this information into state, TAF and NPIAS forecasts for future years. Ultimately, this incorporation would clear another hurdle to possible funding for UAS infrastructure improvements.

Step 6: Determine Fleet Mix Forecast

The purpose of a fleet mix forecast is to determine likely critical aircraft demand and to provide a baseline for identifying existing and future airport infrastructure needs (e.g. runway length, pavement strength, and hangar size). Fleet mix source data can be obtained from official airline guides and T-100 data for commercial aircraft activity in addition to airport users and stakeholders.

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As noted earlier, due to the size of current UAS and operating needs, the majority of UAS do not regularly, if at all, use traditional airport infrastructure. However, the military is actively using larger UAS (i.e., Global Hawk) as well as developing pilot onboard options for UAS, and manufacturers are currently designing and testing larger UAS for various commercial uses (e.g. passenger transport, cargo transport, search and rescue, and firefighting). Thus, it is expected, given current development and testing of larger UAS, that as UAS become larger and more sophisticated, their use of airport facilities will likely increase.

Based upon information provided by UAS manufacturers such as Boeing, Airbus, Northrop Grumman, and Lockheed, platforms for UAS greater than 100 lbs are based upon existing manned aircraft airframes. Therefore, using weight and operating requirements, UAS were categorized as shown in Table C-3.

Using UAS size information and existing fleet mix data obtained from DOD and other aviation models (e.g. Teal Forecasts, Business Insider, and IBISworld) as well as historical data from airports currently supporting UAS activity, a breakdown of likely UAS operations by aircraft type was created. Historical and forecast data provided by governments and industry show that airport use by small UAS (4.5 to 55 lbs) will represent the majority of

Table C-3. UAS fleet mix category descriptions.

Type of UAS	Weight (lbs.)	Estimated Size (ft.)	Approximate Wingspan/Rotor Blades (Ft)	Mission Speed (MPH)	Mission Altitude (ft above surface)	Mission Radius (Miles)
Nano	<1	0.98-1.64	<1.15	Varies	<30 feet	< 1
Micro **(based on Black Hornet 3)**	<4.5	< 4	< 8	<13	<400	< 1.5
Small UAS	4.5 to 55	4<10	8' > 20' /8" to 40"	50 to 75	<10,000	5 to 25
Ultralight UAS Airframe	55 to 255	<30	9' > 40'	75 to 150	<15,000	50 to 100
Light Sport UAS Airframe	255 to 1,320	<45	15' >45'	75 to 150	<18,000	100 to 200
Small UAS Airframe/Rotorcraft	1,320 to 12,500	<60	<79'/<40'	100 to 200	<25,000	100 to 200
Medium UAS Airframe	12,500 to 41,000	60 > 95	79'>139'/40'>55'	Est. 100 to 200	<100,000	TBD
Large UAS Airframe	>41,000	>95	>139'*/> 55'	Est. >140	>100,000	TBD

Notes: 139-foot wingspan information was obtained from discussion with Mr. Michael Hainsey AAE, Executive Director, Golden Triangle Regional Airport. Current tenant Northrop Grumman is currently building and testing these UAS/Drone wings at his facility.

Sources: U.S. DOT UAS Forecast, Table 8, Volpe UAS Activity Forecasts, FAA Aircraft Data, ultralight manufacturer aircraft data, UAS rotorcraft blade sellers, and UAS manufacturers, 2018, U.S. DOD Report, Drone Report 2018, Military Drones Specifications, and Commercial Industry UAS criteria, Department of Geography at Penn State University, and Astrid Aviation and Aerospace 2018.

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UAS operations in the near term. However, as the size of UAS aircraft increase and resemble traditional aircraft, use of airports is also expected to increase. The likely breakdown of UAS activity at airports is provided in Table C-4.

Using the data in Table C-4, each type of UAS may be divided by the percent of the total estimated UAS fleet anticipated to regularly use an airport. For example, small UAS using airports in 2017 is estimated at 6 percent.

Dividing that 6 percent by 6.38 percent, the total percentage of the UAS fleet likely to operate at an airport, results in an estimate that 94 percent of those UAS operations will be associated with small UAS activity.

Likely small UAS operating at an Airport = 6%/6.38% = 94% of total UAS Airport Operations would be Small UAS

Table C-5 demonstrates the likely breakdown of UAS by type during the planning period.

Note, this is an estimate given current market conditions, UAS industry trends and current UAS data. This analysis also considered that airports support some level of military operations. Therefore, the percentage of large UAS estimated may be higher than other similarly sized airports that don't support substantial military operations.

Table C-4. Estimated percentage of total forecast UAS operations likely to occur at an airport.

Type of UAS	2017	2018	2023	2028	2033	2038
Nano	0%	0%	0%	0%	0%	0%
Micro	0%	0%	0%	0%	0%	0%
Small UAS*	6%	6%	7%	8.00%	8.30%	8.60%
Ultralight UAS Airframe**	0.38%	0.54%	4.00%	5.00%	5.00%	5.00%
Light Sport UAS Airframe**	0.00%	0.03%	4.00%	6.00%	7.00%	8.00%
Small UAS Airframe/Rotorcraft**	0.00%	0.00%	3.00%	6.50%	6.50%	6.50%
Medium UAS Airframe**	0.00%	0.00%	2.00%	6.00%	8.00%	8.00%
Large UAS Airframe**	0.00%	0.00%	0.00%	4.00%	4.00%	6.00%
Estimated Percent of Total UAS Operations likely to occur at U.S. Airports	6.38%	6.57%	20.00%	35.50%	38.80%	42.10%

Notes:

* Small UAS is expected to be used based upon historical and current data by airport operators as well as for UAS training.

** These are in development and are being tested. Some include new designs and others involve conversion of approved aircraft airframes.

Due to rounding, numbers may not add up to total.

Note of sources: These data were gathered from discussions with existing and potential users, FAA test site management, Searchlight Airport Management, GSP Management, FAA Aerospace Forecasts and U.S. DOT UAS Forecast 2015-35, The Teal Group Worldwide UAS Forecast, 2017-2037, Unmanned Aircraft Systems, 2015; U.S. DOT Unmanned Aircraft System (UAS) Service Demand, 2015-2035; Business Insider Tech Forecasts of UAS Demand, Boeing bets big on flying taxis and pilotless planes, and Astrid Aviation and Aerospace 2018.

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Table C-5. Estimated airport UAS fleet mix.

Type of UAS	2017	2018	2023	2028	2033	2038
Small UAS	94%	91%	35%	23%	21%	20%
Ultralight UAS Airframe*	6%	8%	20%	14%	13%	12%
Light Sport UAS Airframe*	0%	0%	20%	17%	18%	19%
Small UAS Airframe/Rotorcraft*	0%	0%	15%	18%	17%	15%
Medium UAS Airframe*	0%	0%	10%	17%	21%	19%
Large UAS Airframe*	0%	0%	0%	11%	10%	14%
TOTAL	100%	100%	100%	100%	100%	100%

Applying this data to total forecast UAS operations, results in an estimate of likely operations by type which may then be used along with manned forecast fleet mix data to determine the critical aircraft or group of critical aircraft requirements.

It is important to note that activity forecasts become less reliable the further out they project due to expected changes in technology, consumer demand and overall aviation activity volatility.

Anticipated growth will also be impacted by regulatory requirements and public acceptance of new technology.

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APPENDIX D

UAS and Airport Operational Guidance

UAS Operations and Air Traffic Control

FAA is implementing LAANC. Based upon discussions with airport sponsors supporting UAS operations as well as FAA personnel, the LAANC system applies to airspace and airport terminal airspace in which procedures have not yet been put into place. According to the Executive Director of Golden Triangle Regional Airport (GTR), LAANC procedures do not apply to GTR since management and the FAA ATCT personnel have completed a safety risk assessment related to UAS operations and established specific operational criteria to safely support manned aircraft and UAS operating within the airport's controlled airspace.[2]

If an airport and community support expansion of UAS and aviation related activity within the airspace above and around the airport, the airport can first request inclusion into the LAANC network of airports and then obtain a COA or operations to support expanded UAS commercial activity.

If an airport supports military operations, additional operating criteria must be implemented. Thus, as part of the COA, an operating manual should be developed with input from the military, ATC personnel, on-airport tenants and stakeholders as well as state DOT and FAA regional Airport personnel and FAA UAS Integration and Flight Safety staff. Note, military UAS activity will require a different set of requirements as required by DOD.

Airspace and Airport Coordination

Established UAS operational procedures should be implemented especially within controlled airport airspace to support safe integration of unmanned and manned aircraft. FAA has provided several waivers allowing UAS operations in Class B, C and D airspace if specific requirements are implemented. For example, in January 2017, the FAA allowed limited UAS operations at Hartsfield-Jackson Atlanta International Airport's (ATL) controlled airspace as part of a major parking garage expansion. However, to avoid potential impacts with manned aircraft, the UAS operator, 3D Robotics, was required to comply with the following requirements:

- The pilot/operator must have a manned pilot's license to steer the drones and have experience speaking with controllers. Currently, commercial drone pilots just need to pass a knowledge-based test which does not include radio communication and procedures training.
- Three observers watched the drones from the ground to ensure that they flew the planned routes.

[2] Based upon conference call and follow-up discussions held on September 21, 2018 with Mr. Mike Hainsey as part of Topic B, Stakeholder Discussions.

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- FAA put an operating ceiling of 200-feet to avoid any conflicts with other traffic.
- The UAS pilot kept in communication with air traffic controllers for each of the 15-minute flights to avoid any potential conflicts or changes in manned aircraft operations (e.g. aborted landings or takeoff and missed approach).
- The drones flew along automated paths but could be redirected by the pilot as needed.
- If the UAS did lose connection with the pilot, they were programmed to automatically return to one of the four launch points. The pilot also had the ability to recall the UAS to their launch points at any time during the flight (UAS Vision, 2017).

According to ongoing waivers and research, UAS operations can be safely integrated into even the busiest of airports if infrastructure, operational, and standard communication protocols are in place. Thus, operational protocols and minimum operating standards should be developed as part of any COA request. Also, the protocols used at Atlanta and other similarly sized airports may be used to facilitate integration and coordination between UAS and manned aircraft operations and can be included in an airport's COA operating standards documentation.

Communication and Coordination with Manned Aircraft

Procedures could be implemented, as they are at Golden Triangle Regional Airport, that would allow for UAS operators and ATC to coordinate both over the radio and via ground communications. This allows ATC personnel and manned aircraft users to be aware of UAS operating in the vicinity and would allow the UAS user to apply the traditional "see and avoid" procedures for collision avoidance.

If an airport will support military UAS operations along with other military and civilian activities, coordination and communication is extremely important. Larger military unmanned aircraft are equipped with navigational aids to avoid collisions with other manned or unmanned aircraft, however, most commercial UAS to date are not that sophisticated. Therefore, to facilitate UAS integration, airport management could:

- Request that the military operator provide its operating frequency to airport management who can then monitor military operations along with manned air traffic;
- Request that the military monitor the manned traffic frequencies, so that it's aware and can give right-of-way to manned traffic; and
- Obtain a memorandum of understanding with regional air traffic control and military ATC if planned operations are to be a regular occurrence.

In addition, airport management should work with airport stakeholders and air traffic personnel to establish coordination, communications, and other operating procedures to avoid conflicts between operations as well as have protocols in place in case of loss of link issues. A temporary mobile communication array system, like used at Searchlight Droneport, could be used initially to address the communication and link issues while the airport works with FAA and FCC to add another frequency to the airport's Automated Surface Observing System (ASOS). Alternatively, the airport could introduce a separate ground-based surveillance and communication such as the system currently being tested by the FAA Technical Center at Egg Harbor New Jersey.

Eventually, it is anticipated that UAS will operate like manned aircraft. Therefore, plans should be considered for developing coordination and communications procedures to address both varying size and use of manned and unmanned aircraft along with potential ATC automation and monitoring.

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Data Management

Data management like communications requires an unencumbered, flexible and secure system to allow the transfer of data between a UAS vehicle and its operator. Due to commercial, civil, and military demands on the wireless network, loss of link and data either through lost signal or cyber-attack is possible. Therefore, a secondary back-up and encrypted system can be requested from the FCC. As noted, the FAA Technology Center along with its academic partners are working on the spectrum capacity issue. However, both the commercial and military UAS segments have stated that demand outweighs capacity. Therefore, alternative data collection efforts are being evaluated by both the FAA and the ASSURE Team.

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APPENDIX E

"How to" Tools

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How to Include UAS in an Airport Master Plan

How to Include UAS in an Airport Master Plan

Contact and coordinate with FAA ADO Program and Planning Managers and State DOT Representative to obtain latest guidance and any update on funding eligibility on UAS

Format - Include a UAS Chapter or Appendix in the near term because to date AIP funding is not available for UAS associated infrastructure. Also allows for ease in updating data due to evolving technology

Typical Master Plan Steps	How to Integrate UAS
Initial Need Determination	• Document UAS Commercial Operator interest • Identify UAS testing resources • Coordinate with stakeholders, local government and the public • Conduct UAS and Airport Business and Economic Study
Study Design	Include the following in the scope of work: • Airport Strategic Vision and Objectives • Existing Conditions • Aviation Forecasts • Alternative(s) Development and Evaluation • Airport Layout Plan • Facility Implementation Plan and Financial Feasibility Analysis • Airport Capacity Evaluation • UAS Facility and Infrastructure Needs • UAS and Airport Operational Guidance • Airport Environmental and Land Use Analysis • Critical Aircraft Analysis

Facility Requirements

Use existing FAA and DOD design standards based upon available UAS operating criteria and comparably sized conventional aircraft to provide suggested design criteria for airfield requirements and support facilities.

Airfield Considerations

- Type of UAS launch and recovery systems
- Taxiway holding bays to avoid capacity delays associated with UAS Avionics calibration.
- Markings, lighting and signage to alert operators as to location of UAS activity/UAS Holding Bays/Pads

Support Facilities Considerations

- Fuel storage and handling
- UAV/vehicle storage and equipment needs
- Secured communication/control and backup power systems
- Additional wind cones, segmented circle(s) and magnetic wind rose near or adjacent to small UAS operating areas
- Additional security and safety systems to protect cable, power and other critical utilities
- Self-driving land and air vehicles and growth of electric passenger aircraft and large UAS cargo operations

Ground Access, Circulation and Parking Considerations

- Facilities that offer flexibility to support anticipated changes in technology and user demand. For example, air taxi UAS helipads could be constructed on top of the parking garage, thus providing easy access to the terminal facilities.
- Other options may include developing secure areas landside to support short-hop market.

Public Involvement Program	Refer to Volume 1 Managing and Engaging Stakeholders on Unmanned Aircraft Systems in the Vicinity of Airports
Existing Conditions	• Identify available hangar and apron space • Document airspace including nearby COAs, MOAs, restricted areas and nearby airport airspace • Identify existing air traffic control procedures and aircraft operating procedures
Aviation Forecast	• Refer to UAS Forecast How To • Identify critical aircraft characteristics by considering "right-sizing" airport infrastructure and facilities to support the most critical aircraft and/or UAS that 'regularly' utilizes those
Alternatives Development and Evaluation	• For the Short-Term, segregate small UAS from larger manned aircraft since most do not require same facilities • For the Mid & Long-Term, depending upon size, operating requirements and mission, UAS likely to integrate and use existing airfield facilities as manned aircraft • UAS environmental impacts and land use implications
Airport Layout Plan	Show: • Proposed UAS facilities • UAS Approach and departure procedures • Land Use and Zoning
Facility Implementation Plan	• Identify short, mid and long-term improvements • Primarily focus on near term and short-term recommendations and continuously reevaluate and modify suggested projects to address demand • Include flexibility into some airfield and landside projects (i.e. parking garage, terminal facilities, fuel storage, etc.) to support future UAS integration and expansion
Financial Feasibility Analysis	Since UAS development is not currently eligible for AIP funding consider: • Federal and state economic funding to support aviation related development (US Commerce Grant), • Academic or FAA Technology Center research grants, • Industry participation (public private partnerships), • Potential US DOD funds, and • Local government investments.

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How to Prepare a Stand-Alone Plan to Incorporate UAS at an Airport

General Support Facilities

On-airport UAS activity requires different support services than traditional manned aircraft operations

Fuel Storage and Handling

UAVs currently use a variety of fuel systems

Growth in electric and hybrid UAV engines will require charging stations to be located either adjacent to parking aprons or a designated fueling area should be constructed

UAV/Vehicle Storage and Equipment Needs

Hangar and storage building needs based upon UAS User needs (i.e. UAV storage, mobile communication trailer storage, parts storage, on-site office, etc.)

Additional UAS Support Facilities

Communications: airport infrastructure may be needed to accommodate secured communication/ control and backup power systems

NAVAIDs: For small UAS, additional wind cones, segmented circle(s) and magnetic wind rose near or adjacent to the UAS operating areas

Security - site development for fixed UAS facilities as well as pads for mobile operating centers should integrate additional security and safety systems to protect cable, power and other critical utilities

Office/ Administration -space requirements are dependent upon the needs of the operator

Mobile Operations Center: could be located on or adjacent to an existing apron or on separate UAS pads to segregate traffic

Fixed Administrative Facilities: could be housed above or within large hangar facilities or in an adjacent building

GA and Commercial Terminal Facilities

Consider how to accommodate new self-driving land and air vehicles and growth of electric passenger aircraft and large UAS cargo operations

Landside

Ground Access, Circulation and Parking

With the growth of UAS and automation, both on the ground and in the air, it is recommended that development of new landside facilities consider unmanned system operations when calculating demand as well as design facilities that offer flexibility to support anticipated changes in technology and user demand. Since the growth of personal air taxi is based upon helicopter technology, air taxi UAS helipads could be constructed on top of the parking garage, thus providing easy access to the terminal facilities.

Prepare Facility Recommendation Summary

- Identify short, mid and long-term improvements
- Primarily focus on near-term and short-term recommendations and continuously reevaluate and modify suggested projects to address demand
- Include flexibility into some airfield and landside projects (i.e. parking garage, terminal facilities, fuel storage, etc.)

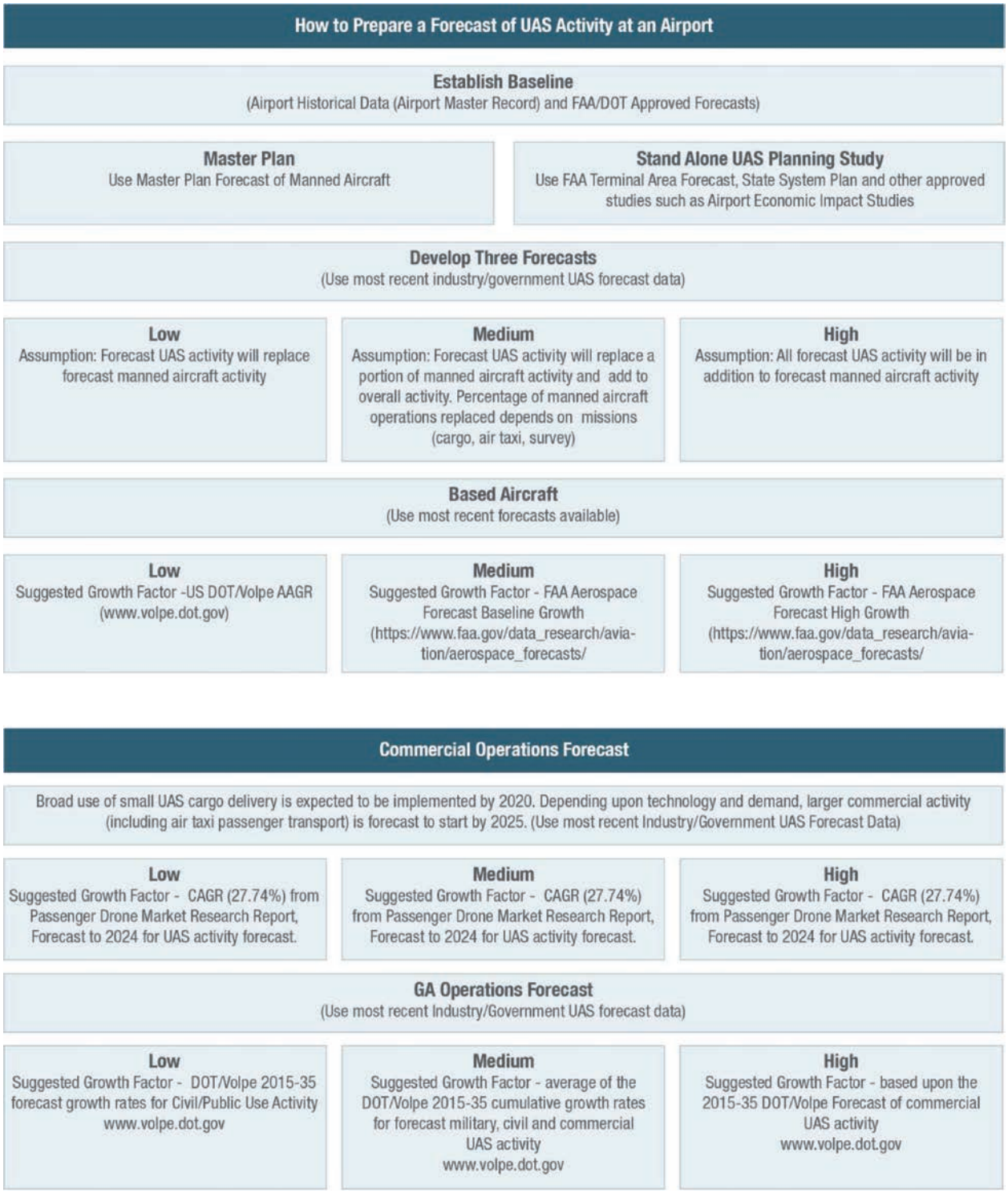
Develop Proposed Implementation and Funding Plan

Since UAS development is not currently eligible for AIP funding consider:

- Federal and state economic funding to support aviation related development (US Commerce Grant)
- Academic or FAA Technology Center research grants,
- Industry participation (public private partnerships),
- Potential US DOD funds,
- Local government investments
- Private funding/investment by tenant or community

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How to Prepare a Forecast of UAS Activity at an Airport



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Military Operations Forecast

U.S. DOD anticipates that 70 percent of their operational aircraft fleet will consist of drones by the year 2035. This includes unmanned autonomous and unmanned with optional pilot augmentation.

Low	Medium	High
Suggested Growth Factor - request information from U.S. DOD and Military	Suggested Growth Factor - request information from U.S. DOD and Military	Suggested Growth Factor - request information from U.S. DOD and Military

Itinerant and Local Operations

- Review historical data (based on aircraft, operations, and fleet mix)
- Estimate operations by based aircraft
- Consider that near term UAS operations will be local (within 20 NM of airport), but as size and technology improves likely to shift toward itinerant operations

UAS Fleet Mix Forecasts

Develop table of UAS likely to use/operate at an Airport based upon discussions with existing and potential tenants and industry data – apply to low, medium and high based aircraft and operational forecasts like manned aircraft fleet mix estimates

- Near-Term - likely small quadcopter or fixed wing drones
- Mid to Long-Term - may include larger UAS similar in size to manned fixed wing and rotor wing aircraft. However, implementation will depend upon the overall UAS market, commercial and military demand, technology and regulation

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APPENDIX F

Case Studies

Case studies were conducted to develop and evaluate the planning strategies described in this Guidebook. These case studies demonstrated how to integrate UAS into airport infrastructure planning for a variety of "real life" airport environments. The case studies also demonstrated how to integrate UAS into the development of a master plan and how to develop a stand-alone UAS planning study. The subject airports included Atlantic City International Airport, Middle Georgia Regional Airport and Sebring Regional Airport. Airport management from these three airports provided input and suggestions based upon their current experiences and planned development. Table F-1 presents a summary of the focus of each of the case studies. For brevity, the case studies are only summarized here.

The basic premise for the case studies was that a UAS is an aircraft. This premise was based on the following FAA and ICAO definitions of UAS.

> *"An unmanned aircraft system (UAS), sometimes called a drone, is an aircraft without a human pilot onboard – instead, the UAS is controlled from an operator on the ground."* (FAA, 2018)

> *"Unmanned Aircraft System. An aircraft and its associated elements which are operated with no pilot on board."* (ICAO, 2011)

Case Study 1: Atlantic City International Airport

The South Jersey Transportation Authority (SJTA) is seeking to integrate international and domestic manned commercial, general aviation, and military operations with unmanned commercial and civil aviation expansion at ACY. ACY already supports scheduled commercial service and military operations as well as UAS testing, training and research. Spirit provides direct passenger airline service to and from ten major cities. ACY is a base for the New Jersey Air National Guard's 177th Fighter Wing and the United States Coast Guard Air Station Atlantic City. ACY is also home to the FAA TC and the National Aviation and Technology Research Park who conduct UAS testing, training and research in addition to other aviation research and development. ACY is also one of several airports included in New Jersey's aviation plans to attract and support UAS developers and operators by creating a UAS/Aviation Zone. Thus, this case study provided infrastructure examples for integration of UAS commercial, civil, and military operations at a joint-use airport.

Case Study 2: Middle Georgia Regional Airport

Middle Georgia Regional Airport (MCN) supports limited passenger commercial operations as well as supports several large Maintenance Repair and Overhaul (MRO) and aircraft manufacturing operations. The City of Macon, Georgia owns and operates both the MCN and Macon

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Table F-1. Case study airports and focus areas.

Airport	Operating Classification	Airspace Class	Case Study Focus
Atlantic City International Airport (ACY), Atlantic City, New Jersey	Primary Commercial Service - Small Hub	Class C	Integrate civil, commercial/business (e.g. air taxi), and cargo UAS with commercial passenger and military activities Support UAS training in addition to academic and federal research and development programs Develop stand-alone UAS infrastructure planning study
Middle Georgia Regional Airport (MCN), Macon, Georgia	Non-Primary Commercial Service	Class D	Support military operations (Robins Air Force Base) Establish R&D and commercial UAS park Support expansion of Maintenance, Repair and Overhaul facilities to support UAS demand. Develop stand-alone UAS infrastructure planning study
Sebring Regional Airport (SEF), Sebring, Florida	Non-Primary Regional General Aviation	Class E	Attract commercial UAS tenants Supporting R&D, Avon Park Air Force Range, and academic UAS development Promote educational and UAS support opportunities as part of new logistics center Integrate UAS infrastructure planning into 2018 airport master plan update.
Notes: MCN regained commercial service as of late 2017. October 2018 enplanements exceeded 10,000.			

Downtown Airport. The Airport is also located approximately 4 nautical miles from Warner Robins Air Force Base, which supports military UAS maintenance and training operations. Discussions between the Air Force and the City are ongoing regarding using MCN as another site for UAS research, maintenance, manufacturing and other needs. Thus, this case study considered the impacts and opportunities associated with integrating research, civilian and military UAS operations with large manned commercial operations.

Case Study 3: Sebring Regional Airport

This case study will focus on a regional, rural general aviation airport that is already supporting limited UAS operations in addition to manned sport aircraft, general aviation, and corporate operations. Sebring Regional Airport (SEF) also supports an industrial park as well as on-site international and national automobile racing facilities. The City of Sebring and the Airport Authority originally received its FAA 333 Certification in 2015, and is in the process of teaming with several local universities and colleges to attract UAS operations and tenants to the airport. This case study identified facilities and operational examples to support research, training, and commercial manufacturing of UAS at a general aviation airport.

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APPENDIX G

Other Resources

The following is a sampling of resources that airport sponsors and management may review to obtain the latest legislation, technology, funding, and regulatory guidance related to UAS development and integration. This is not an exhaustive list, but most sites also provide additional links to other governmental (i.e., FCC, U.S. Department of Homeland Security, and U.S. DOT), academic or business-related resources.

Governmental Organizations

Federal Aviation Administration Unmanned Aircraft Systems (FAA UAS)

Description: The FAA's UAS website contains up-to-date information on UAS regulations, certification and operating requirements, ongoing programs and partnerships, as well as frequently asked questions and links to other key resources.

Why it is a resource: This is the definitive source for UAS regulatory, operational and licensing requirements in the United States. The site provides the most up-to-date data on current regulations and rules related to UAS operations and integration with NAS.

Link: https://www.faa.gov/uas/

International Civil Aviation Organization (ICAO)

Description: ICAO creates Standards and Recommended Practices (SARPs) as well as Procedures for Air Navigation (PANS) for the "safe, efficient, secure, economically sustainable and environmentally responsible civil aviation sector" (ICAO, n.d.).

Why it is a resource: The ICAO website contains a "how to" guide to developing SARPs and PANS; information on the Unmanned Aircraft Systems Advisory Group (UAS-AG) which develops guidance on the harmonization of domestic UAS operations in order to maintain safety standards for manned aircraft as part of the UTM system; a UAS toolkit; and information about the Remotely Piloted Aircraft Systems Panel (RPASP) which develops material to facilitate the integration of RPAs into non-segregated airspace and airports. The UAS toolkit also provides access to 35 different countries, including the United States, state drone regulations to support international cooperation and business development.

Link: https://www.icao.int/safety/UA/Pages/default.aspx

European Union Aviation Safety Agency (EASA)

Description: The EASA is responsible for the certification, regulation, and creation of standards for the civil aviation industry in the European Union.

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Why it is a resource: EASA's website contains news and content related to the operation of UAS in the European airspace, as well as updates on the status of EASA's proposed amendment to provide standardized regulatory framework for UAS operations throughout Europe.

Link: https://www.easa.europa.eu/easa-and-you/civil-drones-rpas

Florida DOT—Aviation and Spaceports Office

Description: Contains information on how to build a new airport to assist planning efforts through the Florida Aviation System Plan. Also contains description of several studies about air demand/capacity, economic impact, air cargo, as well as aviation forecasts and the Guidebook for Airport Master Planning. POC information is available in addition to links and downloadable documents.

Why it is a resource: A list of references to FAA ACs and other publications is available to assist during the planning process. Users can find information on the travel patterns, demand for service, non-stop destinations and factors that affect the cost of fuel. The Airport Demand/Capacity study contains forecasts for years 2020 through 2060.

Link: http://www.fdot.gov/aviation/planning.shtm

North Carolina DOT

Description: The North Carolina Department of Transportation aims at ensuring unmanned aircraft are flown safely and responsibly by implementing and managing regulations related to the operation of drones in the airspace of North Carolina.

Why it is a resource: The North Carolina Department of Transportation's website contains safety and regulatory resources pertaining to the operation of drones in general, as well as federal and state requirements and guidelines for recreational, commercial, and government operators. The website also contains laws, regulations, statutes, and guides to the N.C. DOT's UAS Operator's Knowledge Test.

Link: https://www.ncdot.gov/divisions/aviation/uas/Pages/default.aspx

National Conference of State Legislatures (NCSL)

Description: NCSL is a national organization that supports, defends and strengthens state legislatures. Since 2013, state lawmakers have debated if and how UAS technology should be regulated to address privacy concerns and economic impacts. To date, more than 41 states have enacted laws and/or resolutions regarding UAS activity.

Why it is a resource: The NCSL website provides links and information regarding state UAS legislation (2013–2017), federal UAS legislation, state resources, NCSL resources in addition to other resources such as Congressional Reports. The site also provides recent information regarding public policy, government fiscal policy as well as trends in transportation.

Link: http://www.ncsl.org/research/transportation/current-unmanned-aircraft-state-law-landscape.aspx

UAS Organizations

Association for Unmanned Vehicle Systems International (AUVSI)

Description: AUVSI is made of corporations and industry, government, and academia professionals from over 60 countries, making it the world's largest non-profit organization.

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It promotes collective interest in unmanned systems technologies in the recent emergence of commercial and demand in consumer applications.

Why it is a resource: AUVSI members benefit from access to innovative resources in this fast-paced and continually evolving industry.

Link: https://www.auvsi.org/

National Business Aviation Association (NBAA)

Description: NBAA is a major aviation organization which supports the business aviation industry's diverse interests. The association represents more than 11,000 companies and provides a variety of services and products.

Why is it a resource: NBAA works with the aviation business community to address certification, operational and safety needs related to UAS integration. The Association advocates for the business aviation community and engages with federal and state regulatory agencies to develop recommendations and guidance for UAS development.

Link: https://www.nbaa.org/ops/uas/

UAV Systems Association (UAVSA)

Description: UAVSA was created in 2014 to support the sUAS and drone community. UAVSA provides its members a variety of tools and resources including insurance, compliance and networking to support small UAS businesses.

Why is it a resource: The site provides a variety of domestic and international networking opportunities, compliance information, on-site digital log books, UAS insurance information, industry collaborations, and blog highlighting industry trends and new legislation. Services specifically available to members include Part 107 and 333 Exemption assistance, business development and insurance solutions.

Link: http://www.uavsa.org/

Law Firms

Rupprecht Law

Description: Jonathan Rupprecht, JD, Esq., CFI, CFI-I is a lawyer, FAA-certified commercial pilot with single and multi-engine ratings, flight instructor, and adjunct professor at Embry-Riddle Aeronautical University for the Unmanned Systems Legal and Regulatory Compliance class. As part of his attorney practice Rupprecht Law, P.A., he has successfully obtained 333 exceptions to FAA Part 107.

Why it is a resource: Drone law is one of the main practice areas of Rupprecht Law, with special emphasis on obtaining FAA approval for commercial drone operations. The website also contains articles, guidebook, and information about drone laws as well as lawsuits involving drones.

Link: https://jrupprechtlaw.com/

Hogan Lovells Law Firm

Description: One of the main capabilities of the Washington D.C.-based law firm Hogan Lovells is unmanned aircraft systems. The firm provides counsel related to drone regulation in various countries; assistance to secure FAA waivers to fly UAS over the public; advises customers

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on flying UAS beyond line of sight, in urban areas, at night, or outside of Part 107 boundaries; and counsels companies about privacy issues related to the use of drones.

Why it is a resource: The website contains publications, events, firm news, and blog articles concerning recent cases and regulations related to UAS activities.

Link: https://www.hoganlovells.com/en/service/unmanned-aircraft-systems

Magazines and Journals

DroneWorld

Description: DroneWorld is an authorized DJI Dealer and 5-star member of independent customer review site Trust Pilot.

Why it is a resource: Customers will find various brands and models of UAV for sale, as well as associated accessories and spare parts.

Link: https://www.drone-world.com/

UAS Magazine

Description: UAS Magazine is an online publication whose audience is commercial manufacturers and operators, as well as those wanting to expand their knowledge of unmanned aircraft operations. The articles cover the latest technologies for the civil, defense, and commercial markets of unmanned aerial systems.

Why it is a resource: UAS Magazine covers the latest news on the topics of manufacturing & distribution, operations, engineering and design, policy and regulation, payload.

Link: http://www.uasmagazine.com/

Journal of Unmanned Aerial Systems

Description: The Journal of Unmanned Aerial Systems is a peer-reviewed and open-source UAS online resource related to UAS applications and technologies.

Why it is a resource: Readers can download volumes 1, 2, and 3 of the Journal of Unmanned Aerial Systems at no cost.

Link: http://uasjournal.org/

Top Drone Magazines

Description: This website contains a list of the following major magazines related to drones: RotorDrone Magazine, Unmanned Vehicles, Unmanned Systems, Inside Unmanned Systems, Drone Magazine (Australia), Drone User Magazine, Drone Magazine (UK), DRONES Monthly, Drone MULTIROTOR, Journal of Unmanned Vehicle Systems, International Journal of Unmanned Systems Engineering. The list contains basic information for each magazine, as well as a description of their content.

Why it is a resource: Readers will find articles on a variety of topics, from aerial, ground, and maritime unmanned systems, to reviews for first-time buyers, and updates on new technologies and upcoming rules and regulations.

Link: https://skytango.com/top-drone-magazines/

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DroneLife

Description: DRONELIFE shares the latest updates on drone-related news, new technologies and products, videos, and legal precedents.

Why it is a resource: Readers can find product reviews, information and articles about a variety of industries (e.g. agriculture, inspection, police and fire, delivery, real estate, surveying, and construction), as well as articles about businesses investing in drone technology.

Link: https://dronelife.com/

Other Resources

Embry-Riddle Aeronautical University Library Resources

Description: Embry-Riddle Aeronautical University is one of a few four-year universities that offer bachelor and master degrees in unmanned aerial systems in addition to providing extensive research and development in the field.

Why it is a resource: The ERAU library provides access to a variety of resources including books, e-books, latest versions of electronic aeronautical information publications, UAS Law, International UAS regulations, links to various UAS organizations and other useful websites.

Link: https://erau.libguides.com/UAS

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Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TDC	Transit Development Corporation
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

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